Rscript for the suspension analysis

Pipeline following the python processing of data

Libraries and Data

In [1]:
```
library(Matrix)
library(matrixStats)
library(anndata)
library(rhdf5)
library(parallel)
library(reformulas)
library(variancePartition)
```

```
Loading required package: ggplot2

Loading required package: limma

Loading required package: BiocParallel


Attaching package: ‘variancePartition’


The following objects are masked from ‘package:limma’:

    eBayes, topTable
```

In [2]:
```
setwd("../")
```

In [3]:
```
func.dir="functions"
source("functions/load_functions.r")
```

```
Loading required package: RColorBrewer
```

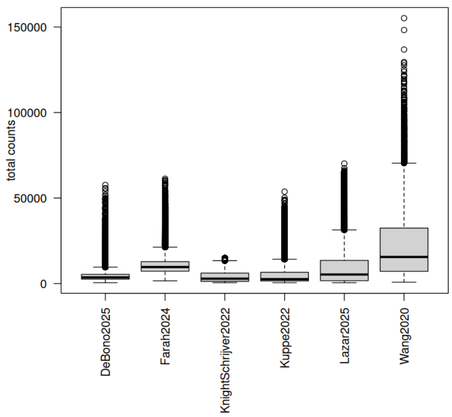
```
The following functions contained errors and were not loaded properly, please check the code:
```

In [4]:
```
# Read the anndata
adata = anndata::read_h5ad("data/suspension/anndata_markersOnly.h5ad")
```

In [5]:
```
# pick up the raw counts matrix:
x = adata$layers[["counts"]]
```

In [6]:
```
# Pick up the metadata
metadata = adata$obs
 # re-level classes
   metadata$new_clusters = factor(
     as.character(metadata$new_clusters),
     levels=levels(metadata$new_clusters)[c(20,18,2,17,15,7,9,8,3,4,5,6,10,11,13,14,12,1,16,19)]
   )

umap = adata$obsm$X_umap
```

In [7]:
```
# check the depths of all datasets
counts_df = data.frame(
  "y" = adata$obs$total_counts,
  "x" = adata$obs$dataset
)
par(oma=c(5,1,1,1))
boxplot(y~x, counts_df, las=2, xlab="", ylab="total counts")
```



Detection Thresholds above ambient RNA using Poisson Null distribution

We noted that the depths of our datasets differed. With the well-based assay had a higher depth in general. I decided to approach "detection" of EDN* transcripts using a detection threshold under the assumption that ambient RNA was present in reasonable proportions. This was seen in a few datasets, DeBono2025 CM genes for example.

This was modelled on a Poisson Null distribution for each gene, in each dataset where the Lambda was assumed to be the mean of all counts for each gene in each dataset. The threshold for detection for each gene in each dataset was set to the 0.95 quantile, i.e. above the count value at which only 5 % of cells would have this value by chance.

In [8]:
```
# Detection thresholds:
  dataset_thresholds = lapply(
    levels(adata$obs$dataset),
    function(dataset){
       dataset_metadata = adata$obs[adata$obs$dataset == dataset,]
       x_dataset = x[adata$obs$dataset == dataset,]

       set.seed(42)
       sampled_cells = sample.equal(
         rownames(x_dataset),
         dataset_metadata$new_clusters,
         300,
         override=T
       )
       counts = t(x_dataset[sampled_cells,])

      # Using a Poisson Null Distribution for the selected genes
      # Calculate a 'Noise Rate' (lambda) for each gene - the mean as a parameter for "ambient RNA".
        gene_lambdas <- rowMeans(counts)

      # Find the 95% confidence threshold:
      # "What is the count value where only 5% of cells would have this many counts by pure random chance?"
        objective_thresholds <- qpois(0.95, lambda = gene_lambdas)
        objective_thresholds[is.na(objective_thresholds)] = 0

      return(
        list(
          "gene_lambdas" = gene_lambdas,
          "obj_thresholds" = objective_thresholds
        )
      )
    }
  )
  names(dataset_thresholds) = levels(adata$obs$dataset)
```

In [9]:
```
dataset_thresholds
```

$DeBono2025	**$gene_lambdas**	**TNNT2:** 5.42138888888889 **MYH7:** 8.22888888888889 **MYH6:** 4.19666666666667 **NR2F2:** 0.4325 **SHOX2:** 0.0480555555555556 **VSNL1:** 0.122222222222222 **HCN4:** 0.163888888888889 **MYH11:** 0.246944444444444 **ACTA2:** 0.455277777777778 **RGS5:** 0.387777777777778 **CYGB:** 0.0491666666666667 **PECAM1:** 0.568055555555556 **NPR3:** 0.323611111111111 **ADGRG6:** 0.410555555555556 **ACKR1:** 0.00194444444444444 **HEY1:** 0.0172222222222222 **FBLN5:** 0.226944444444444 **RGCC:** 0.0369444444444444 **TFF3:** 0.00361111111111111 **NFATC1:** 0.271111111111111 **ITLN1:** 0.0669444444444444 **UPK3B:** 0.0161111111111111 **LUM:** 0.0486111111111111 **COL1A1:** 1.39 **NRXN1:** 1.50972222222222 **ENO2:** 0.0275 **CHGA:** 0.00277777777777778 **CHGB:** 0.00388888888888889 **MPZ:** 0.0161111111111111 **S100B:** 0.0127777777777778 **PTPRC:** 0.0677777777777778 **ADIPOQ:** 0.000277777777777778 **HBB:** 0.153611111111111 **ITGB3:** 0.140555555555556 **ITGA2B:** 0.0222222222222222 **EDNRA:** 0.937777777777778 **EDNRB:** 0.133333333333333 **EDN1:** 0.110555555555556 **EDN2:** 0 **EDN3:** 0.00666666666666667 **ECE1:** 0.6375 **ECE2:** 0.00444444444444444
	$obj_thresholds	**TNNT2:** 9 **MYH7:** 13 **MYH6:** 8 **NR2F2:** 2 **SHOX2:** 0 **VSNL1:** 1 **HCN4:** 1 **MYH11:** 1 **ACTA2:** 2 **RGS5:** 2 **CYGB:** 0 **PECAM1:** 2 **NPR3:** 1 **ADGRG6:** 2 **ACKR1:** 0 **HEY1:** 0 **FBLN5:** 1 **RGCC:** 0 **TFF3:** 0 **NFATC1:** 1 **ITLN1:** 1 **UPK3B:** 0 **LUM:** 0 **COL1A1:** 4 **NRXN1:** 4 **ENO2:** 0 **CHGA:** 0 **CHGB:** 0 **MPZ:** 0 **S100B:** 0 **PTPRC:** 1 **ADIPOQ:** 0 **HBB:** 1 **ITGB3:** 1 **ITGA2B:** 0 **EDNRA:** 3 **EDNRB:** 1 **EDN1:** 1 **EDN2:** 0 **EDN3:** 0 **ECE1:** 2 **ECE2:** 0
$Farah2024	**$gene_lambdas**	**TNNT2:** 15.5549705139006 **MYH7:** 5.71714406065712 **MYH6:** 6.49010109519798 **NR2F2:** 2.04254422914912 **SHOX2:** 0.201558550968829 **VSNL1:** 0.208719460825611 **HCN4:** 0.0827716933445661 **MYH11:** 0.244102780117944 **ACTA2:** 6.82434709351306 **RGS5:** 4.60699241786015 **CYGB:** 1.04401853411963 **PECAM1:** 2.42396798652064 **NPR3:** 0.530117944397641 **ADGRG6:** 0.24262847514743 **ACKR1:** 0.0553917438921651 **HEY1:** 0.367733782645324 **FBLN5:** 0.856149957877001 **RGCC:** 1.44945240101095 **TFF3:** 1.91069924178602 **NFATC1:** 0.219039595619208 **ITLN1:** 7.54591406908172 **UPK3B:** 0.50737152485257 **LUM:** 1.74115417017692 **COL1A1:** 16.7049283909014 **NRXN1:** 0.32413647851727 **ENO2:** 0.198820556023589 **CHGA:** 0.0857203032855939 **CHGB:** 0.0819292333614153 **MPZ:** 0.365627632687447 **S100B:** 1.61878685762426 **PTPRC:** 0.0911962931760741 **ADIPOQ:** 0 **HBB:** 0.198188711036226 **ITGB3:** 0.0162173546756529 **ITGA2B:** 0.0553917438921651 **EDNRA:** 0.413437236731255 **EDNRB:** 0.639427127211457 **EDN1:** 0.580244313395114 **EDN2:** 0.0040016849199663 **EDN3:** 0.0132687447346251 **ECE1:** 0.582350463352991 **ECE2:** 0
	$obj_thresholds	**TNNT2:** 22 **MYH7:** 10 **MYH6:** 11 **NR2F2:** 5 **SHOX2:** 1 **VSNL1:** 1 **HCN4:** 1 **MYH11:** 1 **ACTA2:** 11 **RGS5:** 8 **CYGB:** 3 **PECAM1:** 5 **NPR3:** 2 **ADGRG6:** 1 **ACKR1:** 1 **HEY1:** 2 **FBLN5:** 3 **RGCC:** 4 **TFF3:** 4 **NFATC1:** 1 **ITLN1:** 12 **UPK3B:** 2 **LUM:** 4 **COL1A1:** 24 **NRXN1:** 1 **ENO2:** 1 **CHGA:** 1 **CHGB:** 1 **MPZ:** 2 **S100B:** 4 **PTPRC:** 1 **ADIPOQ:** 0 **HBB:** 1 **ITGB3:** 0 **ITGA2B:** 1 **EDNRA:** 2 **EDNRB:** 2 **EDN1:** 2 **EDN2:** 0 **EDN3:** 0 **ECE1:** 2 **ECE2:** 0
$KnightSchrijver2022	**$gene_lambdas**	**TNNT2:** 4.3323709852039 **MYH7:** 1.00180440274269 **MYH6:** 1.59852038975099 **NR2F2:** 0.98159509202454 **SHOX2:** 0.119451461566222 **VSNL1:** 0.0990617105738001 **HCN4:** 0.023276795380729 **MYH11:** 0.365211115120895 **ACTA2:** 2.76380368098159 **RGS5:** 1.35709130277878 **CYGB:** 0.16600505232768 **PECAM1:** 0.848430169613858 **NPR3:** 0.217791411042945 **ADGRG6:** 0.155359076145796 **ACKR1:** 0.299711295561169 **HEY1:** 0.17286178274991 **FBLN5:** 0.669433417538795 **RGCC:** 0.415734391916276 **TFF3:** 0.339588596174666 **NFATC1:** 0.0759653554673403 **ITLN1:** 1.47942980873331 **UPK3B:** 0.168531216167449 **LUM:** 0.540057740887766 **COL1A1:** 3.95579213280404 **NRXN1:** 0.35258029592205 **ENO2:** 0.0660411403825334 **CHGA:** 0.156802598339949 **CHGB:** 0.310176831468784 **MPZ:** 0.147600144352219 **S100B:** 0.4976542764345 **PTPRC:** 0.111512089498376 **ADIPOQ:** 0.0647780584626489 **HBB:** 0.673222663298448 **ITGB3:** 0.00848069289065319 **ITGA2B:** 0.0102850956333454 **EDNRA:** 0.192710212919524 **EDNRB:** 0.288523998556478 **EDN1:** 0.557019126669072 **EDN2:** 0.00613496932515337 **EDN3:** 0.0153374233128834 **ECE1:** 0.371706964994587 **ECE2:** 0.00162396246842295
	$obj_thresholds	**TNNT2:** 8 **MYH7:** 3 **MYH6:** 4 **NR2F2:** 3 **SHOX2:** 1 **VSNL1:** 1 **HCN4:** 0 **MYH11:** 2 **ACTA2:** 6 **RGS5:** 3 **CYGB:** 1 **PECAM1:** 3 **NPR3:** 1 **ADGRG6:** 1 **ACKR1:** 1 **HEY1:** 1 **FBLN5:** 2 **RGCC:** 2 **TFF3:** 1 **NFATC1:** 1 **ITLN1:** 4 **UPK3B:** 1 **LUM:** 2 **COL1A1:** 7 **NRXN1:** 1 **ENO2:** 1 **CHGA:** 1 **CHGB:** 1 **MPZ:** 1 **S100B:** 2 **PTPRC:** 1 **ADIPOQ:** 1 **HBB:** 2 **ITGB3:** 0 **ITGA2B:** 0 **EDNRA:** 1 **EDNRB:** 1 **EDN1:** 2 **EDN2:** 0 **EDN3:** 0 **ECE1:** 2 **ECE2:** 0
$Kuppe2022	**$gene_lambdas**	**TNNT2:** 2.98155440414508 **MYH7:** 2.14134715025907 **MYH6:** 2.24704663212435 **NR2F2:** 0.184870466321244 **SHOX2:** 0.00849740932642487 **VSNL1:** 0.0478756476683938 **HCN4:** 0.0126424870466321 **MYH11:** 0.294507772020725 **ACTA2:** 0.197720207253886 **RGS5:** 0.227979274611399 **CYGB:** 0.0281865284974093 **PECAM1:** 0.740103626943005 **NPR3:** 0.0870466321243523 **ADGRG6:** 0.123523316062176 **ACKR1:** 0.0232124352331606 **HEY1:** 0.0286010362694301 **FBLN5:** 0.13699481865285 **RGCC:** 0.195854922279793 **TFF3:** 0.00808290155440415 **NFATC1:** 0.168704663212435 **ITLN1:** 0.00103626943005181 **UPK3B:** 0.00227979274611399 **LUM:** 0.0992746113989637 **COL1A1:** 0.340103626943005 **NRXN1:** 2.1560621761658 **ENO2:** 0.0161658031088083 **CHGA:** 0 **CHGB:** 0.00248704663212435 **MPZ:** 0.0184455958549223 **S100B:** 0.0225906735751295 **PTPRC:** 0.155854922279793 **ADIPOQ:** 0.532849740932642 **HBB:** 0.0304663212435233 **ITGB3:** 0.0540932642487047 **ITGA2B:** 0.0190673575129534 **EDNRA:** 0.246424870466321 **EDNRB:** 0.135336787564767 **EDN1:** 0.0265284974093264 **EDN2:** 0.00103626943005181 **EDN3:** 0 **ECE1:** 0.419067357512953 **ECE2:** 0.000414507772020725
	$obj_thresholds	**TNNT2:** 6 **MYH7:** 5 **MYH6:** 5 **NR2F2:** 1 **SHOX2:** 0 **VSNL1:** 0 **HCN4:** 0 **MYH11:** 1 **ACTA2:** 1 **RGS5:** 1 **CYGB:** 0 **PECAM1:** 2 **NPR3:** 1 **ADGRG6:** 1 **ACKR1:** 0 **HEY1:** 0 **FBLN5:** 1 **RGCC:** 1 **TFF3:** 0 **NFATC1:** 1 **ITLN1:** 0 **UPK3B:** 0 **LUM:** 1 **COL1A1:** 1 **NRXN1:** 5 **ENO2:** 0 **CHGA:** 0 **CHGB:** 0 **MPZ:** 0 **S100B:** 0 **PTPRC:** 1 **ADIPOQ:** 2 **HBB:** 0 **ITGB3:** 1 **ITGA2B:** 0 **EDNRA:** 1 **EDNRB:** 1 **EDN1:** 0 **EDN2:** 0 **EDN3:** 0 **ECE1:** 2 **ECE2:** 0
$Lazar2025	**$gene_lambdas**	**TNNT2:** 7.43859649122807 **MYH7:** 2.03052631578947 **MYH6:** 4.39280701754386 **NR2F2:** 3.21877192982456 **SHOX2:** 0.146315789473684 **VSNL1:** 0.172456140350877 **HCN4:** 0.0768421052631579 **MYH11:** 0.564912280701754 **ACTA2:** 4.45 **RGS5:** 3.87350877192982 **CYGB:** 0.764035087719298 **PECAM1:** 4.77912280701754 **NPR3:** 0.873684210526316 **ADGRG6:** 0.90280701754386 **ACKR1:** 0.0833333333333333 **HEY1:** 0.752982456140351 **FBLN5:** 1.5719298245614 **RGCC:** 3.70684210526316 **TFF3:** 1.78122807017544 **NFATC1:** 1.42877192982456 **ITLN1:**

3.25614035087719 **UPK3B:** 0.30719298245614 **LUM:** 1.84666666666667 **COL1A1:** 17.1321052631579 **$obj_thresholds** **$Wang2020**
NRXN1: 4.74947368421053 **ENO2:** 0.195964912280702 **CHGA:** 0.107719298245614 **CHGB:** 0.268070175438597 **MPZ:** 0.640877192982456 **S100B:** 2.2540350877193 **PTPRC:** 0.281754385964912
ADIPOQ: 0.00894736842105263 **HBB:** 1.97052631578947 **ITGB3:** 0.00192982456140351 **ITGA2B:** 0.153333333333333 **EDNRA:** 0.83280701754386 **EDNRB:** 1.56561403508772 **EDN1:** 0.858421052631579 **EDN2:** 0.00842105263157895 **EDN3:** 0.0349122807017544 **ECE1:** 1.75684210526316 **ECE2:** 0.0107017543859649
TNNT2: 12 **MYH7:** 5 **MYH6:** 8 **NR2F2:** 6 **SHOX2:** 1 **VSNL1:** 1 **HCN4:** 1 **MYH11:** 2 **ACTA2:** 8 **RGS5:** 7 **CYGB:** 2 **PECAM1:** 9 **NPR3:** 3 **ADGRG6:** 3 **ACKR1:** 1 **HEY1:** 2 **FBLN5:** 4 **RGCC:** 7 **TFF3:** 4 **NFATC1:** 4 **ITLN1:** 6
UPK3B: 1 **LUM:** 4 **COL1A1:** 24 **NRXN1:** 9 **ENO2:** 1 **CHGA:** 1 **CHGB:** 1 **MPZ:** 2 **S100B:** 5 **PTPRC:** 1
ADIPOQ: 0 **HBB:** 5 **ITGB3:** 0 **ITGA2B:** 1 **EDNRA:** 3 **EDNRB:** 4 **EDN1:** 3 **EDN2:** 0 **EDN3:** 0 **ECE1:** 4 **ECE2:** 0

$gene_lambdas **TNNT2:** 92.7935540069686 **MYH7:** 57.3379790940767 **MYH6:** 80.3969221835076 **NR2F2:** 2.1506968641115 **SHOX2:** 0.0484901277584204 **VSNL1:** 0.292392566782811 **HCN4:** 0.0479094076655052 **MYH11:** 4.92479674796748 **ACTA2:** 1.4027293844367 **RGS5:** 11.1054006968641
CYGB: 0.470383275261324 **PECAM1:** 6.90360046457607 **NPR3:** 0.921893147502904 **ADGRG6:** 0.383565621370499 **ACKR1:** 2.22735191637631 **HEY1:** 0.777293844367015 **FBLN5:** 0.878339140534262 **RGCC:** 5.05516840882695 **TFF3:** 2.50609756097561 **NFATC1:** 0.276132404181185
ITLN1: 24.862950058072 **UPK3B:** 0.0421022067363531 **LUM:** 2.31010452961672 **COL1A1:** 0.603077816492451 **NRXN1:** 0.219802555168409 **ENO2:** 0.264227642276423 **CHGA:** 0.0165505226480836 **CHGB:** 1.01480836236934 **MPZ:** 0.0116144018583043 **S100B:** 0.127758420441347 **PTPRC:** 1.46631823461092 **ADIPOQ:** 0.00232288037166086 **HBB:** 0.861788617886179 **ITGB3:** 0.0275842044134727 **ITGA2B:** 0.0223577235772358 **EDNRA:** 0.519163763066202 **EDNRB:** 1.56533101045296 **EDN1:** 0.697154471544715 **EDN2:** 0.00232288037166086 **EDN3:** 0.00174216027874564 **ECE1:** 1.1965737514518 **ECE2:** 0.0481997677119628

$obj_thresholds **TNNT2:** 109 **MYH7:** 70 **MYH6:** 95 **NR2F2:** 5 **SHOX2:** 0 **VSNL1:** 1 **HCN4:** 0 **MYH11:** 9 **ACTA2:** 4 **RGS5:** 17
CYGB: 2 **PECAM1:** 11 **NPR3:** 3 **ADGRG6:** 2 **ACKR1:** 5 **HEY1:** 2 **FBLN5:** 3 **RGCC:** 9 **TFF3:** 5 **NFATC1:** 1
ITLN1: 33 **UPK3B:** 0 **LUM:** 5 **COL1A1:** 2 **NRXN1:** 1 **ENO2:** 1 **CHGA:** 0 **CHGB:** 3 **MPZ:** 0 **S100B:** 1 **PTPRC:** 4
ADIPOQ: 0 **HBB:** 3 **ITGB3:** 0 **ITGA2B:** 0 **EDNRA:** 2 **EDNRB:** 4 **EDN1:** 2 **EDN2:** 0 **EDN3:** 0 **ECE1:** 3 **ECE2:** 0

Processing data matrices and Metadata

In [10]:
```
# generate a new "cut matrix" for reporting
# apply a cut onto the object
  x_thresholded = as.matrix(x)

  for(dataset in levels(adata$obs$dataset)){
    x_dataset = as.matrix(x[adata$obs$dataset == dataset,])

    for(gene in colnames(x_dataset)){
      x_dataset[x_dataset[,gene] < dataset_thresholds[[dataset]][["obj_thresholds"]][gene],gene] = 0
    }

    x_thresholded[adata$obs$dataset == dataset,] = x_dataset
  }
```

In [11]:
```
# Normalisation / depth / Counts per 10K reads / log transformation
# make new "norm" object
x_norm = x/metadata$total_counts # divide by total counts
x_norm = x*1e4                   # multiply by 10 k
x_norm = log(x+1)                # natural log transformation, with a pseudocount of 1

x_thresholded_norm = x_thresholded/metadata$total_counts      # divide by total counts
x_thresholded_norm = x_thresholded_norm*1e4                   # multiply by 10 k
x_thresholded_norm = log(x_thresholded_norm+1)                # natural log transformation, with a pseudocount of 1
```

In [12]:
```
# Define Gene sets
# forming expression matrices
genes_EDN = c(
  "EDN1",
  "EDN2",
  "EDN3",
  "EDNRA",
  "EDNRB",
  "ECE1",
  "ECE2"
)

genes_markers = c(
  "TNNT2",
  "MYH7",
  "MYH6", "NR2F2",
  "SHOX2",
  "VSNL1", "HCN4",
  "MYH11", "ACTA2",
  "RGS5", "CYGB",
  "PECAM1",
  "NPR3", "ADGRG6",
  "ACKR1",
  "HEY1", "FBLN5",
  "RGCC",
  "TFF3",
  "NFATC1",
  "ITLN1", "UPK3B",
  "LUM", "COL1A1",
  "NRXN1", "ENO2",
  "CHGA", "CHGB",
  "MPZ", "S100B",
```

```
    "PTPRC",
    "ADIPOQ",
    "HBB",
    "ITGA2B", "ITGB3"
)
```

In [13]:
```
# some colours to save for use here
 cluster.cols = data.frame(
    "cluster.id" = levels(metadata$new_clusters),
    "color" = sample(Discrete.U2,length(levels(metadata$new_clusters)))#[1:length(levels(metadata$new_clusters))]
  )
  cluster.cols$merged_labels = cluster.cols[,1]
  #cluster.cols$merged_labels[grep("EC_Lymphatic", cluster.cols$merged_labels)] = "otherCM"
  #cluster.cols$merged_labels[grep("EC_", cluster.cols$merged_labels)] = "BEC"
  #cluster.cols$merged_labels[grep("EC_", cluster.cols$merged_labels)] = "BEC"
  #cluster.cols$merged_labels[grep("Schwann", cluster.cols$merged_labels)] = "SchwannGlial"
  #cluster.cols$merged_labels[grep("Neurons_M", cluster.cols$merged_labels)] = "Neurons"

  #rownames(cluster.cols) = cluster.cols[,1]
  # #  write.table(cluster.cols, file="cluster_colours.csv", sep=",", col.names=T, row.names=F, quote=F) # Written, do not overwrite
  cluster.cols = read.csv("cluster_colours.csv")
  rownames(cluster.cols) = cluster.cols[,1]
```

Technical threshold plots

In [14]:
```
counts = t(as.matrix(x))
```

In [15]:
```
dataset = "DeBono2025"
dataset_mask = metadata$dataset %in% dataset
table(dataset_mask)
```

```
dataset_mask
 FALSE   TRUE
461903  49738
```

In [16]:
```
CM_mask = metadata$new_clusters %in% c("aCM", "vCM", "CCS")
table(CM_mask)
```

```
CM_mask
 FALSE   TRUE
330806 180835
```

In [17]:
```
# DeBono2025 TNNT2 threshold for Figure 1, example of thresholding
gene_name <- "TNNT2"
gene_counts <- counts[gene_name, dataset_mask]
lambda_est <- mean(gene_counts)
# thresh <- qpois(0.95, lambda = lambda_est)
thresh = dataset_thresholds[[dataset]][["obj_thresholds"]][gene_name]

dens_obs = density(gene_counts, from=0, to= thresh * 4)
x_axis <- 0:max(gene_counts)
prob_null <- dpois(x_axis, lambda = lambda_est)
y_max <- max(c(dens_obs$y, prob_null))

pdf("figures/panels/technical/SignalThreshold_DeBono2025_TNNT2.pdf", width=6, height=6)
# Plot
plot(dens_obs, main = paste("Signal to Noise: ", gene_name, dataset),
     xlab = "Count Value", ylab = "Density", lwd = 5, col = "blue",
     xlim = c(0, thresh * 4),
     ylim=c(0,0.3))


CM_counts <- counts[gene_name, dataset_mask & CM_mask]
nonCM_counts <- counts[gene_name, dataset_mask & !CM_mask]

dens_obs_CM = density(CM_counts, from=0, to= thresh * 4, bw=1)
dens_obs_nonCM = density(nonCM_counts, from=0, to= thresh * 4, bw=1)

lines(dens_obs_CM, col="red", lwd=5)
lines(dens_obs_nonCM, col="orange2", lwd=5)

# Overlay the Poisson Null
lines(x_axis, prob_null, type = "h", col = rgb(0,0,0,1), lwd = 4)

# Add the Threshold Line
abline(v = thresh, col = "grey", lty = 2, lwd = 2)

# Add Labels
text(thresh, max(prob_null), "95% Noise Threshold", pos = 4, col = "grey", cex = 0.8)
legend("topright", legend = c("Observed (All cells)", "Observed (Cardiomyocytes)", "Observed (non-Cardiomyocytes)", "Poisson Null (A
       col = c("blue", "red", "orange2", "black"), lwd = 2, bty = "n")

dev.off()
```

agg_record_1276517633: 2

In [18]:
```
# Effect of thresholding on expression matrices / Dot Plots

# Fetal - Figure 1
options(repr.plot.width = 20)
options(repr.plot.height = 20)

stage = c("fetal")
disease_condition = c("healthy")
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
```

```
                 #metadata[,"disease_subcategory"] == "dHF"

matrix2plot = x_thresholded_norm

matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

genes = genes_markers
#stratification_column = "stage"
#stratification = "adult"
cluster_column = "new_clusters"

mem = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)

matrix2plot1 = mem_norm

dm = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)

dot2plot = dm

dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]

bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit


# color matrices
col_matrix1 = matrix(plotcols(matrix2plot1, c("grey95", "blue")), nrow(matrix2plot1), ncol(matrix2plot1))
col_matrix2 = matrix(plotcols(matrix2plot1, c("grey95", "red")), nrow(matrix2plot1), ncol(matrix2plot1))
col_matrix3 = matrix(plotcols(matrix2plot1, c("grey95", "black")), nrow(matrix2plot1), ncol(matrix2plot1))

col2plot = col_matrix1

col2plot1 = col2plot
mem_norm1 = mem_norm
dot2plot1 = dot2plot

matrix2plot = x_norm

matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

genes = genes_markers
#stratification_column = "stage"
#stratification = "adult"
cluster_column = "new_clusters"

mem = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)

matrix2plot2 = mem_norm

dm = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
```

```
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)

dot2plot = dm

dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]

bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit


# color matrices
col_matrix1 = matrix(plotcols(matrix2plot2, c("grey95", "blue")), nrow(matrix2plot2), ncol(matrix2plot2))
col_matrix2 = matrix(plotcols(matrix2plot2, c("grey95", "red")), nrow(matrix2plot2), ncol(matrix2plot2))
col_matrix3 = matrix(plotcols(matrix2plot2, c("grey95", "black")), nrow(matrix2plot2), ncol(matrix2plot2))

col2plot = col_matrix1

col2plot2 = col2plot
mem_norm2 = mem_norm
dot2plot2 = dot2plot


# combine both pre and post correction
mem_norm = cbind(mem_norm1[,1], mem_norm2[,1])
matrix2plot = cbind(matrix2plot1[,1], matrix2plot2[,1])
col2plot = cbind(col2plot1[,1], col2plot2[,1])
dot2plot = cbind(dot2plot1[,1], dot2plot2[,1])

pdf("figures/panels/technical/MatrixPlot_Markergenes_Fetal_corrections.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(matrix2plot, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=col2plot)
#par(new=T)
#matrix.plot(mem_norm, cex.multiplier=3, round.n=2, ylab="", xlab="", pch=0, col="grey", new=T)
dev.off()

pdf("figures/panels/technical/MatrixPlot_Markergenes_Fetal_corrections_DotPlot.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=15, col=col2plot)
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=0, col="grey90", new=T)
    points(1:6, rep(-2, 6), pch=0, cex = 2.5, xpd=NA, col="grey90")
    points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=15)
    text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

    points(seq(10,15, length=50), rep(-1.75, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "blue")))
    text(
      seq(10,15, length=3),
      rep(-3, 3),
      seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
dev.off()

# options(repr.plot.width = 21.2)
# options(repr.plot.height = 16.6)
# par(oma=c(2,10,10,2), cex=1.5)
# matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "blue"))
```

agg_record_88211539: 2

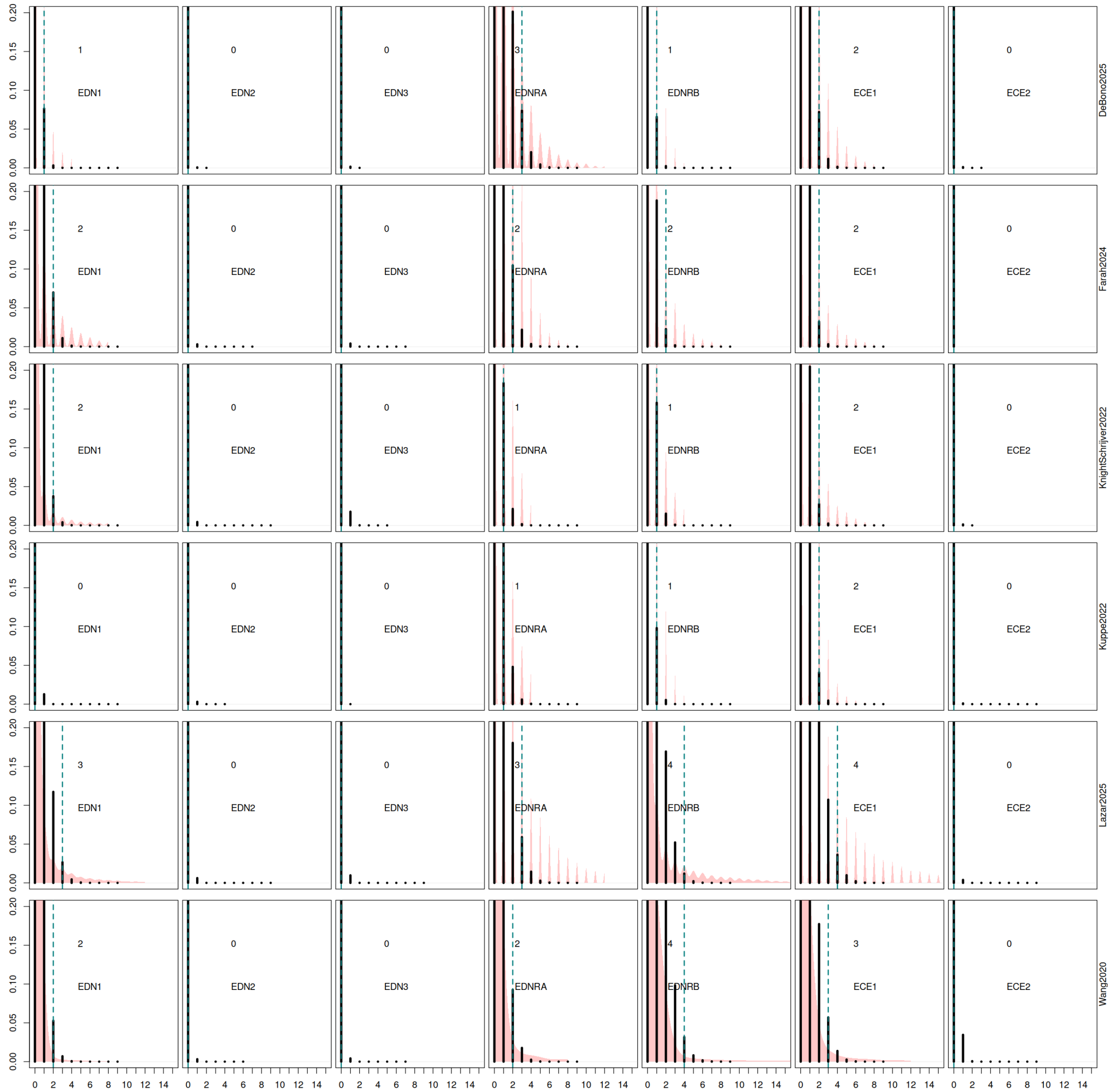
agg_record_88211539: 2

In [19]:

```
# EDN* gene thresholds
datasets = levels(metadata$dataset)

pdf("figures/panels/technical/SignalThreshold_allEDNs.pdf", width=14, height=12)
par(mfrow=c(length(datasets), length(genes_EDN)), mar=c(0.2,0.2,0.2,0.2), cex=2, oma=c(0,0,0,0))

for(dataset in datasets){
    for(gene in genes_EDN){
        gene_name = gene

        dataset_mask = metadata$dataset %in% dataset
        gene_counts <- counts[gene_name, dataset_mask]

        lambda_est <- mean(gene_counts)
        thresh = dataset_thresholds[[dataset]][["obj_thresholds"]][gene_name]

        dens_obs = density(gene_counts, from=0, to= thresh * 4)
        x_axis <- 0:max(gene_counts)
        prob_null <- dpois(x_axis, lambda = lambda_est)
        y_max <- max(c(dens_obs$y, prob_null))

        plot(dens_obs, main = paste(gene_name),
             xlab = "Count Value", ylab = "Density", lwd = 5, col = NA,
             xlim = c(0, 15),
             ylim=c(0,0.2), xaxt="n", yaxt="n")
        polygon(
            c(dens_obs$x, rev(dens_obs$x)),
```

```
            c(dens_obs$y, rep(0, length(dens_obs$y))),
            col=rgb(1,0,0,0.2),
            border=NA
            )

        # Overlay the Poisson Null
        lines(x_axis[1:10], prob_null[1:10], type = "h", col = rgb(0,0,0,1), lwd = 4)

        # Add the Threshold Line
        abline(v = thresh, col = rgb(0,0.5,0.5,1), lty = 2, lwd = 2)
        legend("topright", bty="n", legend=thresh)
    }
}
dev.off()
# Fetal - Figure 1
options(repr.plot.width = 20)
options(repr.plot.height = 20)

par(mfrow=c(length(datasets), length(genes_EDN)), mar=c(0.4,0.2,0.6,0.2), cex=1, oma=c(2,2,2,2))
for(dataset in datasets){
    for(gene in genes_EDN){
        gene_name = gene

        dataset_mask = metadata$dataset %in% dataset
        gene_counts <- counts[gene_name, dataset_mask]

        lambda_est <- mean(gene_counts)
        thresh = dataset_thresholds[[dataset]][["obj_thresholds"]][gene_name]

        dens_obs = density(gene_counts, from=0, to= thresh * 4)
        x_axis <- 0:max(gene_counts)
        prob_null <- dpois(x_axis, lambda = lambda_est)
        y_max <- max(c(dens_obs$y, prob_null))

        plot(dens_obs, main = "",
             xlab = "", ylab = "", lwd = 5, col = NA,
             xlim = c(0, 15),
             ylim=c(0,0.2), xaxt="n", yaxt="n")
        polygon(
            c(dens_obs$x, rev(dens_obs$x)),
            c(dens_obs$y, rep(0, length(dens_obs$y))),
            col=rgb(1,0,0,0.2),
            border=NA
            )

        # Overlay the Poisson Null
        lines(x_axis[1:10], prob_null[1:10], type = "h", col = rgb(0,0,0,1), lwd = 4)

        # Add the Threshold Line
        abline(v = thresh, col = rgb(0,0.5,0.5,1), lty = 2, lwd = 2)
        legend("topright", bty="n", legend=c(thresh, gene_name))
        if(dataset == "Wang2020"){
            axis(1, 0:15)
        }
        if(gene_name == "EDN1"){
            axis(2, c(0, 0.05, 0.1, 0.15, 0.2))
        }
    }
    mtext(dataset, 4)
}
```

agg_record_1914021070: 2



Analysis and plots

Initial UMAPs, of annotation, integration, technical parts, supplemental

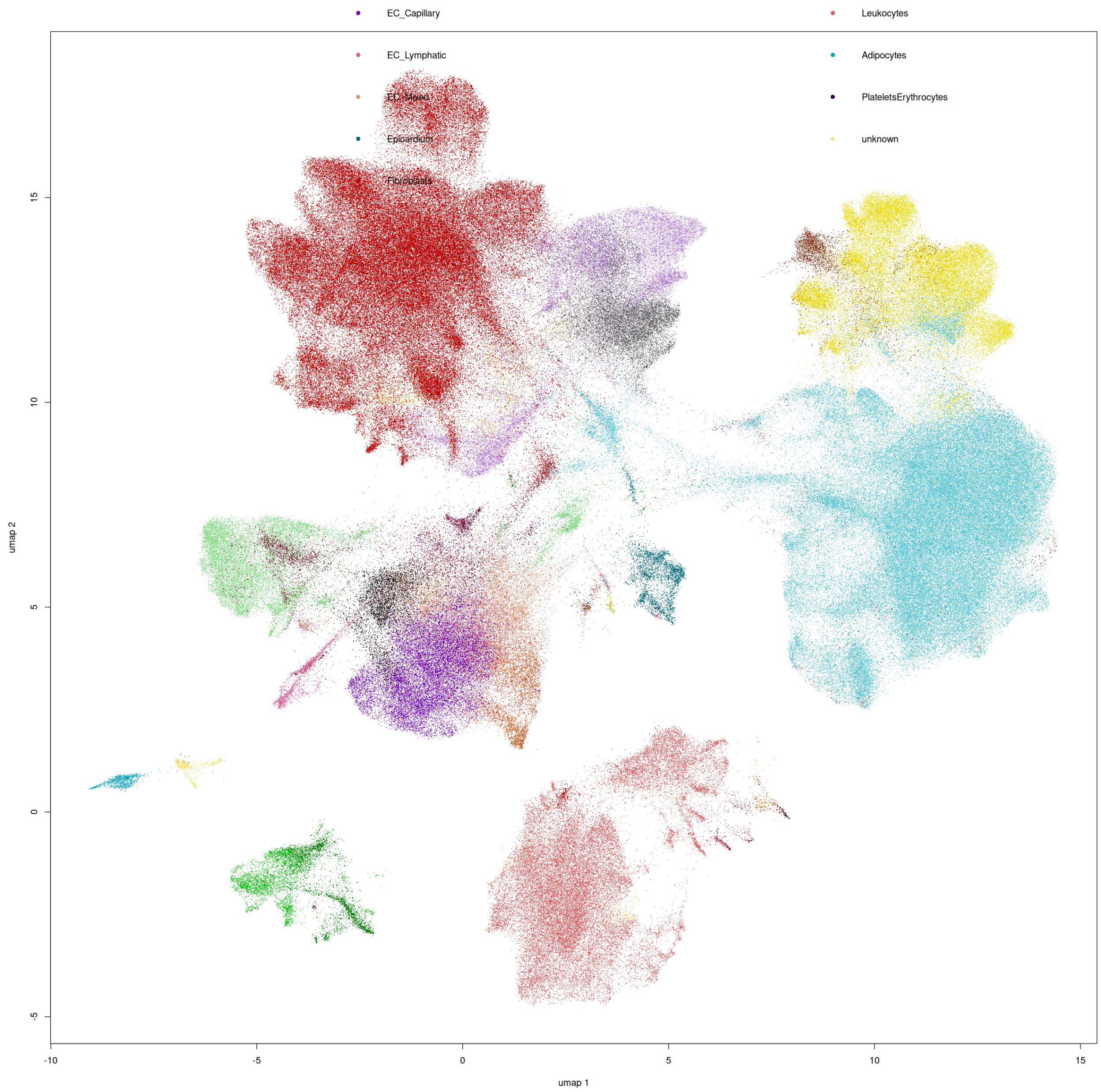
In [25]:
```
# New Clustering

  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=cluster.cols[metadata$new_clusters,2],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    cluster.cols[,1],
    col=cluster.cols[,2],
    ncol=3,
    yjust=1.5,
    pch=16
  )
pdf("figures/panels/integration/UMAP_NewClusters.pdf", height=12, width=12)
  par(oma=c(5,5,5,5), mar=c(5,5,5,5))

  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=cluster.cols[metadata$new_clusters,2],
    xlab="umap 1",
    ylab="umap 2"
  )
```

```r
  legend.3(
    cluster.cols[,1],
    col=cluster.cols[,2],
    ncol=3,
    yjust=1.5,
    pch=16
  )
dev.off()

pdf("figures/panels/integration/UMAP_NewClusters_LegendOnly.pdf", height=12, width=12)
  par(oma=c(5,5,5,5), mar=c(5,5,5,5))

  plot(
    umap[,1],
    umap[,2],
    pch=NA,
    col=cluster.cols[metadata$new_clusters,2],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    cluster.cols[,1],
    col=cluster.cols[,2],
    ncol=3,
    yjust=1.5,
    pch=16
  )
dev.off()

png("figures/panels/integration/UMAP_NewClusters_PanelOnly.png", height=1200, width=1200)
  par(oma=c(0,0,0,0), mar=c(0,0,0,0))

  plot(
    umap[,1],
    umap[,2],
    pch=16,
    col=cluster.cols[metadata$new_clusters,2],
    xlab="umap 1",
    ylab="umap 2",
    cex=0.25
  )

  legend.3(
    cluster.cols[,1],
    col=cluster.cols[,2],
    ncol=3,
    yjust=1.5,
    pch=16
  )
dev.off()
```

agg_record_699161170: 2

agg_record_699161170: 2

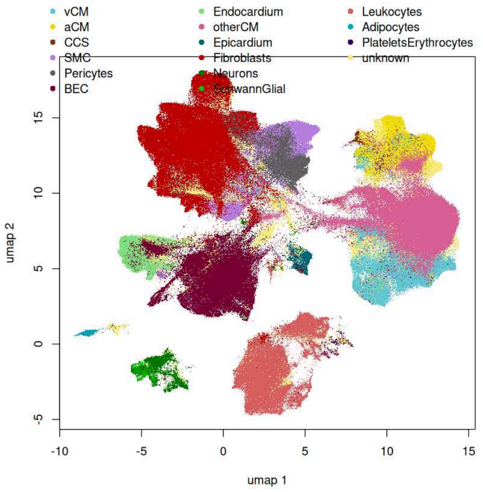
agg_record_699161170: 2



In [20]:

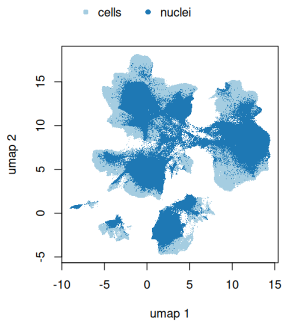
```
# Old Clustering
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=cluster.cols[match(metadata$merged_labels, cluster.cols[,3]),2],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    unique(cluster.cols[,3]),
    col=cluster.cols[match(unique(cluster.cols[,3]), cluster.cols[,3]),2],
    ncol=3,
    yjust=1.5,
    pch=16
  )
pdf("figures/panels/integration/UMAP_MergedLabels.pdf", height=12, width=12)
  par(oma=c(5,5,5,5), mar=c(5,5,5,5))

  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=cluster.cols[match(metadata$merged_labels, cluster.cols[,3]),2],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    unique(cluster.cols[,3]),
    col=cluster.cols[match(unique(cluster.cols[,3]), cluster.cols[,3]),2],
    ncol=3,
    yjust=1.5,
    pch=16
  )
dev.off()
```

agg_record_1348154365: 2



In [20]:
```
# Cell Vs Nuclei
par(oma=c(5,5,5,5), mar=c(5,5,5,5))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=Discrete12[metadata$cell_or_nuclei],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    levels(metadata$cell_or_nuclei),
    col=Discrete12[1:2],
    ncol=3,
    yjust=1.5,
    pch=16
  )

png("figures/panels/integration/UMAP_CellNuclei_PanelOnly.png", height=1200, width=1200)
    par(oma=c(0,0,0,0), mar=c(0,0,0,0))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=Discrete12[metadata$cell_or_nuclei],
    xlab="umap 1",
    ylab="umap 2"
  )
dev.off()
```

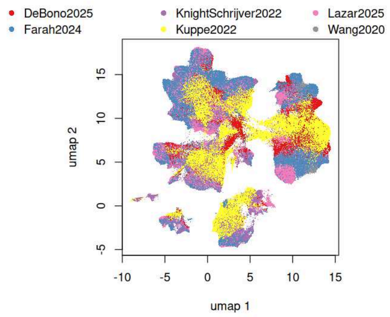
agg_record_277156924: 2



In [109…
```
# Dataset
# ran.ord
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )
plot.order = sample(1:nrow(umap))
par(oma=c(5,5,5,5), mar=c(5,5,5,5))
  plot(
    umap[plot.order,1],
    umap[plot.order,2],
```

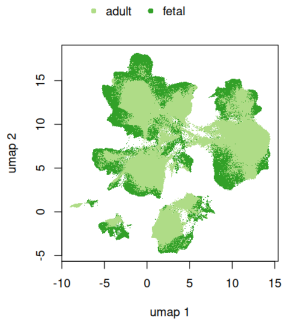
```
    pch=".",
    col=dataset_cols[metadata$dataset[plot.order]],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    levels(metadata$dataset),
    col=dataset_cols,
    ncol=3,
    yjust=1.5,
    pch=16
  )

png("figures/panels/integration/UMAP_Dataset_PanelOnly.png", height=1200, width=1200)
    par(oma=c(0,0,0,0), mar=c(0,0,0,0))
  plot(
    umap[plot.order,1],
    umap[plot.order,2],
    pch=".",
    col=dataset_cols[metadata$dataset[plot.order]],
    xlab="umap 1",
    ylab="umap 2"
  )
dev.off()
```

agg_record_948929138: 2



In [97]:
```
# Stage
options(repr.plot.width = 7)
options(repr.plot.height = 7)
par(oma=c(5,5,5,5), mar=c(5,5,5,5))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=Discrete12[3:4][metadata$stage],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    levels(metadata$stage),
    col=Discrete12[3:4],
    ncol=3,
    yjust=1.5,
    pch=16
  )

png("figures/panels/integration/UMAP_Stage_PanelOnly.png", height=1200, width=1200)
    par(oma=c(0,0,0,0), mar=c(0,0,0,0))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=Discrete12[3:4][metadata$stage],
    xlab="umap 1",
    ylab="umap 2"
  )
dev.off()
```

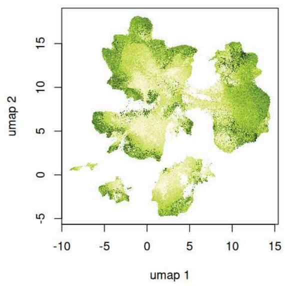
agg_record_1190025925: 2



In [22]:
```
# UMIs
options(repr.plot.width = 7)
options(repr.plot.height = 7)
par(oma=c(5,5,5,5), mar=c(5,5,5,5))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=plotcols(metadata$log1p_total_counts, col=rev(verdant)),
    xlab="umap 1",
    ylab="umap 2"
  )

png("figures/panels/integration/UMAP_depth_PanelOnly.png", height=1200, width=1200)
    par(oma=c(0,0,0,0), mar=c(0,0,0,0))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=plotcols(metadata$log1p_total_counts, col=rev(verdant)),
    xlab="umap 1",
    ylab="umap 2"
  )
dev.off()
range(metadata$log1p_total_counts)
```

agg_record_603126728: 2

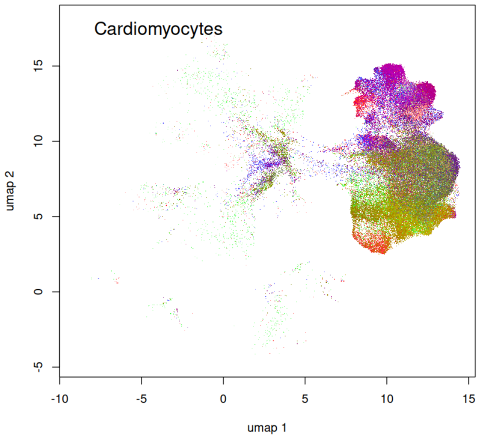
6.27287721633911 · 11.9521932601929



In [23]:
```
# Top Markers
 # Cardiomyocytes
  col2plot = whiteRGB(
    normalise(x_thresholded_norm[,"TNNT2"]),
    normalise(x_thresholded_norm[,"MYH7"]),
    normalise(x_thresholded_norm[,"MYH6"])
  )

  plot.light()
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=col2plot,
    xlab="umap 1",
    ylab="umap 2"
  )
  legend("topleft", "Cardiomyocytes", bty="n", cex=1.5)

 png("figures/panels/integration/UMAP_CellTypeMarkers_Cardiomyocyte.png", height=1500, width=1500)
```

```
par(oma=c(0,0,0,0), mar=c(5,5,5,5), cex=3)
 col2plot = whiteRGB(
   normalise(x_thresholded_norm[,"TNNT2"]),
   normalise(x_thresholded_norm[,"MYH7"]),
   normalise(x_thresholded_norm[,"MYH6"])
 )

 plot.light()
 plot(
   umap[,1],
   umap[,2],
   pch=".",
   col=col2plot,
   xlab="umap 1",
   ylab="umap 2"
 )
 legend("topleft", "Cardiomyocytes", bty="n", cex=1.5)

 dev.off()
```

agg_record_1348918604: 2

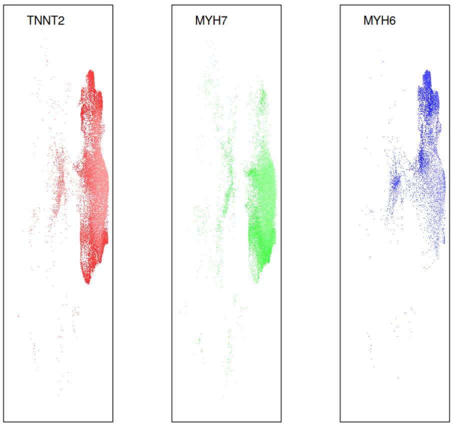


In [26]:

```
# Cardiomyocyte SubPanels
par(mfrow=c(1,3))
    gene = "TNNT2"
    col2plot = plotcols(x_thresholded_norm[,gene], col=c("white", "red"))
      plot(
        umap[,1],
        umap[,2],
        pch=".",
        col=col2plot,
        xlab="",
        ylab="",
        xaxt="n",
        yaxt="n"
      )
      legend("topleft", gene, bty="n", cex=1.5)

      gene = "MYH7"
      col2plot = plotcols(x_thresholded_norm[,gene], col=c("white", "green"))
      plot(
        umap[,1],
        umap[,2],
        pch=".",
        col=col2plot,
        xlab="",
        ylab="",
        xaxt="n",
        yaxt="n"
      )
      legend("topleft", gene, bty="n", cex=1.5)

      gene = "MYH6"
      col2plot = plotcols(x_thresholded_norm[,gene], col=c("white", "blue"))
      plot(
        umap[,1],
        umap[,2],
        pch=".",
        col=col2plot,
        xlab="",
        ylab="",
        xaxt="n",
        yaxt="n"
      )
      legend("topleft", gene, bty="n", cex=1.5)

  png("figures/panels/integration/UMAP_CellTypeMarkers_Cardiomyocyte_SubPanels.png", height=500, width=1500)
  par(oma=c(0,0,0,0), mar=c(0,0,0,0), mfrow=c(1,3), cex=3)
    gene = "TNNT2"
    col2plot = plotcols(x_thresholded_norm[,gene], col=c("white", "red"))
      plot(
        umap[,1],
        umap[,2],
        pch=".",
        col=col2plot,
```

```
        xlab="",
        ylab="",
        xaxt="n",
        yaxt="n"
      )
      legend("topleft", gene, bty="n", cex=1.5)

      gene = "MYH7"
      col2plot = plotcols(x_thresholded_norm[,gene], col=c("white", "green"))
      plot(
        umap[,1],
        umap[,2],
        pch=".",
        col=col2plot,
        xlab="",
        ylab="",
        xaxt="n",
        yaxt="n"
      )
      legend("topleft", gene, bty="n", cex=1.5)

      gene = "MYH6"
      col2plot = plotcols(x_thresholded_norm[,gene], col=c("white", "blue"))
      plot(
        umap[,1],
        umap[,2],
        pch=".",
        col=col2plot,
        xlab="",
        ylab="",
        xaxt="n",
        yaxt="n"
      )
      legend("topleft", gene, bty="n", cex=1.5)
    dev.off()
```

agg_record_1518497825: 2



In [34]:
```
# all markers across UMAP grid
  png("figures/panels/integration/UMAP_CellTypeMarkers_AllMarkers.png", height=3500, width=2500)
    par(oma=c(0,0,0,0), mar=c(0,0,0,0), mfrow=c(7,5), cex=3)
      for(gene in genes_markers){
        col2plot = plotcols(x_norm[,gene], col=rev(verdant))
        plot(
          umap[,1],
          umap[,2],
          pch=".",
          col=col2plot,
          xlab="",
          ylab="",
          xaxt="n",
          yaxt="n"
        )
        legend("bottomright", gene, bty="n", cex=1.2)
      }
  dev.off()
## Check the PNG file generated to save on plotting many points and saving a large script here
```

agg_record_579468568: 2

In [43]:
```
# Supplemental information on new clustering IDs
  H = bar.matrix(metadata$new_clusters, metadata$cell_type_mid, scale=T) # just using as a function to format the
  matrix2plot = t(H$cell.id.matrix)
  matrix2plot = matrix2plot[order(rownames(matrix2plot)),]
  pdf("figures/panels/integration/MatrixPlot_CellTypevsNewClusters.pdf", height=30, width=10)
    par(oma=c(1,15,15,10), mar=c(0,0,0,0), cex=1)

    matrix.plot(matrix2plot, round.n=2)
  dev.off()
```

agg_record_327572093: 2

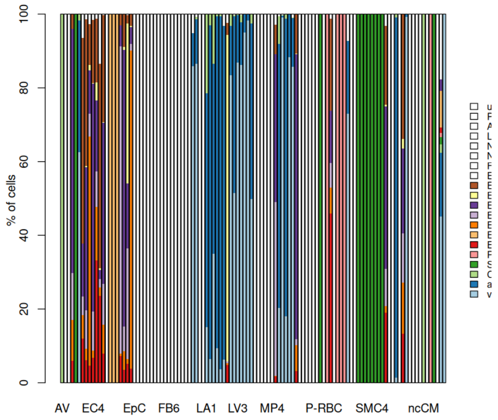


Figure 1

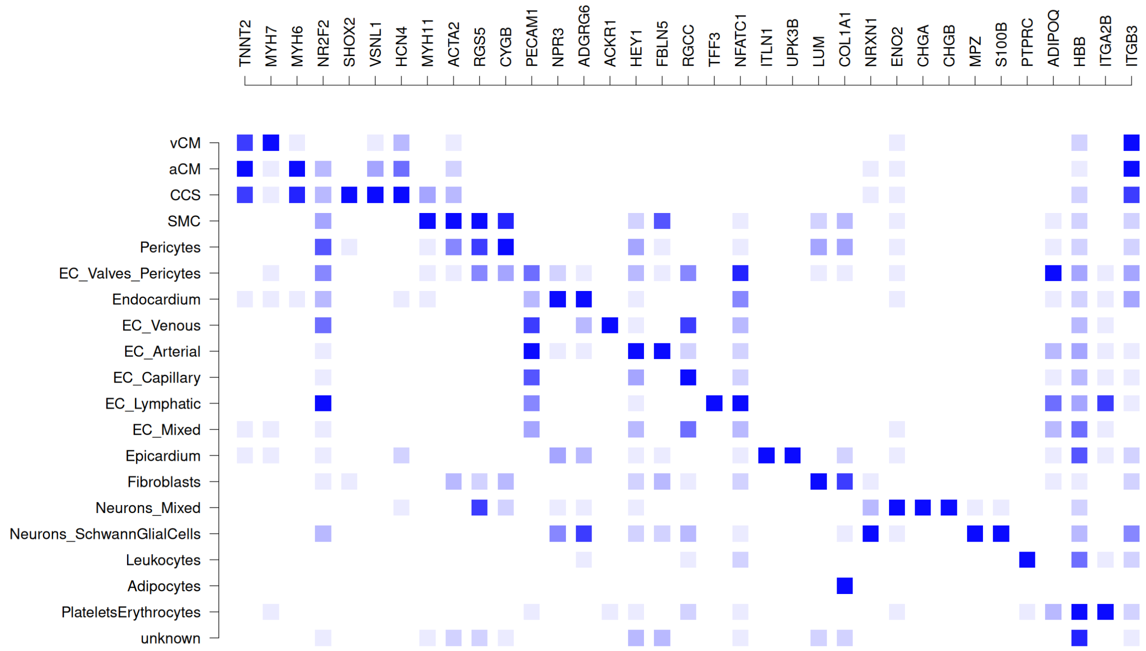
In [128…

```
# Fetal - Figure 1
options(repr.plot.width = 20)
options(repr.plot.height = 20)

stage = c("fetal")
disease_condition = c("healthy")
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"

matrix2plot = x_thresholded_norm

matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

genes = genes_markers
#stratification_column = "stage"
#stratification = "adult"
cluster_column = "new_clusters"

mem = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)

matrix2plot = mem_norm

dm = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)

dot2plot = dm

dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]

bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit


# color matrices
col_matrix1 = matrix(plotcols(matrix2plot, c("grey95", "blue")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix2 = matrix(plotcols(matrix2plot, c("grey95", "red")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix3 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))
```

```
col2plot = col_matrix1

pdf("figures/panels/results/MatrixPlot_Markergenes_Fetal.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "blue"))
#par(new=T)
#matrix.plot(mem_norm, cex.multiplier=3, round.n=2, ylab="", xlab="", pch=0, col="grey", new=T)
dev.off()

pdf("figures/panels/results/MatrixPlot_Markergenes_Fetal_DotPlot.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=15, col=c("white", "blue"))
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=0, col="grey90", new=T)
    points(1:6, rep(-2, 6), pch=0, cex = 2.5, xpd=NA, col="grey90")
    points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=15)
    text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

    points(seq(10,15, length=50), rep(-1.75, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "blue")))
    text(
      seq(10,15, length=3),
      rep(-3, 3),
      seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
dev.off()

options(repr.plot.width = 21.2)
options(repr.plot.height = 16.6)
par(oma=c(2,10,10,2), cex=1.5)
matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "blue"))
```

agg_record_1131218898: 2

agg_record_1131218898: 2



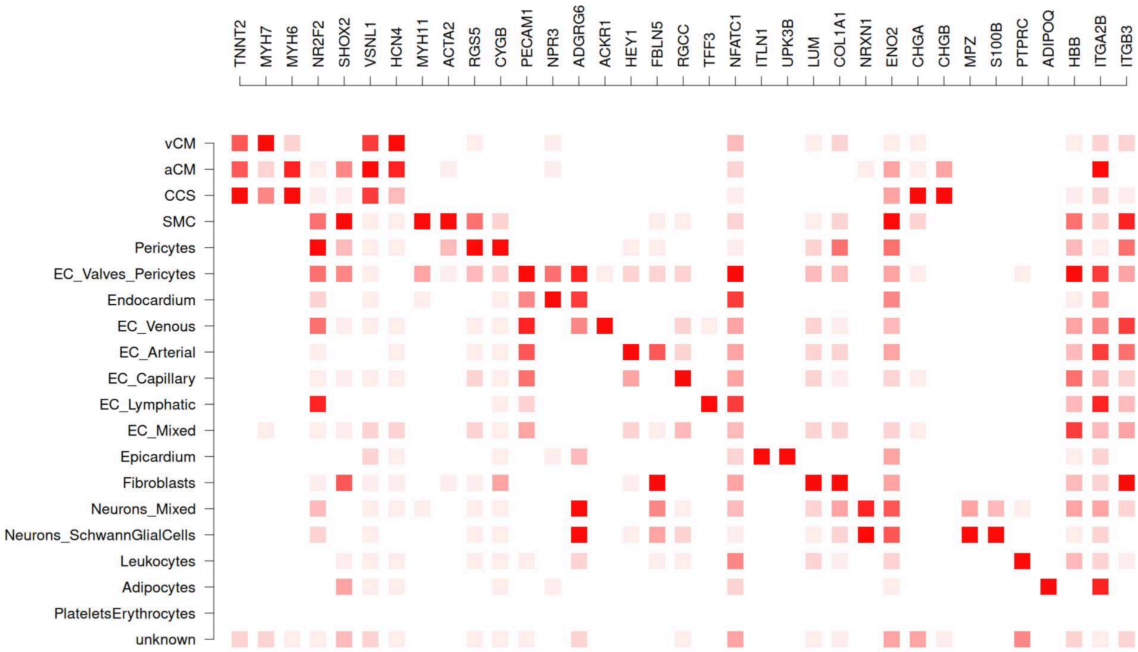
In [119…

```
# Adult - Figure 1
options(repr.plot.width = 20)
options(repr.plot.height = 20)

stage = c("adult")
disease_condition = c("healthy")
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"
```

```
matrix2plot = x_thresholded_norm

matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

genes = genes_markers
#stratification_column = "stage"
#stratification = "adult"
cluster_column = "new_clusters"

mem = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)

matrix2plot = mem_norm

dm = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)

dot2plot = dm

dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]

bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit


# color matrices
col_matrix1 = matrix(plotcols(matrix2plot, c("grey95", "blue")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix2 = matrix(plotcols(matrix2plot, c("grey95", "red")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix3 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))

col2plot = col_matrix2

pdf("figures/panels/results/MatrixPlot_Markergenes_Adult.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "red"))
#par(new=T)
#matrix.plot(mem_norm, cex.multiplier=3, round.n=2, ylab="", xlab="", pch=0, col="grey", new=T)
dev.off()

pdf("figures/panels/results/MatrixPlot_Markergenes_Adult_DotPlot.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=15, col=c("white", "red"))
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=0, col="grey90", new=T)
    points(1:6, rep(-2, 6), pch=0, cex = 2.5, xpd=NA, col="grey90")
    points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=15)
    text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

    points(seq(10,15, length=50), rep(-2.25, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "red")))
    text(
      seq(10,15, length=3),
      rep(-3, 3),
      seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
dev.off()

options(repr.plot.width = 21.2)
options(repr.plot.height = 16.6)
par(oma=c(2,10,10,2), cex=1.5)
matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "red"))
```

png: 2

png: 2



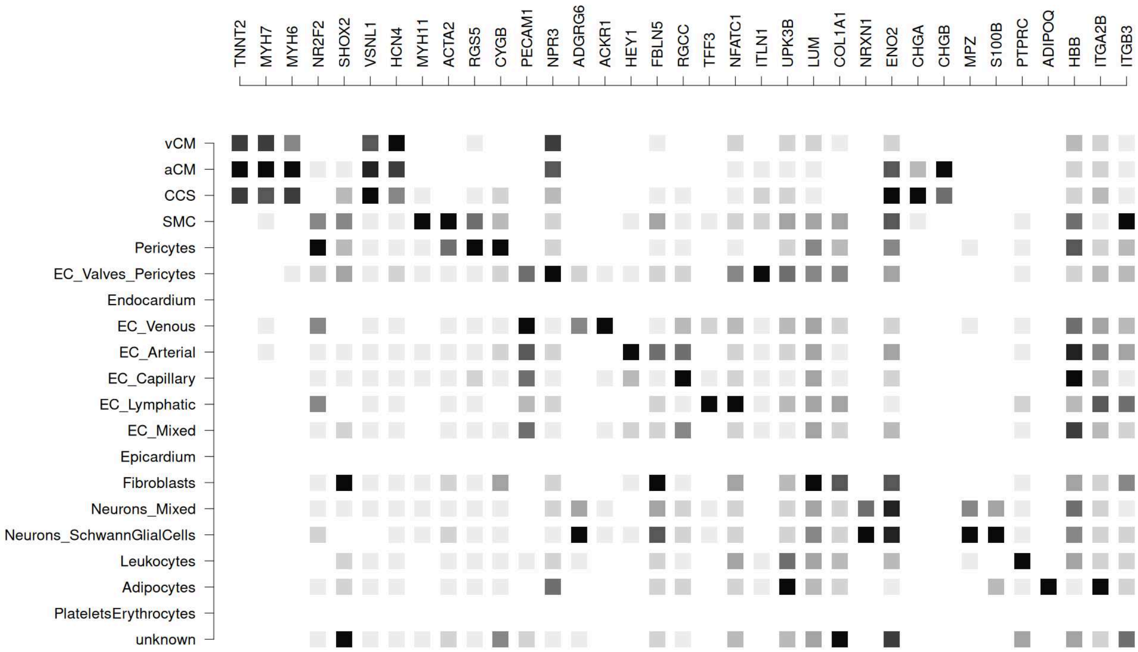
In [121…

```
# Adult DISEASE - Figure 1
options(repr.plot.width = 20)
options(repr.plot.height = 20)

stage = c("adult")
disease_condition = c("disease")
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"

matrix2plot = x_thresholded_norm

matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

genes = genes_markers
#stratification_column = "stage"
#stratification = "adult"
cluster_column = "new_clusters"

mem = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)

matrix2plot = mem_norm

dm = sapply(
  levels(metadata_stratification$new_clusters),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
```

```
  }
)
dm = t(dm)

dot2plot = dm

dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]

bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit


# color matrices
col_matrix1 = matrix(plotcols(matrix2plot, c("grey95", "blue")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix2 = matrix(plotcols(matrix2plot, c("grey95", "red")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix3 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))

col2plot = col_matrix3

pdf("figures/panels/results/MatrixPlot_Markergenes_Disease.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "black"))
#par(new=T)
#matrix.plot(mem_norm, cex.multiplier=3, round.n=2, ylab="", xlab="", pch=0, col="grey", new=T)
dev.off()

pdf("figures/panels/results/MatrixPlot_Markergenes_Disease_DotPlot.pdf", height=8.3, width=10.6)
par(oma=c(2,10,10,2), cex=1)
matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=15, col=c("white", "black"))
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=0, col="grey90", new=T)
    points(1:6, rep(-2, 6), pch=0, cex = 2.5, xpd=NA, col="grey90")
    points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=15)
    text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

    points(seq(10,15, length=50), rep(-2.25, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "black")))
    text(
      seq(10,15, length=3),
      rep(-3, 3),
      seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
dev.off()

options(repr.plot.width = 21.2)
options(repr.plot.height = 16.6)
par(oma=c(2,10,10,2), cex=1.5)
matrix.plot(mem_norm, cex.multiplier=2.4, round.n=2, ylab="", xlab="", pch=15, col=c("white", "black"))
```

png: 2

png: 2



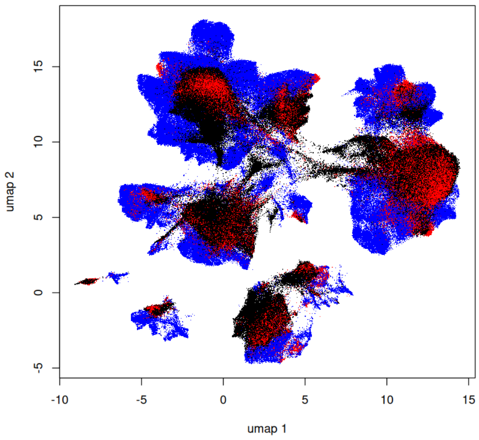
In [14]:
```
# donor info colours
  class2plot = factor(paste(
      metadata$stage,
      metadata$disease_condition,
      sep="_"
  ))
levels(class2plot)
```

'adult_disease' · 'adult_healthy' · 'fetal_healthy'

In [15]:
```
# donor info colours
col2plot = c("black", "red", "blue")
```

In [41]:
```
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=col2plot[class2plot],
    xlab="umap 1",
    ylab="umap 2"
  )

  # legend.3(
  #   cluster.cols[,1],
  #   col=cluster.cols[,2],
  #   ncol=3,
  #   yjust=1.5,
  #   pch=16
  # )

png("figures/panels/integration/UMAP_Donors_PanelOnly.png", height=1200, width=1200)
  par(oma=c(0,0,0,0), mar=c(0,0,0,0))
  plot(
    umap[,1],
    umap[,2],
    pch=".",
    col=col2plot[class2plot],
    xlab="umap 1",
    ylab="umap 2"
  )

  legend.3(
    levels(class2plot),
    col=col2plot,
    ncol=3,
    yjust=1.5,
    pch=16
  )
dev.off()
```

agg_record_2032637574: 2



In [66]:
```
datasets
```

'DeBono2025' · 'Farah2024' · 'KnightSchrijver2022' · 'Kuppe2022' · 'Lazar2025' · 'Wang2020'

In [67]:
```
genes_EDN
```

'EDN1' · 'EDN2' · 'EDN3' · 'EDNRA' · 'EDNRB' · 'ECE1' · 'ECE2'

Figure 2

EDN* dotplots

In [68]:
```
# Fetal vs Adult
matrix2plot = x_thresholded_norm

# data 2 remove
stage = c("fetal", "adult")
disease_condition = c("healthy") # remove DISEASE samples
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"



matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

# new factor object:
  combined_factor = paste(metadata_stratification[,"new_clusters"], metadata_stratification[,"stage"], sep="_")
  metadata_stratification$combined_factor = factor(
    combined_factor,
    levels=c(paste(rep(levels(metadata_stratification[,"new_clusters"]),each=2), levels(metadata_stratification[,"stage"]),sep="_"))
  )

  genes = genes_EDN
  #stratification_column = "stage"
  #stratification = "adult"
  cluster_column = "combined_factor"

mem = sapply(
  levels(metadata_stratification[,cluster_column]),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
#mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)


dm = sapply(
  levels(metadata_stratification[,cluster_column] ),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
```

```r
dm = t(dm)


fetal_matrix = t(mem_norm)[,1:20*2]
adult_matrix = t(mem_norm)[,1:20*2-1]
fetal_dm = t(dm)[,1:20*2]
adult_dm = t(dm)[,1:20*2-1]

matrix2plot = rbind(
  fetal_matrix[1,],
  adult_matrix[1,],
  fetal_matrix[2,],
  adult_matrix[2,],
  fetal_matrix[3,],
  adult_matrix[3,],
  fetal_matrix[4,],
  adult_matrix[4,],
  fetal_matrix[5,],
  adult_matrix[5,],
  fetal_matrix[6,],
  adult_matrix[6,],
  fetal_matrix[7,],
  adult_matrix[7,]
  )
  rownames(matrix2plot) = paste(rep(c("fetal", "adult"),7),rep(genes_EDN,each=2))

dot2plot = rbind(
  fetal_dm[1,],
  adult_dm[1,],
  fetal_dm[2,],
  adult_dm[2,],
  fetal_dm[3,],
  adult_dm[3,],
  fetal_dm[4,],
  adult_dm[4,],
  fetal_dm[5,],
  adult_dm[5,],
  fetal_dm[6,],
  adult_dm[6,],
  fetal_dm[7,],
  adult_dm[7,]
  )
dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]


bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit

# color matrices
col_matrix1 = matrix(plotcols(matrix2plot, c("grey95", "blue")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix2 = matrix(plotcols(matrix2plot, c("grey95", "red")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix3 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))

col2plot =  rbind(
  col_matrix1[1,],
  col_matrix2[2,],
  col_matrix1[3,],
  col_matrix2[4,],
  col_matrix1[5,],
  col_matrix2[6,],
  col_matrix1[7,],
  col_matrix2[8,],
  col_matrix1[9,],
  col_matrix2[10,],
  col_matrix1[11,],
  col_matrix2[12,],
  col_matrix1[13,],
  col_matrix2[14,]
  )

colnames(matrix2plot) = gsub("_fetal", "", colnames(matrix2plot))
matrix2plot = matrix2plot[,1:19]
dot2plot = dot2plot[,1:19]
col2plot = col2plot[,1:19]

pdf("figures/panels/results/MatrixPlot_EDNgenes_Adult_Fetal.pdf", height=7, width=7.3)
  par(oma=c(2,10,10,2))
  matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=16, col=col2plot)
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=1, col="grey90", new=T)
points(1:6, rep(-2, 6), pch=1, cex = 2.5, xpd=NA, col="grey90")
points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=16)
text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

points(seq(10,15, length=50), rep(-1.75, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "blue")))
points(seq(10,15, length=50), rep(-2.25, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "red")))
text(
  seq(10,15, length=3),
  rep(-3, 3),
  seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
  xpd=NA,
  srt=90,
  adj=1,
  cex=0.6
)
```

```
dev.off()

colnames(fraction_table) = colnames(matrix2plot)
fraction_table = data.frame(
  "condition_gene" = rownames(matrix2plot),
  fraction_table
)
write.table(fraction_table, file="source_data/results/positiveFraction_celltype_adult_fetal.csv", sep=",", quote=F, col.names=T, row
```

agg_record_919233902: 2

In [69]:
```
#  Adult  Healthy vs Disease

matrix2plot = x_thresholded_norm

# data 2 remove
stage = c("adult")
disease_condition = c("healthy", "disease")
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"



matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

# new factor object:
  combined_factor = paste(metadata_stratification[,"new_clusters"], metadata_stratification[,"disease_condition"], sep="_")
  metadata_stratification$combined_factor = factor(
    combined_factor,
    levels=c(paste(rep(levels(metadata_stratification[,"new_clusters"]),each=2), levels(metadata_stratification[,"disease_condition"
  )

  genes = genes_EDN
  #stratification_column = "stage"
  #stratification = "adult"
  cluster_column = "combined_factor"

mem = sapply(
  levels(metadata_stratification[,cluster_column]),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
#mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)


dm = sapply(
  levels(metadata_stratification[,cluster_column] ),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)


healthy_matrix = t(mem_norm)[,1:20*2]
disease_matrix = t(mem_norm)[,1:20*2-1]
healthy_dm = t(dm)[,1:20*2]
disease_dm = t(dm)[,1:20*2-1]

matrix2plot = rbind(
  healthy_matrix[1,],
  disease_matrix[1,],
  healthy_matrix[2,],
  disease_matrix[2,],
  healthy_matrix[3,],
  disease_matrix[3,],
  healthy_matrix[4,],
  disease_matrix[4,],
  healthy_matrix[5,],
  disease_matrix[5,],
  healthy_matrix[6,],
  disease_matrix[6,],
  healthy_matrix[7,],
  disease_matrix[7,]
  )
  rownames(matrix2plot) = paste(rep(c("healthy", "disease"),7),rep(genes_EDN,each=2))

dot2plot = rbind(
  healthy_dm[1,],
  disease_dm[1,],
  healthy_dm[2,],
```

```
  disease_dm[2,],
  healthy_dm[3,],
  disease_dm[3,],
  healthy_dm[4,],
  disease_dm[4,],
  healthy_dm[5,],
  disease_dm[5,],
  healthy_dm[6,],
  disease_dm[6,],
  healthy_dm[7,],
  disease_dm[7,]
  )
dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]

bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit

# color matrices
col_matrix1 = matrix(plotcols(matrix2plot, c("grey95", "red")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix2 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix3 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))

col2plot =  rbind(
  col_matrix1[1,],
  col_matrix2[2,],
  col_matrix1[3,],
  col_matrix2[4,],
  col_matrix1[5,],
  col_matrix2[6,],
  col_matrix1[7,],
  col_matrix2[8,],
  col_matrix1[9,],
  col_matrix2[10,],
  col_matrix1[11,],
  col_matrix2[12,],
  col_matrix1[13,],
  col_matrix2[14,]
  )

colnames(matrix2plot) = gsub("_healthy", "", colnames(matrix2plot))
matrix2plot = matrix2plot[,1:19]
dot2plot = dot2plot[,1:19]
col2plot = col2plot[,1:19]

pdf("figures/panels/results/MatrixPlot_EDNgenes_Adult_HealthyvsDisease.pdf", height=7, width=7.3)
  par(oma=c(2,10,10,2))
  matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=16, col=col2plot)
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=1, col="grey90", new=T)
  points(1:6, rep(-2, 6), pch=1, cex = 2.5, xpd=NA, col="grey90")
  points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=16)
  text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

  points(seq(10,15, length=50), rep(-1.75, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "red")))
  points(seq(10,15, length=50), rep(-2.25, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "black")))
  text(
    seq(10,15, length=3),
    rep(-3, 3),
    seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
    xpd=NA,
    srt=90,
    adj=1,
    cex=0.6
  )

dev.off()


colnames(fraction_table) = colnames(matrix2plot)
fraction_table = data.frame(
  "condition_gene" = rownames(matrix2plot),
  fraction_table
)
write.table(fraction_table, file="source_data/results/positiveFraction_celltype_adult_healthydisease.csv", sep=",", quote=F, col.name
```

agg_record_1400693519: 2

In [36]:

```
# Fetal vs Adult vs Disease
matrix2plot = x_thresholded_norm

# data 2 remove
stage = c("fetal", "adult")
disease_condition = c("healthy", "disease") #
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"



matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

# new factor object:
```

```
  combined_factor = paste(metadata_stratification[,"new_clusters"], metadata_stratification[,"stage"], metadata_stratification[,"dis
  metadata_stratification$combined_factor = factor(
    combined_factor,
    levels=paste(rep(levels(metadata_stratification[,"new_clusters"]),each=3), rep(rev(levels(metadata_stratification[,"stage"])),c(
  )

  genes = genes_EDN
  #stratification_column = "stage"
  #stratification = "adult"
  cluster_column = "combined_factor"

mem = sapply(
  levels(metadata_stratification[,cluster_column]),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
#mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)


dm = sapply(
  levels(metadata_stratification[,cluster_column] ),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)

fetal_matrix = t(mem_norm)[,1:20*3-2]
adult_matrix = t(mem_norm)[,1:20*3-1]
disease_matrix = t(mem_norm)[,1:20*3]
fetal_dm = t(dm)[,1:20*3-2]
adult_dm = t(dm)[,1:20*3-1]
disease_dm = t(dm)[,1:20*3]

matrix2plot = rbind(
  fetal_matrix[1,],
  adult_matrix[1,],
  disease_matrix[1,],
  fetal_matrix[2,],
  adult_matrix[2,],
  disease_matrix[2,],
  fetal_matrix[3,],
  adult_matrix[3,],
  disease_matrix[3,],
  fetal_matrix[4,],
  adult_matrix[4,],
  disease_matrix[4,],
  fetal_matrix[5,],
  adult_matrix[5,],
  disease_matrix[5,],
  fetal_matrix[6,],
  adult_matrix[6,],
  disease_matrix[6,],
  fetal_matrix[7,],
  adult_matrix[7,],
  disease_matrix[7,]
  )
  rownames(matrix2plot) = paste(rep(c("fetal", "adult", "disease"),7),rep(genes_EDN,each=3))

dot2plot = rbind(
  fetal_dm[1,],
  adult_dm[1,],
  disease_dm[1,],
  fetal_dm[2,],
  adult_dm[2,],
  disease_dm[2,],
  fetal_dm[3,],
  adult_dm[3,],
  disease_dm[3,],
  fetal_dm[4,],
  adult_dm[4,],
  disease_dm[4,],
  fetal_dm[5,],
  adult_dm[5,],
  disease_dm[5,],
  fetal_dm[6,],
  adult_dm[6,],
  disease_dm[6,],
  fetal_dm[7,],
  adult_dm[7,],
  disease_dm[7,]
  )
dot2plot[is.na(dot2plot)] = 0
fraction_table = round(dot2plot,1)[,1:19]
```

```
bottom_limit = 0.1
cex.value = 2.4
m.ratio = (cex.value-bottom_limit)/log(3)
dot2plot = m.ratio*log((dot2plot / 50)+1)
dot2plot = dot2plot+bottom_limit

# color matrices
col_matrix1 = matrix(plotcols(matrix2plot, c("grey95", "blue")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix2 = matrix(plotcols(matrix2plot, c("grey95", "red")), nrow(matrix2plot), ncol(matrix2plot))
col_matrix3 = matrix(plotcols(matrix2plot, c("grey95", "black")), nrow(matrix2plot), ncol(matrix2plot))

col2plot =  rbind(
  col_matrix1[1,],
  col_matrix2[2,],
  col_matrix3[3,],
  col_matrix1[4,],
  col_matrix2[5,],
  col_matrix3[6,],
  col_matrix1[7,],
  col_matrix2[8,],
  col_matrix3[9,],
  col_matrix1[10,],
  col_matrix2[11,],
  col_matrix3[12,],
  col_matrix1[13,],
  col_matrix2[14,],
  col_matrix3[15,],
  col_matrix1[16,],
  col_matrix2[17,],
  col_matrix3[18,],
  col_matrix1[19,],
  col_matrix2[20,],
  col_matrix3[21,]
  )

colnames(matrix2plot) = gsub("_fetal|_healthy", "", colnames(matrix2plot))
matrix2plot = matrix2plot[,1:19]
dot2plot = dot2plot[,1:19]
col2plot = col2plot[,1:19]

pdf("figures/panels/results/MatrixPlot_EDNgenes_Adult_Fetal_Disease.pdf", height=8.2, width=7.3)
  par(oma=c(2,10,10,2))
  matrix.plot(matrix2plot, cex.multiplier=dot2plot, round.n=2, ylab="", xlab="", pch=16, col=col2plot)
  par(new=T)
  matrix.plot(matrix2plot, cex.multiplier=cex.value, round.n=2, ylab="", xlab="", x.axis=F, y.axis=F, pch=1, col="grey90", new=T)
points(1:6, rep(-2, 6), pch=1, cex = 2.5, xpd=NA, col="grey90")
points(1:6, rep(-2, 6),  cex = bottom_limit+m.ratio*log((seq(0,100,length=6)/50)+1), xpd=NA, col="grey50", pch=16)
text(1:6, rep(-3, 6), c(0,20,40,60,80,100), xpd=NA, srt=90, adj=1, cex=0.6)

points(seq(10,15, length=50), rep(-1.75, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "blue")))
points(seq(10,15, length=50), rep(-2.25, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "red")))
points(seq(10,15, length=50), rep(-2.75, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:51, c("white", "black")))
text(
  seq(10,15, length=3),
  rep(-3.5, 3),
  seq(signif(min(matrix2plot, na.rm=T),2), signif(max(matrix2plot,na.rm=T),2), length=3),
  xpd=NA,
  srt=90,
  adj=1,
  cex=0.6
)

dev.off()
```

agg_record_1551058576: 2

In [37]:

```
colnames(fraction_table) = colnames(matrix2plot)
fraction_table = data.frame(
  "condition_gene" = rownames(matrix2plot),
  fraction_table
)
write.table(fraction_table, file="source_data/results/positiveFraction_celltype_fetal_adult_disease.csv", sep=",", quote=F, col.name
```

In [34]:

```
# Fetal vs Adult vs Disease
matrix2plot = x_thresholded_norm

# data 2 remove
stage = c("fetal", "adult")
disease_condition = c("healthy", "disease") #
cell_or_nuclei = c("cells", "nuclei")

stratification = metadata[,"stage"] %in% stage &
                 metadata[,"disease_condition"] %in% disease_condition &
                 metadata[,"cell_or_nuclei"] %in% cell_or_nuclei #&
                 #metadata[,"disease_subcategory"] == "dHF"



matrix2plot_stratification = matrix2plot[stratification,]
metadata_stratification = metadata[stratification,]

# new factor object:
  combined_factor = paste(metadata_stratification[,"new_clusters"], metadata_stratification[,"stage"], metadata_stratification[,"dis
  metadata_stratification$combined_factor = factor(
    combined_factor,
    levels=paste(rep(levels(metadata_stratification[,"new_clusters"]),each=3), rep(rev(levels(metadata_stratification[,"stage"])),c(
  )

  genes = genes_EDN
  #stratification_column = "stage"
```

```
  #stratification = "adult"
  cluster_column = "combined_factor"

mem = sapply(
  levels(metadata_stratification[,cluster_column]),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colMeans(matrix2plot_stratification[cells,genes,drop=F], na.rm=T))
    } else {
      return(rep(NA, length(genes)))
   }
  }
)

mem_norm = mem
#mem_norm = mem / rowMaxs(mem, na.rm=T)
mem_norm = t(mem_norm)



dm = sapply(
  levels(metadata_stratification[,cluster_column] ),
  function(k){
    cells = rownames(metadata_stratification)[metadata_stratification[,cluster_column] == k]
    if(length(cells) > 0){
      return(colSums(matrix2plot_stratification[cells,genes,drop=F]>0, na.rm=T) / length(cells)*100)
    } else {
      return(rep(NA, length(genes)))
    }
  }
)
dm = t(dm)


fetal_matrix = t(mem_norm)[,1:20*3-2]
adult_matrix = t(mem_norm)[,1:20*3-1]
disease_matrix = t(mem_norm)[,1:20*3]
fetal_dm = t(dm)[,1:20*3-2]
adult_dm = t(dm)[,1:20*3-1]
disease_dm = t(dm)[,1:20*3]
```

In [35]:

```
fetal_dm
```

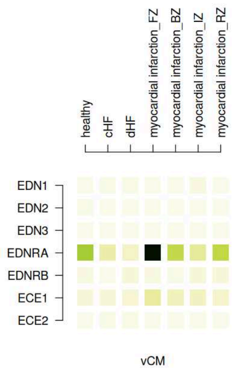
	vCM_fetal_healthy	aCM_fetal_healthy	CCS_fetal_healthy	SMC_fetal_healthy	Pericytes_fetal_healthy	EC_Valves_Pericytes_fetal_healthy	Endocardium_fe
EDN1	0.67346008	0.40417952	0.33905967	1.16613660	1.95788088	19.3661972	2
EDN2	0.47201023	0.20957456	0.74593128	0.26250694	0.06581112	0.4527163	
EDN3	0.03382005	0.01796353	0.18083183	1.13079913	0.44422507	0.4527163	
EDNRA	9.54313527	24.37950960	13.17811935	37.16997324	42.79368213	21.6297787	
EDNRB	0.79991766	0.52393641	0.65551537	12.12580140	12.37249095	36.3179074	3
ECE1	2.28799976	1.44307057	1.69529837	6.12852744	12.05988812	44.5171026	5
ECE2	0.02793830	0.01796353	0.02260398	0.04543389	0.16452781	0.4527163	

EDN in disease subcategories

In [40]:

```
# EDN genes, each cell type, all disease subcategories
celltypes2test = levels(metadata$new_clusters)[!levels(metadata$new_clusters) %in% c("PlateletsErythrocytes", "unknown")]

for(celltype in celltypes2test){
    adult_metadata = metadata[metadata$stage == "adult",]
    adult_metadata = adult_metadata[adult_metadata$new_clusters==celltype,]
    temp_conditions = paste(adult_metadata$disease_subcategory)

    temp_matrix = sapply(1:length(unique(temp_conditions)), function(k){colMeans(x_thresholded_norm[rownames(adult_metadata)[temp_co
    colnames(temp_matrix) = unique(temp_conditions)
    if(celltype == "vCM"){
        m2 = temp_matrix
        colnames(m2) = paste(celltype, colnames(temp_matrix), sep="_")
        #m2 = m2/max(m2, na.rm=T)
        temp_full_matrix = m2
        }
    if(celltype != "vCM"){
        m2 = temp_matrix
        colnames(m2) = paste(celltype, colnames(temp_matrix), sep="_")
        #m2 = m2/max(m2, na.rm=T)
        temp_full_matrix = cbind(temp_full_matrix,NA,m2)
        }

    pdf(paste("figures/panels/results/diseaseSubCat_matrix_", celltype,".pdf"), width=4.8, height=6.3)
        par(mar=c(1,5,20,5))
        matrix.plot(temp_matrix, round.n=3, col=rev(verdant), ylab="", xlab="")
        dev.off()
    }
# e.g.
options(repr.plot.width = 4.5)
options(repr.plot.height = 7)
 adult_metadata = metadata[metadata$stage == "adult",]
    adult_metadata = adult_metadata[adult_metadata$new_clusters=="vCM",]
    temp_conditions = paste(adult_metadata$disease_subcategory)

    temp_matrix = sapply(1:length(unique(temp_conditions)), function(k){colMeans(x_thresholded_norm[rownames(adult_metadata)[temp_co
    colnames(temp_matrix) = unique(temp_conditions)
```

```
par(mar=c(1,5,20,5))
        matrix.plot(temp_matrix, round.n=3, col=rev(verdant), ylab="", xlab="")
        mtext("vCM", 1)
```



In [41]:

```
# Figure of all EDN genes, and all disease subcategories across all cell types

options(repr.plot.width = 40)
options(repr.plot.height = 15)
pdf(paste("figures/panels/results/diseaseSubCat_matrix_allCellTypes.pdf"), width=34, height=6.3)
par(mar=c(1,5,20,5))
    matrix.plot(temp_full_matrix, round.n=3, cex.multiplier = 3, col=rev(verdant), ylab="", xlab="")
    points(seq(10,15, length=50), rep(21, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:50, rev(verdant)))
    text(
      seq(10,15, length=3),
      rep(20, 3),
      seq(signif(min(temp_full_matrix, na.rm=T),3), signif(max(temp_full_matrix,na.rm=T),3), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
dev.off()
    par(oma=c(0,0,52,0))
 matrix.plot(temp_full_matrix, round.n=3, cex.multiplier = 3, col=rev(verdant), ylab="", xlab="")
    points(seq(10,15, length=50), rep(21, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:50, rev(verdant)))
    text(
      seq(10,15, length=3),
      rep(20, 3),
      seq(signif(min(temp_full_matrix, na.rm=T),3), signif(max(temp_full_matrix,na.rm=T),3), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
```

agg_record_2032637574: 2

Create psuedobulk samples

In [336…

```
# pseudobulk?
bulk_class = factor(paste(metadata$new_clusters, metadata$stage, metadata$disease_subcategory, metadata$donor,  metadata$cell_or_nuc
levels(bulk_class)
```

'aCM__adult__cHF__C1__cells' · 'aCM__adult__dHF__D1__cells' · 'aCM__adult__dHF__D2__cells' · 'aCM__adult__dHF__D4__cells' ·
'aCM__adult__healthy__D2__nuclei' · 'aCM__adult__healthy__D3__cells' · 'aCM__adult__healthy__D3__nuclei' · 'aCM__adult__healthy__D4__cells' ·
'aCM__adult__healthy__D4__nuclei' · 'aCM__adult__healthy__D5__cells' · 'aCM__adult__healthy__D5__nuclei' · 'aCM__adult__healthy__D6__cells' ·
'aCM__adult__healthy__D6__nuclei' · 'aCM__adult__healthy__D7__cells' · 'aCM__adult__healthy__D7__nuclei' · 'aCM__adult__healthy__N10__cells' ·
'aCM__adult__healthy__N11__cells' · 'aCM__adult__healthy__N12__cells' · 'aCM__adult__healthy__N13__cells' · 'aCM__adult__healthy__N14__cells' ·
'aCM__adult__healthy__N6__cells' · 'aCM__adult__healthy__N7__cells' · 'aCM__adult__healthy__N8__cells' · 'aCM__adult__healthy__N9__cells' ·
'aCM__adult__healthy__P1__nuclei' · 'aCM__adult__healthy__P17__nuclei' · 'aCM__adult__healthy__P7__nuclei' · 'aCM__adult__healthy__P8__nuclei' ·
'aCM__adult__myocardial infarction_BZ__P12__nuclei' · 'aCM__adult__myocardial infarction_BZ__P2__nuclei' ·
'aCM__adult__myocardial infarction_BZ__P3__nuclei' · 'aCM__adult__myocardial infarction_FZ__P14__nuclei' ·
'aCM__adult__myocardial infarction_FZ__P18__nuclei' · 'aCM__adult__myocardial infarction_FZ__P19__nuclei' ·
'aCM__adult__myocardial infarction_FZ__P20__nuclei' · 'aCM__adult__myocardial infarction_FZ__P4__nuclei' · 'aCM__adult__myocardial infarction_FZ__P5__nuclei' ·
'aCM__adult__myocardial infarction_IZ__P10__nuclei' · 'aCM__adult__myocardial infarction_IZ__P9__nuclei' ·
'aCM__adult__myocardial infarction_RZ__P11__nuclei' · 'aCM__adult__myocardial infarction_RZ__P2__nuclei' ·
'aCM__adult__myocardial infarction_RZ__P6__nuclei' · 'aCM__adult__myocardial infarction_RZ__P9__nuclei' · 'aCM__fetal__healthy__09W0D__cells' ·
'aCM__fetal__healthy__09W6D__cells' · 'aCM__fetal__healthy__10X127__cells' · 'aCM__fetal__healthy__10X142__cells' · 'aCM__fetal__healthy__10X151__cells' ·
'aCM__fetal__healthy__10X158__cells' · 'aCM__fetal__healthy__10X166__cells' · 'aCM__fetal__healthy__10X246__cells' · 'aCM__fetal__healthy__10X253__cells' ·
'aCM__fetal__healthy__10X256__cells' · 'aCM__fetal__healthy__10X263__cells' · 'aCM__fetal__healthy__10X274__cells' · 'aCM__fetal__healthy__10X285__cells' ·
'aCM__fetal__healthy__10X289__cells' · 'aCM__fetal__healthy__10X299__cells' · 'aCM__fetal__healthy__10X300__cells' · 'aCM__fetal__healthy__10X303__cells' ·
'aCM__fetal__healthy__11W1D__cells' · 'aCM__fetal__healthy__11W4D__cells' · 'aCM__fetal__healthy__13W3D__cells' · 'aCM__fetal__healthy__15W0D__cells' ·
'aCM__fetal__healthy__15W2D__cells' · 'aCM__fetal__healthy__alexsc__cells' · 'aCM__fetal__healthy__BRC2251__cells' · 'aCM__fetal__healthy__BRC2252__cells' ·
'aCM__fetal__healthy__BRC2256__cells' · 'aCM__fetal__healthy__BRC2260__cells' · 'aCM__fetal__healthy__BRC2262__cells' ·
'aCM__fetal__healthy__BRC2263__cells' · 'aCM__fetal__healthy__F8W4D__nuclei' · 'aCM__fetal__healthy__M10W5D__nuclei' ·
'aCM__fetal__healthy__M9W0D__nuclei' · 'Adipocytes__adult__healthy__D2__nuclei' · 'Adipocytes__adult__healthy__D3__nuclei' ·
'Adipocytes__adult__healthy__D4__nuclei' · 'Adipocytes__adult__healthy__D5__nuclei' · 'Adipocytes__adult__healthy__D6__nuclei' ·
'Adipocytes__adult__healthy__D7__nuclei' · 'Adipocytes__adult__healthy__P1__nuclei' · 'Adipocytes__adult__healthy__P17__nuclei' ·
'Adipocytes__adult__healthy__P7__nuclei' · 'Adipocytes__adult__healthy__P8__nuclei' · 'Adipocytes__adult__myocardial infarction_BZ__P12__nuclei' ·
'Adipocytes__adult__myocardial infarction_BZ__P2__nuclei' · 'Adipocytes__adult__myocardial infarction_BZ__P3__nuclei' ·
'Adipocytes__adult__myocardial infarction_FZ__P14__nuclei' · 'Adipocytes__adult__myocardial infarction_FZ__P18__nuclei' ·
'Adipocytes__adult__myocardial infarction_FZ__P19__nuclei' · 'Adipocytes__adult__myocardial infarction_FZ__P20__nuclei' ·
'Adipocytes__adult__myocardial infarction_FZ__P4__nuclei' · 'Adipocytes__adult__myocardial infarction_FZ__P5__nuclei' ·
'Adipocytes__adult__myocardial infarction_IZ__P10__nuclei' · 'Adipocytes__adult__myocardial infarction_IZ__P13__nuclei' ·
'Adipocytes__adult__myocardial infarction_IZ__P15__nuclei' · 'Adipocytes__adult__myocardial infarction_IZ__P16__nuclei' ·
'Adipocytes__adult__myocardial infarction_IZ__P2__nuclei' · 'Adipocytes__adult__myocardial infarction_IZ__P3__nuclei' ·
'Adipocytes__adult__myocardial infarction_IZ__P9__nuclei' · 'Adipocytes__adult__myocardial infarction_RZ__P11__nuclei' ·
'Adipocytes__adult__myocardial infarction_RZ__P2__nuclei' · 'Adipocytes__adult__myocardial infarction_RZ__P3__nuclei' ·
'Adipocytes__adult__myocardial infarction_RZ__P6__nuclei' · 'Adipocytes__adult__myocardial infarction_RZ__P9__nuclei' ·
'Adipocytes__fetal__healthy__BRC2251__cells' · 'CCS__adult__cHF__C1__cells' · 'CCS__adult__cHF__C2__cells' · 'CCS__adult__dHF__D1__cells' ·
'CCS__adult__dHF__D2__cells' · 'CCS__adult__dHF__D4__cells' · 'CCS__adult__dHF__D5__cells' · 'CCS__adult__healthy__D2__nuclei' ·
'CCS__adult__healthy__D3__nuclei' · 'CCS__adult__healthy__D4__cells' · 'CCS__adult__healthy__D5__nuclei' · 'CCS__adult__healthy__D7__nuclei' ·
'CCS__adult__healthy__N1__cells' · 'CCS__adult__healthy__N10__cells' · 'CCS__adult__healthy__N11__cells' · 'CCS__adult__healthy__N12__cells' ·
'CCS__adult__healthy__N13__cells' · 'CCS__adult__healthy__N14__cells' · 'CCS__adult__healthy__N2__cells' · 'CCS__adult__healthy__N3__cells' ·
'CCS__adult__healthy__N4__cells' · 'CCS__adult__healthy__N5__cells' · 'CCS__adult__healthy__N6__cells' · 'CCS__adult__healthy__N7__cells' ·
'CCS__adult__healthy__N8__cells' · 'CCS__adult__healthy__N9__cells' · 'CCS__adult__healthy__P1__nuclei' · 'CCS__adult__myocardial infarction_BZ__P2__nuclei' ·
'CCS__adult__myocardial infarction_BZ__P3__nuclei' · 'CCS__adult__myocardial infarction_FZ__P14__nuclei' · 'CCS__adult__myocardial infarction_FZ__P18__nuclei' ·
'CCS__adult__myocardial infarction_FZ__P20__nuclei' · 'CCS__adult__myocardial infarction_FZ__P5__nuclei' · 'CCS__adult__myocardial infarction_RZ__P2__nuclei' ·
'CCS__fetal__healthy__09W0D__cells' · 'CCS__fetal__healthy__09W6D__cells' · 'CCS__fetal__healthy__10X127__cells' · 'CCS__fetal__healthy__10X142__cells' ·
'CCS__fetal__healthy__10X151__cells' · 'CCS__fetal__healthy__10X158__cells' · 'CCS__fetal__healthy__10X166__cells' · 'CCS__fetal__healthy__10X246__cells' ·
'CCS__fetal__healthy__10X253__cells' · 'CCS__fetal__healthy__10X256__cells' · 'CCS__fetal__healthy__10X263__cells' · 'CCS__fetal__healthy__10X274__cells' ·
'CCS__fetal__healthy__10X285__cells' · 'CCS__fetal__healthy__10X289__cells' · 'CCS__fetal__healthy__10X299__cells' · 'CCS__fetal__healthy__10X300__cells' ·
'CCS__fetal__healthy__10X303__cells' · 'CCS__fetal__healthy__11W1D__cells' · 'CCS__fetal__healthy__11W4D__cells' · 'CCS__fetal__healthy__13W3D__cells' ·
'CCS__fetal__healthy__15W0D__cells' · 'CCS__fetal__healthy__15W2D__cells' · 'CCS__fetal__healthy__BRC2251__cells' · 'CCS__fetal__healthy__BRC2252__cells' ·
'CCS__fetal__healthy__BRC2256__cells' · 'CCS__fetal__healthy__BRC2260__cells' · 'CCS__fetal__healthy__BRC2262__cells' · 'CCS__fetal__healthy__BRC2263__cells' ·
'CCS__fetal__healthy__F8W4D__nuclei' · 'CCS__fetal__healthy__M10W5D__nuclei' · 'CCS__fetal__healthy__M9W0D__nuclei' · 'EC_Arterial__adult__cHF__C1__cells' ·
'EC_Arterial__adult__cHF__C2__cells' · 'EC_Arterial__adult__dHF__D1__cells' · 'EC_Arterial__adult__dHF__D2__cells' · 'EC_Arterial__adult__dHF__D4__cells' ·
'EC_Arterial__adult__healthy__D2__nuclei' · 'EC_Arterial__adult__healthy__D3__cells' · 'EC_Arterial__adult__healthy__D3__nuclei' ·
'EC_Arterial__adult__healthy__D4__cells' · 'EC_Arterial__adult__healthy__D4__nuclei' · 'EC_Arterial__adult__healthy__D5__cells' ·
'EC_Arterial__adult__healthy__D5__nuclei' · 'EC_Arterial__adult__healthy__D6__cells' · 'EC_Arterial__adult__healthy__D6__nuclei' ·
'EC_Arterial__adult__healthy__D7__cells' · 'EC_Arterial__adult__healthy__D7__nuclei' · 'EC_Arterial__adult__healthy__N1__cells' ·
'EC_Arterial__adult__healthy__N10__cells' · 'EC_Arterial__adult__healthy__N11__cells' · 'EC_Arterial__adult__healthy__N12__cells' ·
'EC_Arterial__adult__healthy__N2__cells' · 'EC_Arterial__adult__healthy__N4__cells' · 'EC_Arterial__adult__healthy__N5__cells' ·
'EC_Arterial__adult__healthy__N6__cells' · 'EC_Arterial__adult__healthy__N7__cells' · 'EC_Arterial__adult__healthy__N8__cells' ·
'EC_Arterial__adult__healthy__N9__cells' · 'EC_Arterial__adult__healthy__P1__nuclei' · 'EC_Arterial__adult__healthy__P17__nuclei' · ⋯ ·
'SMC__fetal__healthy__10X127__cells' · 'SMC__fetal__healthy__10X142__cells' · 'SMC__fetal__healthy__10X151__cells' · 'SMC__fetal__healthy__10X158__cells' ·
'SMC__fetal__healthy__10X166__cells' · 'SMC__fetal__healthy__10X246__cells' · 'SMC__fetal__healthy__10X253__cells' · 'SMC__fetal__healthy__10X256__cells' ·
'SMC__fetal__healthy__10X263__cells' · 'SMC__fetal__healthy__10X274__cells' · 'SMC__fetal__healthy__10X285__cells' · 'SMC__fetal__healthy__10X289__cells' ·
'SMC__fetal__healthy__10X299__cells' · 'SMC__fetal__healthy__10X300__cells' · 'SMC__fetal__healthy__10X303__cells' · 'SMC__fetal__healthy__11W1D__cells' ·
'SMC__fetal__healthy__11W4D__cells' · 'SMC__fetal__healthy__13W3D__cells' · 'SMC__fetal__healthy__15W0D__cells' · 'SMC__fetal__healthy__15W2D__cells' ·
'SMC__fetal__healthy__alexsc__cells' · 'SMC__fetal__healthy__BRC2251__cells' · 'SMC__fetal__healthy__BRC2252__cells' · 'SMC__fetal__healthy__BRC2256__cells' ·
'SMC__fetal__healthy__BRC2260__cells' · 'SMC__fetal__healthy__BRC2262__cells' · 'SMC__fetal__healthy__BRC2263__cells' ·
'SMC__fetal__healthy__F8W4D__nuclei' · 'SMC__fetal__healthy__M10W5D__nuclei' · 'SMC__fetal__healthy__M9W0D__nuclei' · 'unknown__adult__cHF__C1__cells' ·
'unknown__adult__cHF__C2__cells' · 'unknown__adult__dHF__D1__cells' · 'unknown__adult__dHF__D2__cells' · 'unknown__adult__dHF__D5__cells' ·
'unknown__adult__healthy__D2__nuclei' · 'unknown__adult__healthy__D3__cells' · 'unknown__adult__healthy__D3__nuclei' ·
'unknown__adult__healthy__D4__cells' · 'unknown__adult__healthy__D4__nuclei' · 'unknown__adult__healthy__D5__cells' ·
'unknown__adult__healthy__D5__nuclei' · 'unknown__adult__healthy__D6__cells' · 'unknown__adult__healthy__D6__nuclei' ·
'unknown__adult__healthy__D7__cells' · 'unknown__adult__healthy__D7__nuclei' · 'unknown__adult__healthy__N1__cells' ·
'unknown__adult__healthy__N10__cells' · 'unknown__adult__healthy__N11__cells' · 'unknown__adult__healthy__N13__cells' ·

'unknown__adult__healthy__N14__cells' · 'unknown__adult__healthy__N2__cells' · 'unknown__adult__healthy__N3__cells' · 'unknown__adult__healthy__N5__cells' · 'unknown__adult__healthy__N6__cells' · 'unknown__adult__healthy__N7__cells' · 'unknown__adult__healthy__N8__cells' · 'unknown__adult__healthy__N9__cells' · 'unknown__adult__healthy__P1__nuclei' · 'unknown__adult__healthy__P17__nuclei' · 'unknown__adult__healthy__P7__nuclei' · 'unknown__adult__healthy__P8__nuclei' · 'unknown__adult__myocardial infarction_BZ__P12__nuclei' · 'unknown__adult__myocardial infarction_BZ__P2__nuclei' · 'unknown__adult__myocardial infarction_BZ__P3__nuclei' · 'unknown__adult__myocardial infarction_FZ__P14__nuclei' · 'unknown__adult__myocardial infarction_FZ__P18__nuclei' · 'unknown__adult__myocardial infarction_FZ__P19__nuclei' · 'unknown__adult__myocardial infarction_FZ__P20__nuclei' · 'unknown__adult__myocardial infarction_FZ__P4__nuclei' · 'unknown__adult__myocardial infarction_FZ__P5__nuclei' · 'unknown__adult__myocardial infarction_IZ__P10__nuclei' · 'unknown__adult__myocardial infarction_IZ__P13__nuclei' · 'unknown__adult__myocardial infarction_IZ__P15__nuclei' · 'unknown__adult__myocardial infarction_IZ__P16__nuclei' · 'unknown__adult__myocardial infarction_IZ__P2__nuclei' · 'unknown__adult__myocardial infarction_IZ__P3__nuclei' · 'unknown__adult__myocardial infarction_IZ__P9__nuclei' · 'unknown__adult__myocardial infarction_RZ__P11__nuclei' · 'unknown__adult__myocardial infarction_RZ__P2__nuclei' · 'unknown__adult__myocardial infarction_RZ__P3__nuclei' · 'unknown__adult__myocardial infarction_RZ__P6__nuclei' · 'unknown__adult__myocardial infarction_RZ__P9__nuclei' · 'unknown__fetal__healthy__09W0D__cells' · 'unknown__fetal__healthy__09W6D__cells' · 'unknown__fetal__healthy__10X127__cells' · 'unknown__fetal__healthy__10X142__cells' · 'unknown__fetal__healthy__10X151__cells' · 'unknown__fetal__healthy__10X158__cells' · 'unknown__fetal__healthy__10X166__cells' · 'unknown__fetal__healthy__10X246__cells' · 'unknown__fetal__healthy__10X253__cells' · 'unknown__fetal__healthy__10X256__cells' · 'unknown__fetal__healthy__10X263__cells' · 'unknown__fetal__healthy__10X274__cells' · 'unknown__fetal__healthy__10X285__cells' · 'unknown__fetal__healthy__10X289__cells' · 'unknown__fetal__healthy__10X299__cells' · 'unknown__fetal__healthy__10X300__cells' · 'unknown__fetal__healthy__10X303__cells' · 'unknown__fetal__healthy__11W1D__cells' · 'unknown__fetal__healthy__11W4D__cells' · 'unknown__fetal__healthy__13W3D__cells' · 'unknown__fetal__healthy__15W0D__cells' · 'unknown__fetal__healthy__15W2D__cells' · 'unknown__fetal__healthy__alexsc__cells' · 'unknown__fetal__healthy__BRC2251__cells' · 'unknown__fetal__healthy__BRC2252__cells' · 'unknown__fetal__healthy__BRC2256__cells' · 'unknown__fetal__healthy__BRC2260__cells' · 'unknown__fetal__healthy__BRC2262__cells' · 'unknown__fetal__healthy__BRC2263__cells' · 'vCM__adult__cHF__C1__cells' · 'vCM__adult__cHF__C2__cells' · 'vCM__adult__dHF__D1__cells' · 'vCM__adult__dHF__D2__cells' · 'vCM__adult__dHF__D4__cells' · 'vCM__adult__dHF__D5__cells' · 'vCM__adult__healthy__D2__nuclei' · 'vCM__adult__healthy__D3__cells' · 'vCM__adult__healthy__D3__nuclei' · 'vCM__adult__healthy__D4__cells' · 'vCM__adult__healthy__D4__nuclei' · 'vCM__adult__healthy__D5__cells' · 'vCM__adult__healthy__D5__nuclei' · 'vCM__adult__healthy__D6__cells' · 'vCM__adult__healthy__D6__nuclei' · 'vCM__adult__healthy__D7__cells' · 'vCM__adult__healthy__D7__nuclei' · 'vCM__adult__healthy__N1__cells' · 'vCM__adult__healthy__N10__cells' · 'vCM__adult__healthy__N11__cells' · 'vCM__adult__healthy__N12__cells' · 'vCM__adult__healthy__N13__cells' · 'vCM__adult__healthy__N14__cells' · 'vCM__adult__healthy__N2__cells' · 'vCM__adult__healthy__N3__cells' · 'vCM__adult__healthy__N4__cells' · 'vCM__adult__healthy__N5__cells' · 'vCM__adult__healthy__N6__cells' · 'vCM__adult__healthy__N7__cells' · 'vCM__adult__healthy__N8__cells' · 'vCM__adult__healthy__N9__cells' · 'vCM__adult__healthy__P1__nuclei' · 'vCM__adult__healthy__P17__nuclei' · 'vCM__adult__healthy__P7__nuclei' · 'vCM__adult__healthy__P8__nuclei' · 'vCM__adult__myocardial infarction_BZ__P12__nuclei' · 'vCM__adult__myocardial infarction_BZ__P2__nuclei' · 'vCM__adult__myocardial infarction_BZ__P3__nuclei' · 'vCM__adult__myocardial infarction_FZ__P14__nuclei' · 'vCM__adult__myocardial infarction_FZ__P18__nuclei' · 'vCM__adult__myocardial infarction_FZ__P19__nuclei' · 'vCM__adult__myocardial infarction_FZ__P20__nuclei' · 'vCM__adult__myocardial infarction_FZ__P4__nuclei' · 'vCM__adult__myocardial infarction_FZ__P5__nuclei' · 'vCM__adult__myocardial infarction_IZ__P10__nuclei' · 'vCM__adult__myocardial infarction_IZ__P13__nuclei' · 'vCM__adult__myocardial infarction_IZ__P15__nuclei' · 'vCM__adult__myocardial infarction_IZ__P16__nuclei' · 'vCM__adult__myocardial infarction_IZ__P2__nuclei' · 'vCM__adult__myocardial infarction_IZ__P3__nuclei' · 'vCM__adult__myocardial infarction_IZ__P9__nuclei' · 'vCM__adult__myocardial infarction_RZ__P11__nuclei' · 'vCM__adult__myocardial infarction_RZ__P2__nuclei' · 'vCM__adult__myocardial infarction_RZ__P3__nuclei' · 'vCM__adult__myocardial infarction_RZ__P6__nuclei' · 'vCM__adult__myocardial infarction_RZ__P9__nuclei' · 'vCM__fetal__healthy__09W0D__cells' · 'vCM__fetal__healthy__09W6D__cells' · 'vCM__fetal__healthy__10X127__cells' · 'vCM__fetal__healthy__10X142__cells' · 'vCM__fetal__healthy__10X151__cells' · 'vCM__fetal__healthy__10X158__cells' · 'vCM__fetal__healthy__10X166__cells' · 'vCM__fetal__healthy__10X246__cells' · 'vCM__fetal__healthy__10X253__cells' · 'vCM__fetal__healthy__10X256__cells' · 'vCM__fetal__healthy__10X263__cells' · 'vCM__fetal__healthy__10X274__cells' · 'vCM__fetal__healthy__10X285__cells' · 'vCM__fetal__healthy__10X289__cells' · 'vCM__fetal__healthy__10X299__cells' · 'vCM__fetal__healthy__10X300__cells' · 'vCM__fetal__healthy__10X303__cells' · 'vCM__fetal__healthy__11W1D__cells' · 'vCM__fetal__healthy__11W4D__cells' · 'vCM__fetal__healthy__13W3D__cells' · 'vCM__fetal__healthy__15W0D__cells' · 'vCM__fetal__healthy__15W2D__cells' · 'vCM__fetal__healthy__alexsc__cells' · 'vCM__fetal__healthy__BRC2251__cells' · 'vCM__fetal__healthy__BRC2252__cells' · 'vCM__fetal__healthy__BRC2256__cells' · 'vCM__fetal__healthy__BRC2260__cells' · 'vCM__fetal__healthy__BRC2262__cells' · 'vCM__fetal__healthy__BRC2263__cells' · 'vCM__fetal__healthy__F8W4D__nuclei' · 'vCM__fetal__healthy__M10W5D__nuclei' · 'vCM__fetal__healthy__M9W0D__nuclei'

In [340…
```
bulk_class_cellnums = table(bulk_class)
```

In [341…
```
bulk_class_cellnums
```

bulk_class	
aCM__adult__cHF__C1__cells	183
aCM__adult__dHF__D1__cells	120
aCM__adult__dHF__D2__cells	138
aCM__adult__dHF__D4__cells	223
aCM__adult__healthy__D2__nuclei	389
aCM__adult__healthy__D3__cells	2
aCM__adult__healthy__D3__nuclei	371
aCM__adult__healthy__D4__cells	8
aCM__adult__healthy__D4__nuclei	429
aCM__adult__healthy__D5__cells	13
aCM__adult__healthy__D5__nuclei	432
aCM__adult__healthy__D6__cells	6
aCM__adult__healthy__D6__nuclei	428
aCM__adult__healthy__D7__cells	9
aCM__adult__healthy__D7__nuclei	437
aCM__adult__healthy__N10__cells	88
aCM__adult__healthy__N11__cells	120
aCM__adult__healthy__N12__cells	187
aCM__adult__healthy__N13__cells	154
aCM__adult__healthy__N14__cells	171
aCM__adult__healthy__N6__cells	56
aCM__adult__healthy__N7__cells	38
aCM__adult__healthy__N8__cells	32
aCM__adult__healthy__N9__cells	184
aCM__adult__healthy__P1__nuclei	22
aCM__adult__healthy__P17__nuclei	13
aCM__adult__healthy__P7__nuclei	3
aCM__adult__healthy__P8__nuclei	27
aCM__adult__myocardial infarction_BZ__P12__nuclei	3
aCM__adult__myocardial infarction_BZ__P2__nuclei	62
aCM__adult__myocardial infarction_BZ__P3__nuclei	27
aCM__adult__myocardial infarction_FZ__P14__nuclei	25
aCM__adult__myocardial infarction_FZ__P18__nuclei	5
aCM__adult__myocardial infarction_FZ__P19__nuclei	14
aCM__adult__myocardial infarction_FZ__P20__nuclei	7
aCM__adult__myocardial infarction_FZ__P4__nuclei	1
aCM__adult__myocardial infarction_FZ__P5__nuclei	89
aCM__adult__myocardial infarction_IZ__P10__nuclei	3
aCM__adult__myocardial infarction_IZ__P9__nuclei	2
aCM__adult__myocardial infarction_RZ__P11__nuclei	30
aCM__adult__myocardial infarction_RZ__P2__nuclei	87
aCM__adult__myocardial infarction_RZ__P6__nuclei	31
aCM__adult__myocardial infarction_RZ__P9__nuclei	38
aCM__fetal__healthy__09W0D__cells	314
aCM__fetal__healthy__09W6D__cells	780
aCM__fetal__healthy__10X127__cells	374
aCM__fetal__healthy__10X142__cells	457
aCM__fetal__healthy__10X151__cells	248
aCM__fetal__healthy__10X158__cells	1515
aCM__fetal__healthy__10X166__cells	

659
aCM__fetal__healthy__10X246__cells
386
aCM__fetal__healthy__10X253__cells
402
aCM__fetal__healthy__10X256__cells
240
aCM__fetal__healthy__10X263__cells
255
aCM__fetal__healthy__10X274__cells
64
aCM__fetal__healthy__10X285__cells
100
aCM__fetal__healthy__10X289__cells
514
aCM__fetal__healthy__10X299__cells
510
aCM__fetal__healthy__10X300__cells
243
aCM__fetal__healthy__10X303__cells
252
aCM__fetal__healthy__11W1D__cells
3037
aCM__fetal__healthy__11W4D__cells
780
aCM__fetal__healthy__13W3D__cells
7735
aCM__fetal__healthy__15W0D__cells
2941
aCM__fetal__healthy__15W2D__cells
2455
aCM__fetal__healthy__alexsc__cells
3
aCM__fetal__healthy__BRC2251__cells
31
aCM__fetal__healthy__BRC2252__cells
384
aCM__fetal__healthy__BRC2256__cells
97
aCM__fetal__healthy__BRC2260__cells
8
aCM__fetal__healthy__BRC2262__cells
5
aCM__fetal__healthy__BRC2263__cells
9
aCM__fetal__healthy__F8W4D__nuclei
4716
aCM__fetal__healthy__M10W5D__nuclei
2625
aCM__fetal__healthy__M9W0D__nuclei
1262
Adipocytes__adult__healthy__D2__nuclei
168
Adipocytes__adult__healthy__D3__nuclei
95
Adipocytes__adult__healthy__D4__nuclei
131
Adipocytes__adult__healthy__D5__nuclei
91
Adipocytes__adult__healthy__D6__nuclei
132
Adipocytes__adult__healthy__D7__nuclei
73
Adipocytes__adult__healthy__P1__nuclei
2
Adipocytes__adult__healthy__P17__nuclei
2
Adipocytes__adult__healthy__P7__nuclei
3
Adipocytes__adult__healthy__P8__nuclei
8
Adipocytes__adult__myocardial infarction_BZ__P12__nuclei
2
Adipocytes__adult__myocardial infarction_BZ__P2__nuclei
4
Adipocytes__adult__myocardial infarction_BZ__P3__nuclei
11
Adipocytes__adult__myocardial infarction_FZ__P14__nuclei
12
Adipocytes__adult__myocardial infarction_FZ__P18__nuclei
175
Adipocytes__adult__myocardial infarction_FZ__P19__nuclei
2
Adipocytes__adult__myocardial infarction_FZ__P20__nuclei
31
Adipocytes__adult__myocardial infarction_FZ__P4__nuclei
4
Adipocytes__adult__myocardial infarction_FZ__P5__nuclei
7
Adipocytes__adult__myocardial infarction_IZ__P10__nuclei
4
Adipocytes__adult__myocardial infarction_IZ__P13__nuclei
3
Adipocytes__adult__myocardial infarction_IZ__P15__nuclei
22
Adipocytes__adult__myocardial infarction_IZ__P16__nuclei
16
Adipocytes__adult__myocardial infarction_IZ__P2__nuclei
1
Adipocytes__adult__myocardial infarction_IZ__P3__nuclei

1
Adipocytes__adult__myocardial infarction_IZ__P9__nuclei
11
Adipocytes__adult__myocardial infarction_RZ__P11__nuclei
3
Adipocytes__adult__myocardial infarction_RZ__P2__nuclei
10
Adipocytes__adult__myocardial infarction_RZ__P3__nuclei
175
Adipocytes__adult__myocardial infarction_RZ__P6__nuclei
2
Adipocytes__adult__myocardial infarction_RZ__P9__nuclei
7
Adipocytes__fetal__healthy__BRC2251__cells
1
CCS__adult__cHF__C1__cells
15
CCS__adult__cHF__C2__cells
14
CCS__adult__dHF__D1__cells
31
CCS__adult__dHF__D2__cells
103
CCS__adult__dHF__D4__cells
6
CCS__adult__dHF__D5__cells
6
CCS__adult__healthy__D2__nuclei
5
CCS__adult__healthy__D3__nuclei
5
CCS__adult__healthy__D4__cells
1
CCS__adult__healthy__D5__nuclei
1
CCS__adult__healthy__D7__nuclei
3
CCS__adult__healthy__N1__cells
36
CCS__adult__healthy__N10__cells
8
CCS__adult__healthy__N11__cells
77
CCS__adult__healthy__N12__cells
8
CCS__adult__healthy__N13__cells
32
CCS__adult__healthy__N14__cells
1
CCS__adult__healthy__N2__cells
8
CCS__adult__healthy__N3__cells
4
CCS__adult__healthy__N4__cells
49
CCS__adult__healthy__N5__cells
17
CCS__adult__healthy__N6__cells
288
CCS__adult__healthy__N7__cells
70
CCS__adult__healthy__N8__cells
41
CCS__adult__healthy__N9__cells
46
CCS__adult__healthy__P1__nuclei
2
CCS__adult__myocardial infarction_BZ__P2__nuclei
6
CCS__adult__myocardial infarction_BZ__P3__nuclei
2
CCS__adult__myocardial infarction_FZ__P14__nuclei
4
CCS__adult__myocardial infarction_FZ__P18__nuclei
1
CCS__adult__myocardial infarction_FZ__P20__nuclei
2
CCS__adult__myocardial infarction_FZ__P5__nuclei
4
CCS__adult__myocardial infarction_RZ__P2__nuclei
4
CCS__fetal__healthy__09W0D__cells
1351
CCS__fetal__healthy__09W6D__cells
298
CCS__fetal__healthy__10X127__cells
18
CCS__fetal__healthy__10X142__cells
40
CCS__fetal__healthy__10X151__cells
25
CCS__fetal__healthy__10X158__cells
132
CCS__fetal__healthy__10X166__cells
6
CCS__fetal__healthy__10X246__cells
16
CCS__fetal__healthy__10X253__cells
22
CCS__fetal__healthy__10X256__cells

16
CCS__fetal__healthy__10X263__cells
10
CCS__fetal__healthy__10X274__cells
5
CCS__fetal__healthy__10X285__cells
2
CCS__fetal__healthy__10X289__cells
22
CCS__fetal__healthy__10X299__cells
4
CCS__fetal__healthy__10X300__cells
5
CCS__fetal__healthy__10X303__cells
14
CCS__fetal__healthy__11W1D__cells
432
CCS__fetal__healthy__11W4D__cells
133
CCS__fetal__healthy__13W3D__cells
289
CCS__fetal__healthy__15W0D__cells
202
CCS__fetal__healthy__15W2D__cells
118
CCS__fetal__healthy__BRC2251__cells
35
CCS__fetal__healthy__BRC2252__cells
54
CCS__fetal__healthy__BRC2256__cells
18
CCS__fetal__healthy__BRC2260__cells
4
CCS__fetal__healthy__BRC2262__cells
1
CCS__fetal__healthy__BRC2263__cells
15
CCS__fetal__healthy__F8W4D__nuclei
577
CCS__fetal__healthy__M10W5D__nuclei
472
CCS__fetal__healthy__M9W0D__nuclei
88
EC_Arterial__adult__cHF__C1__cells
28
EC_Arterial__adult__cHF__C2__cells
23
EC_Arterial__adult__dHF__D1__cells
26
EC_Arterial__adult__dHF__D2__cells
57
EC_Arterial__adult__dHF__D4__cells
8
EC_Arterial__adult__healthy__D2__nuclei
89
EC_Arterial__adult__healthy__D3__cells
42
EC_Arterial__adult__healthy__D3__nuclei
43
EC_Arterial__adult__healthy__D4__cells
48
EC_Arterial__adult__healthy__D4__nuclei
63
EC_Arterial__adult__healthy__D5__cells
54
EC_Arterial__adult__healthy__D5__nuclei
46
EC_Arterial__adult__healthy__D6__cells
90
EC_Arterial__adult__healthy__D6__nuclei
59
EC_Arterial__adult__healthy__D7__cells
42
EC_Arterial__adult__healthy__D7__nuclei
88
EC_Arterial__adult__healthy__N1__cells
19
EC_Arterial__adult__healthy__N10__cells
13
EC_Arterial__adult__healthy__N11__cells
9
EC_Arterial__adult__healthy__N12__cells
1
EC_Arterial__adult__healthy__N2__cells
11
EC_Arterial__adult__healthy__N4__cells
7
EC_Arterial__adult__healthy__N5__cells
19
EC_Arterial__adult__healthy__N6__cells
26
EC_Arterial__adult__healthy__N7__cells
1
EC_Arterial__adult__healthy__N8__cells
8
EC_Arterial__adult__healthy__N9__cells
19
EC_Arterial__adult__healthy__P1__nuclei
346
EC_Arterial__adult__healthy__P17__nuclei

149
EC_Arterial__adult__healthy__P7__nuclei
56
EC_Arterial__adult__healthy__P8__nuclei
242
EC_Arterial__adult__myocardial infarction_BZ__P12__nuclei
141
EC_Arterial__adult__myocardial infarction_BZ__P2__nuclei
476
EC_Arterial__adult__myocardial infarction_BZ__P3__nuclei
104
EC_Arterial__adult__myocardial infarction_FZ__P14__nuclei
336
EC_Arterial__adult__myocardial infarction_FZ__P18__nuclei
168
EC_Arterial__adult__myocardial infarction_FZ__P19__nuclei
140
EC_Arterial__adult__myocardial infarction_FZ__P20__nuclei
89
EC_Arterial__adult__myocardial infarction_FZ__P4__nuclei
260
EC_Arterial__adult__myocardial infarction_FZ__P5__nuclei
430
EC_Arterial__adult__myocardial infarction_IZ__P10__nuclei
15
EC_Arterial__adult__myocardial infarction_IZ__P13__nuclei
36
EC_Arterial__adult__myocardial infarction_IZ__P15__nuclei
34
EC_Arterial__adult__myocardial infarction_IZ__P16__nuclei
9
EC_Arterial__adult__myocardial infarction_IZ__P9__nuclei
59
EC_Arterial__adult__myocardial infarction_RZ__P11__nuclei
135
EC_Arterial__adult__myocardial infarction_RZ__P2__nuclei
582
EC_Arterial__adult__myocardial infarction_RZ__P3__nuclei
121
EC_Arterial__adult__myocardial infarction_RZ__P6__nuclei
268
EC_Arterial__adult__myocardial infarction_RZ__P9__nuclei
92
EC_Arterial__fetal__healthy__09W0D__cells
30
EC_Arterial__fetal__healthy__09W6D__cells
93
EC_Arterial__fetal__healthy__10X127__cells
12
EC_Arterial__fetal__healthy__10X142__cells
40
EC_Arterial__fetal__healthy__10X151__cells
14
EC_Arterial__fetal__healthy__10X158__cells
123
EC_Arterial__fetal__healthy__10X166__cells
32
EC_Arterial__fetal__healthy__10X246__cells
102
EC_Arterial__fetal__healthy__10X253__cells
292
EC_Arterial__fetal__healthy__10X256__cells
171
EC_Arterial__fetal__healthy__10X263__cells
138
EC_Arterial__fetal__healthy__10X274__cells
31
EC_Arterial__fetal__healthy__10X285__cells
52
EC_Arterial__fetal__healthy__10X289__cells
75
EC_Arterial__fetal__healthy__10X299__cells
15
EC_Arterial__fetal__healthy__10X300__cells
26
EC_Arterial__fetal__healthy__10X303__cells
7
EC_Arterial__fetal__healthy__11W1D__cells
110
EC_Arterial__fetal__healthy__11W4D__cells
72
EC_Arterial__fetal__healthy__13W3D__cells
137
EC_Arterial__fetal__healthy__15W0D__cells
77
EC_Arterial__fetal__healthy__15W2D__cells
108
EC_Arterial__fetal__healthy__alexsc__cells
16
EC_Arterial__fetal__healthy__BRC2251__cells
166
EC_Arterial__fetal__healthy__BRC2252__cells
64
EC_Arterial__fetal__healthy__BRC2256__cells
408
EC_Arterial__fetal__healthy__BRC2260__cells
186
EC_Arterial__fetal__healthy__BRC2262__cells
171
EC_Arterial__fetal__healthy__BRC2263__cells

199
EC_Capillary__adult__cHF__C1__cells
121
EC_Capillary__adult__cHF__C2__cells
55
EC_Capillary__adult__dHF__D1__cells
140
EC_Capillary__adult__dHF__D2__cells
133
EC_Capillary__adult__dHF__D4__cells
8
EC_Capillary__adult__dHF__D5__cells
7
EC_Capillary__adult__healthy__D2__nuclei
343
EC_Capillary__adult__healthy__D3__cells
188
EC_Capillary__adult__healthy__D3__nuclei
129
EC_Capillary__adult__healthy__D4__cells
188
EC_Capillary__adult__healthy__D4__nuclei
216
EC_Capillary__adult__healthy__D5__cells
225
EC_Capillary__adult__healthy__D5__nuclei
110
EC_Capillary__adult__healthy__D6__cells
324
EC_Capillary__adult__healthy__D6__nuclei
161
EC_Capillary__adult__healthy__D7__cells
304
EC_Capillary__adult__healthy__D7__nuclei
102
EC_Capillary__adult__healthy__N1__cells
68
EC_Capillary__adult__healthy__N10__cells
32
EC_Capillary__adult__healthy__N11__cells
51
EC_Capillary__adult__healthy__N2__cells
57
EC_Capillary__adult__healthy__N4__cells
36
EC_Capillary__adult__healthy__N5__cells
121
EC_Capillary__adult__healthy__N6__cells
79
EC_Capillary__adult__healthy__N7__cells
4
EC_Capillary__adult__healthy__N8__cells
5
EC_Capillary__adult__healthy__N9__cells
64
EC_Capillary__adult__healthy__P1__nuclei
550
EC_Capillary__adult__healthy__P17__nuclei
418
EC_Capillary__adult__healthy__P7__nuclei
113
EC_Capillary__adult__healthy__P8__nuclei
632
EC_Capillary__adult__myocardial infarction_BZ__P12__nuclei
287
EC_Capillary__adult__myocardial infarction_BZ__P2__nuclei
491
EC_Capillary__adult__myocardial infarction_BZ__P3__nuclei
166
EC_Capillary__adult__myocardial infarction_FZ__P14__nuclei
384
EC_Capillary__adult__myocardial infarction_FZ__P18__nuclei
244
EC_Capillary__adult__myocardial infarction_FZ__P19__nuclei
73
EC_Capillary__adult__myocardial infarction_FZ__P20__nuclei
53
EC_Capillary__adult__myocardial infarction_FZ__P4__nuclei
631
EC_Capillary__adult__myocardial infarction_FZ__P5__nuclei
612
EC_Capillary__adult__myocardial infarction_IZ__P10__nuclei
31
EC_Capillary__adult__myocardial infarction_IZ__P13__nuclei
10
EC_Capillary__adult__myocardial infarction_IZ__P15__nuclei
43
EC_Capillary__adult__myocardial infarction_IZ__P16__nuclei
1
EC_Capillary__adult__myocardial infarction_IZ__P2__nuclei
1
EC_Capillary__adult__myocardial infarction_IZ__P9__nuclei
48
EC_Capillary__adult__myocardial infarction_RZ__P11__nuclei
354
EC_Capillary__adult__myocardial infarction_RZ__P2__nuclei
1122
EC_Capillary__adult__myocardial infarction_RZ__P3__nuclei
139
EC_Capillary__adult__myocardial infarction_RZ__P6__nuclei

873
EC_Capillary__adult__myocardial infarction_RZ__P9__nuclei
81
EC_Capillary__fetal__healthy__09W0D__cells
223
EC_Capillary__fetal__healthy__09W6D__cells
327
EC_Capillary__fetal__healthy__10X127__cells
80
EC_Capillary__fetal__healthy__10X142__cells
72
EC_Capillary__fetal__healthy__10X151__cells
89
EC_Capillary__fetal__healthy__10X158__cells
426
EC_Capillary__fetal__healthy__10X166__cells
35
EC_Capillary__fetal__healthy__10X246__cells
592
EC_Capillary__fetal__healthy__10X253__cells
1436
EC_Capillary__fetal__healthy__10X256__cells
833
EC_Capillary__fetal__healthy__10X263__cells
538
EC_Capillary__fetal__healthy__10X274__cells
193
EC_Capillary__fetal__healthy__10X285__cells
120
EC_Capillary__fetal__healthy__10X289__cells
52
EC_Capillary__fetal__healthy__10X299__cells
24
EC_Capillary__fetal__healthy__10X300__cells
33
EC_Capillary__fetal__healthy__10X303__cells
26
EC_Capillary__fetal__healthy__11W1D__cells
448
EC_Capillary__fetal__healthy__11W4D__cells
338
EC_Capillary__fetal__healthy__13W3D__cells
796
EC_Capillary__fetal__healthy__15W0D__cells
558
EC_Capillary__fetal__healthy__15W2D__cells
599
EC_Capillary__fetal__healthy__alexsc__cells
64
EC_Capillary__fetal__healthy__BRC2251__cells
219
EC_Capillary__fetal__healthy__BRC2252__cells
94
EC_Capillary__fetal__healthy__BRC2256__cells
197
EC_Capillary__fetal__healthy__BRC2260__cells
265
EC_Capillary__fetal__healthy__BRC2262__cells
160
EC_Capillary__fetal__healthy__BRC2263__cells
227
EC_Capillary__fetal__healthy__F8W4D__nuclei
780
EC_Capillary__fetal__healthy__M10W5D__nuclei
519
EC_Capillary__fetal__healthy__M9W0D__nuclei
227
EC_Lymphatic__adult__cHF__C1__cells
3
EC_Lymphatic__adult__cHF__C2__cells
3
EC_Lymphatic__adult__dHF__D1__cells
2
EC_Lymphatic__adult__dHF__D2__cells
4
EC_Lymphatic__adult__healthy__D2__nuclei
128
EC_Lymphatic__adult__healthy__D3__cells
1
EC_Lymphatic__adult__healthy__D3__nuclei
11
EC_Lymphatic__adult__healthy__D4__cells
2
EC_Lymphatic__adult__healthy__D4__nuclei
55
EC_Lymphatic__adult__healthy__D5__cells
3
EC_Lymphatic__adult__healthy__D5__nuclei
29
EC_Lymphatic__adult__healthy__D6__cells
121
EC_Lymphatic__adult__healthy__D6__nuclei
21
EC_Lymphatic__adult__healthy__D7__cells
10
EC_Lymphatic__adult__healthy__D7__nuclei
25
EC_Lymphatic__adult__healthy__N1__cells
4
EC_Lymphatic__adult__healthy__N10__cells

1
EC_Lymphatic__adult__healthy__N11__cells
13
EC_Lymphatic__adult__healthy__N6__cells
4
EC_Lymphatic__adult__healthy__N9__cells
2
EC_Lymphatic__adult__healthy__P1__nuclei
7
EC_Lymphatic__adult__healthy__P17__nuclei
6
EC_Lymphatic__adult__healthy__P7__nuclei
2
EC_Lymphatic__adult__healthy__P8__nuclei
17
EC_Lymphatic__adult__myocardial infarction_BZ__P12__nuclei
1
EC_Lymphatic__adult__myocardial infarction_BZ__P2__nuclei
3
EC_Lymphatic__adult__myocardial infarction_BZ__P3__nuclei
40
EC_Lymphatic__adult__myocardial infarction_FZ__P14__nuclei
17
EC_Lymphatic__adult__myocardial infarction_FZ__P18__nuclei
86
EC_Lymphatic__adult__myocardial infarction_FZ__P19__nuclei
25
EC_Lymphatic__adult__myocardial infarction_FZ__P20__nuclei
68
EC_Lymphatic__adult__myocardial infarction_FZ__P4__nuclei
15
EC_Lymphatic__adult__myocardial infarction_FZ__P5__nuclei
10
EC_Lymphatic__adult__myocardial infarction_IZ__P10__nuclei
16
EC_Lymphatic__adult__myocardial infarction_IZ__P13__nuclei
11
EC_Lymphatic__adult__myocardial infarction_IZ__P15__nuclei
223
EC_Lymphatic__adult__myocardial infarction_IZ__P16__nuclei
33
EC_Lymphatic__adult__myocardial infarction_IZ__P2__nuclei
4
EC_Lymphatic__adult__myocardial infarction_IZ__P3__nuclei
4
EC_Lymphatic__adult__myocardial infarction_IZ__P9__nuclei
57
EC_Lymphatic__adult__myocardial infarction_RZ__P11__nuclei
10
EC_Lymphatic__adult__myocardial infarction_RZ__P3__nuclei
151
EC_Lymphatic__adult__myocardial infarction_RZ__P6__nuclei
2
EC_Lymphatic__adult__myocardial infarction_RZ__P9__nuclei
33
EC_Lymphatic__fetal__healthy__09W0D__cells
43
EC_Lymphatic__fetal__healthy__09W6D__cells
15
EC_Lymphatic__fetal__healthy__10X127__cells
6
EC_Lymphatic__fetal__healthy__10X142__cells
17
EC_Lymphatic__fetal__healthy__10X151__cells
5
EC_Lymphatic__fetal__healthy__10X158__cells
24
EC_Lymphatic__fetal__healthy__10X166__cells
11
EC_Lymphatic__fetal__healthy__10X246__cells
34
EC_Lymphatic__fetal__healthy__10X253__cells
23
EC_Lymphatic__fetal__healthy__10X256__cells
10
EC_Lymphatic__fetal__healthy__10X263__cells
84
EC_Lymphatic__fetal__healthy__10X274__cells
24
EC_Lymphatic__fetal__healthy__10X285__cells
38
EC_Lymphatic__fetal__healthy__10X289__cells
4
EC_Lymphatic__fetal__healthy__10X299__cells
9
EC_Lymphatic__fetal__healthy__10X300__cells
8
EC_Lymphatic__fetal__healthy__10X303__cells
5
EC_Lymphatic__fetal__healthy__11W1D__cells
66
EC_Lymphatic__fetal__healthy__11W4D__cells
15
EC_Lymphatic__fetal__healthy__13W3D__cells
134
EC_Lymphatic__fetal__healthy__15W0D__cells
92
EC_Lymphatic__fetal__healthy__15W2D__cells
119
EC_Lymphatic__fetal__healthy__alexsc__cells

1
EC_Lymphatic__fetal__healthy__BRC2251__cells
59
EC_Lymphatic__fetal__healthy__BRC2252__cells
7
EC_Lymphatic__fetal__healthy__BRC2256__cells
29
EC_Lymphatic__fetal__healthy__BRC2260__cells
113
EC_Lymphatic__fetal__healthy__BRC2262__cells
23
EC_Lymphatic__fetal__healthy__BRC2263__cells
15
EC_Mixed__adult__cHF__C1__cells
100
EC_Mixed__adult__cHF__C2__cells
27
EC_Mixed__adult__dHF__D1__cells
136
EC_Mixed__adult__dHF__D2__cells
119
EC_Mixed__adult__dHF__D4__cells
147
EC_Mixed__adult__dHF__D5__cells
2
EC_Mixed__adult__healthy__D2__nuclei
18
EC_Mixed__adult__healthy__D3__cells
1
EC_Mixed__adult__healthy__D3__nuclei
67
EC_Mixed__adult__healthy__D4__cells
7
EC_Mixed__adult__healthy__D4__nuclei
60
EC_Mixed__adult__healthy__D5__cells
1
EC_Mixed__adult__healthy__D5__nuclei
16
EC_Mixed__adult__healthy__D6__cells
11
EC_Mixed__adult__healthy__D6__nuclei
26
EC_Mixed__adult__healthy__D7__cells
2
EC_Mixed__adult__healthy__D7__nuclei
79
EC_Mixed__adult__healthy__N1__cells
63
EC_Mixed__adult__healthy__N10__cells
81
EC_Mixed__adult__healthy__N11__cells
46
EC_Mixed__adult__healthy__N12__cells
3
EC_Mixed__adult__healthy__N13__cells
10
EC_Mixed__adult__healthy__N14__cells
2
EC_Mixed__adult__healthy__N2__cells
100
EC_Mixed__adult__healthy__N3__cells
17
EC_Mixed__adult__healthy__N4__cells
16
EC_Mixed__adult__healthy__N5__cells
39
EC_Mixed__adult__healthy__N6__cells
112
EC_Mixed__adult__healthy__N7__cells
4
EC_Mixed__adult__healthy__N8__cells
31
EC_Mixed__adult__healthy__N9__cells
55
EC_Mixed__adult__healthy__P1__nuclei
275
EC_Mixed__adult__healthy__P17__nuclei
198
EC_Mixed__adult__healthy__P7__nuclei
781
EC_Mixed__adult__healthy__P8__nuclei
725
EC_Mixed__adult__myocardial infarction_BZ__P12__nuclei
426
EC_Mixed__adult__myocardial infarction_BZ__P2__nuclei
622
EC_Mixed__adult__myocardial infarction_BZ__P3__nuclei
214
EC_Mixed__adult__myocardial infarction_FZ__P14__nuclei
395
EC_Mixed__adult__myocardial infarction_FZ__P18__nuclei
142
EC_Mixed__adult__myocardial infarction_FZ__P19__nuclei
169
EC_Mixed__adult__myocardial infarction_FZ__P20__nuclei
138
EC_Mixed__adult__myocardial infarction_FZ__P4__nuclei
338
EC_Mixed__adult__myocardial infarction_FZ__P5__nuclei

896
EC_Mixed__adult__myocardial infarction_IZ__P10__nuclei
189
EC_Mixed__adult__myocardial infarction_IZ__P13__nuclei
351
EC_Mixed__adult__myocardial infarction_IZ__P15__nuclei
384
EC_Mixed__adult__myocardial infarction_IZ__P16__nuclei
630
EC_Mixed__adult__myocardial infarction_IZ__P2__nuclei
10
EC_Mixed__adult__myocardial infarction_IZ__P3__nuclei
2
EC_Mixed__adult__myocardial infarction_IZ__P9__nuclei
1213
EC_Mixed__adult__myocardial infarction_RZ__P11__nuclei
619
EC_Mixed__adult__myocardial infarction_RZ__P2__nuclei
183
EC_Mixed__adult__myocardial infarction_RZ__P3__nuclei
185
EC_Mixed__adult__myocardial infarction_RZ__P6__nuclei
385
EC_Mixed__adult__myocardial infarction_RZ__P9__nuclei
1405
EC_Mixed__fetal__healthy__09W0D__cells
18
EC_Mixed__fetal__healthy__09W6D__cells
16
EC_Mixed__fetal__healthy__10X127__cells
25
EC_Mixed__fetal__healthy__10X142__cells
6
EC_Mixed__fetal__healthy__10X151__cells
22
EC_Mixed__fetal__healthy__10X158__cells
453
EC_Mixed__fetal__healthy__10X166__cells
32
EC_Mixed__fetal__healthy__10X246__cells
87
EC_Mixed__fetal__healthy__10X253__cells
153
EC_Mixed__fetal__healthy__10X256__cells
63
EC_Mixed__fetal__healthy__10X263__cells
19
EC_Mixed__fetal__healthy__10X274__cells
29
EC_Mixed__fetal__healthy__10X285__cells
86
EC_Mixed__fetal__healthy__10X289__cells
38
EC_Mixed__fetal__healthy__10X299__cells
16
EC_Mixed__fetal__healthy__10X300__cells
11
EC_Mixed__fetal__healthy__10X303__cells
15
EC_Mixed__fetal__healthy__11W1D__cells
30
EC_Mixed__fetal__healthy__11W4D__cells
12
EC_Mixed__fetal__healthy__13W3D__cells
61
EC_Mixed__fetal__healthy__15W0D__cells
29
EC_Mixed__fetal__healthy__15W2D__cells
31
EC_Mixed__fetal__healthy__alexsc__cells
6
EC_Mixed__fetal__healthy__BRC2251__cells
14
EC_Mixed__fetal__healthy__BRC2252__cells
15
EC_Mixed__fetal__healthy__BRC2256__cells
8
EC_Mixed__fetal__healthy__BRC2260__cells
114
EC_Mixed__fetal__healthy__BRC2262__cells
14
EC_Mixed__fetal__healthy__BRC2263__cells
12
EC_Valves_Pericytes__adult__cHF__C1__cells
35
EC_Valves_Pericytes__adult__cHF__C2__cells
24
EC_Valves_Pericytes__adult__dHF__D1__cells
18
EC_Valves_Pericytes__adult__dHF__D2__cells
116
EC_Valves_Pericytes__adult__dHF__D4__cells
14
EC_Valves_Pericytes__adult__dHF__D5__cells
3
EC_Valves_Pericytes__adult__healthy__D2__nuclei
40
EC_Valves_Pericytes__adult__healthy__D3__cells
20
EC_Valves_Pericytes__adult__healthy__D3__nuclei

34
EC_Valves_Pericytes__adult__healthy__D4__cells
14
EC_Valves_Pericytes__adult__healthy__D4__nuclei
15
EC_Valves_Pericytes__adult__healthy__D5__cells
29
EC_Valves_Pericytes__adult__healthy__D5__nuclei
19
EC_Valves_Pericytes__adult__healthy__D6__cells
22
EC_Valves_Pericytes__adult__healthy__D6__nuclei
10
EC_Valves_Pericytes__adult__healthy__D7__cells
12
EC_Valves_Pericytes__adult__healthy__D7__nuclei
18
EC_Valves_Pericytes__adult__healthy__N1__cells
15
EC_Valves_Pericytes__adult__healthy__N10__cells
36
EC_Valves_Pericytes__adult__healthy__N11__cells
18
EC_Valves_Pericytes__adult__healthy__N12__cells
2
EC_Valves_Pericytes__adult__healthy__N13__cells
3
EC_Valves_Pericytes__adult__healthy__N14__cells
1
EC_Valves_Pericytes__adult__healthy__N2__cells
4
EC_Valves_Pericytes__adult__healthy__N4__cells
9
EC_Valves_Pericytes__adult__healthy__N5__cells
10
EC_Valves_Pericytes__adult__healthy__N6__cells
119
EC_Valves_Pericytes__adult__healthy__N7__cells
1
EC_Valves_Pericytes__adult__healthy__N8__cells
6
EC_Valves_Pericytes__adult__healthy__N9__cells
27
EC_Valves_Pericytes__adult__healthy__P1__nuclei
518
EC_Valves_Pericytes__adult__healthy__P17__nuclei
239
EC_Valves_Pericytes__adult__healthy__P7__nuclei
10
EC_Valves_Pericytes__adult__healthy__P8__nuclei
24
EC_Valves_Pericytes__adult__myocardial infarction_BZ__P12__nuclei
25
EC_Valves_Pericytes__adult__myocardial infarction_BZ__P2__nuclei
3
EC_Valves_Pericytes__adult__myocardial infarction_BZ__P3__nuclei
69
EC_Valves_Pericytes__adult__myocardial infarction_FZ__P14__nuclei
40
EC_Valves_Pericytes__adult__myocardial infarction_FZ__P18__nuclei
72
EC_Valves_Pericytes__adult__myocardial infarction_FZ__P19__nuclei
109
EC_Valves_Pericytes__adult__myocardial infarction_FZ__P20__nuclei
108
EC_Valves_Pericytes__adult__myocardial infarction_FZ__P4__nuclei
8
EC_Valves_Pericytes__adult__myocardial infarction_FZ__P5__nuclei
257
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P10__nuclei
453
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P13__nuclei
62
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P15__nuclei
549
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P16__nuclei
81
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P2__nuclei
282
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P3__nuclei
43
EC_Valves_Pericytes__adult__myocardial infarction_IZ__P9__nuclei
677
EC_Valves_Pericytes__adult__myocardial infarction_RZ__P11__nuclei
3
EC_Valves_Pericytes__adult__myocardial infarction_RZ__P2__nuclei
11
EC_Valves_Pericytes__adult__myocardial infarction_RZ__P3__nuclei
393
EC_Valves_Pericytes__adult__myocardial infarction_RZ__P6__nuclei
68
EC_Valves_Pericytes__adult__myocardial infarction_RZ__P9__nuclei
182
EC_Valves_Pericytes__fetal__healthy__09W0D__cells
12
EC_Valves_Pericytes__fetal__healthy__09W6D__cells
12
EC_Valves_Pericytes__fetal__healthy__10X127__cells
18
EC_Valves_Pericytes__fetal__healthy__10X142__cells

54
EC_Valves_Pericytes__fetal__healthy__10X151__cells
14
EC_Valves_Pericytes__fetal__healthy__10X158__cells
66
EC_Valves_Pericytes__fetal__healthy__10X166__cells
105
EC_Valves_Pericytes__fetal__healthy__10X246__cells
93
EC_Valves_Pericytes__fetal__healthy__10X253__cells
103
EC_Valves_Pericytes__fetal__healthy__10X256__cells
52
EC_Valves_Pericytes__fetal__healthy__10X263__cells
60
EC_Valves_Pericytes__fetal__healthy__10X274__cells
70
EC_Valves_Pericytes__fetal__healthy__10X285__cells
136
EC_Valves_Pericytes__fetal__healthy__10X289__cells
23
EC_Valves_Pericytes__fetal__healthy__10X299__cells
4
EC_Valves_Pericytes__fetal__healthy__10X303__cells
16
EC_Valves_Pericytes__fetal__healthy__11W1D__cells
30
EC_Valves_Pericytes__fetal__healthy__11W4D__cells
36
EC_Valves_Pericytes__fetal__healthy__13W3D__cells
76
EC_Valves_Pericytes__fetal__healthy__15W0D__cells
73
EC_Valves_Pericytes__fetal__healthy__15W2D__cells
88
EC_Valves_Pericytes__fetal__healthy__alexsc__cells
1
EC_Valves_Pericytes__fetal__healthy__BRC2251__cells
24
EC_Valves_Pericytes__fetal__healthy__BRC2252__cells
8
EC_Valves_Pericytes__fetal__healthy__BRC2256__cells
20
EC_Valves_Pericytes__fetal__healthy__BRC2260__cells
43
EC_Valves_Pericytes__fetal__healthy__BRC2262__cells
8
EC_Valves_Pericytes__fetal__healthy__BRC2263__cells
17
EC_Valves_Pericytes__fetal__healthy__F8W4D__nuclei
391
EC_Valves_Pericytes__fetal__healthy__M10W5D__nuclei
269
EC_Valves_Pericytes__fetal__healthy__M9W0D__nuclei
66
EC_Venous__adult__cHF__C1__cells
29
EC_Venous__adult__cHF__C2__cells
19
EC_Venous__adult__dHF__D1__cells
22
EC_Venous__adult__dHF__D2__cells
80
EC_Venous__adult__dHF__D4__cells
12
EC_Venous__adult__healthy__D2__nuclei
101
EC_Venous__adult__healthy__D3__cells
48
EC_Venous__adult__healthy__D3__nuclei
59
EC_Venous__adult__healthy__D4__cells
40
EC_Venous__adult__healthy__D4__nuclei
51
EC_Venous__adult__healthy__D5__cells
62
EC_Venous__adult__healthy__D5__nuclei
47
EC_Venous__adult__healthy__D6__cells
57
EC_Venous__adult__healthy__D6__nuclei
51
EC_Venous__adult__healthy__D7__cells
49
EC_Venous__adult__healthy__D7__nuclei
66
EC_Venous__adult__healthy__N1__cells
58
EC_Venous__adult__healthy__N10__cells
30
EC_Venous__adult__healthy__N11__cells
22
EC_Venous__adult__healthy__N12__cells
3
EC_Venous__adult__healthy__N2__cells
46
EC_Venous__adult__healthy__N4__cells
12
EC_Venous__adult__healthy__N5__cells

13
EC_Venous__adult__healthy__N6__cells
52
EC_Venous__adult__healthy__N7__cells
2
EC_Venous__adult__healthy__N8__cells
6
EC_Venous__adult__healthy__N9__cells
24
EC_Venous__adult__healthy__P1__nuclei
223
EC_Venous__adult__healthy__P17__nuclei
131
EC_Venous__adult__healthy__P7__nuclei
196
EC_Venous__adult__healthy__P8__nuclei
300
EC_Venous__adult__myocardial infarction_BZ__P12__nuclei
122
EC_Venous__adult__myocardial infarction_BZ__P2__nuclei
250
EC_Venous__adult__myocardial infarction_BZ__P3__nuclei
157
EC_Venous__adult__myocardial infarction_FZ__P14__nuclei
316
EC_Venous__adult__myocardial infarction_FZ__P18__nuclei
504
EC_Venous__adult__myocardial infarction_FZ__P19__nuclei
167
EC_Venous__adult__myocardial infarction_FZ__P20__nuclei
86
EC_Venous__adult__myocardial infarction_FZ__P4__nuclei
142
EC_Venous__adult__myocardial infarction_FZ__P5__nuclei
221
EC_Venous__adult__myocardial infarction_IZ__P10__nuclei
35
EC_Venous__adult__myocardial infarction_IZ__P13__nuclei
83
EC_Venous__adult__myocardial infarction_IZ__P15__nuclei
167
EC_Venous__adult__myocardial infarction_IZ__P16__nuclei
29
EC_Venous__adult__myocardial infarction_IZ__P2__nuclei
1
EC_Venous__adult__myocardial infarction_IZ__P9__nuclei
126
EC_Venous__adult__myocardial infarction_RZ__P11__nuclei
248
EC_Venous__adult__myocardial infarction_RZ__P2__nuclei
191
EC_Venous__adult__myocardial infarction_RZ__P3__nuclei
34
EC_Venous__adult__myocardial infarction_RZ__P6__nuclei
314
EC_Venous__adult__myocardial infarction_RZ__P9__nuclei
380
EC_Venous__fetal__healthy__09W0D__cells
69
EC_Venous__fetal__healthy__09W6D__cells
81
EC_Venous__fetal__healthy__10X127__cells
8
EC_Venous__fetal__healthy__10X142__cells
13
EC_Venous__fetal__healthy__10X151__cells
17
EC_Venous__fetal__healthy__10X158__cells
63
EC_Venous__fetal__healthy__10X166__cells
5
EC_Venous__fetal__healthy__10X246__cells
58
EC_Venous__fetal__healthy__10X253__cells
67
EC_Venous__fetal__healthy__10X256__cells
32
EC_Venous__fetal__healthy__10X263__cells
66
EC_Venous__fetal__healthy__10X274__cells
25
EC_Venous__fetal__healthy__10X285__cells
27
EC_Venous__fetal__healthy__10X289__cells
9
EC_Venous__fetal__healthy__10X299__cells
8
EC_Venous__fetal__healthy__10X300__cells
4
EC_Venous__fetal__healthy__10X303__cells
4
EC_Venous__fetal__healthy__11W1D__cells
60
EC_Venous__fetal__healthy__11W4D__cells
39
EC_Venous__fetal__healthy__13W3D__cells
75
EC_Venous__fetal__healthy__15W0D__cells
86
EC_Venous__fetal__healthy__15W2D__cells

146
EC_Venous__fetal__healthy__alexsc__cells
11
EC_Venous__fetal__healthy__BRC2251__cells
86
EC_Venous__fetal__healthy__BRC2252__cells
33
EC_Venous__fetal__healthy__BRC2256__cells
51
EC_Venous__fetal__healthy__BRC2260__cells
80
EC_Venous__fetal__healthy__BRC2262__cells
32
EC_Venous__fetal__healthy__BRC2263__cells
35
Endocardium__adult__healthy__D2__nuclei
118
Endocardium__adult__healthy__D3__cells
50
Endocardium__adult__healthy__D3__nuclei
84
Endocardium__adult__healthy__D4__cells
9
Endocardium__adult__healthy__D4__nuclei
93
Endocardium__adult__healthy__D5__cells
78
Endocardium__adult__healthy__D5__nuclei
48
Endocardium__adult__healthy__D6__cells
61
Endocardium__adult__healthy__D6__nuclei
67
Endocardium__adult__healthy__D7__cells
19
Endocardium__adult__healthy__D7__nuclei
103
Endocardium__fetal__healthy__09W0D__cells
1627
Endocardium__fetal__healthy__09W6D__cells
760
Endocardium__fetal__healthy__10X127__cells
326
Endocardium__fetal__healthy__10X142__cells
549
Endocardium__fetal__healthy__10X151__cells
1108
Endocardium__fetal__healthy__10X158__cells
1571
Endocardium__fetal__healthy__10X166__cells
448
Endocardium__fetal__healthy__10X246__cells
101
Endocardium__fetal__healthy__10X253__cells
319
Endocardium__fetal__healthy__10X256__cells
103
Endocardium__fetal__healthy__10X263__cells
201
Endocardium__fetal__healthy__10X274__cells
60
Endocardium__fetal__healthy__10X285__cells
96
Endocardium__fetal__healthy__10X289__cells
522
Endocardium__fetal__healthy__10X299__cells
276
Endocardium__fetal__healthy__10X300__cells
262
Endocardium__fetal__healthy__10X303__cells
634
Endocardium__fetal__healthy__11W1D__cells
981
Endocardium__fetal__healthy__11W4D__cells
856
Endocardium__fetal__healthy__13W3D__cells
1275
Endocardium__fetal__healthy__15W0D__cells
400
Endocardium__fetal__healthy__15W2D__cells
459
Endocardium__fetal__healthy__alexsc__cells
116
Endocardium__fetal__healthy__BRC2251__cells
290
Endocardium__fetal__healthy__BRC2252__cells
233
Endocardium__fetal__healthy__BRC2256__cells
240
Endocardium__fetal__healthy__BRC2260__cells
145
Endocardium__fetal__healthy__BRC2262__cells
339
Endocardium__fetal__healthy__BRC2263__cells
211
Endocardium__fetal__healthy__F8W4D__nuclei
1738
Endocardium__fetal__healthy__M10W5D__nuclei
2381
Endocardium__fetal__healthy__M9W0D__nuclei

884
Epicardium__adult__healthy__D2__nuclei
81
Epicardium__adult__healthy__D3__cells
5
Epicardium__adult__healthy__D3__nuclei
95
Epicardium__adult__healthy__D4__cells
33
Epicardium__adult__healthy__D4__nuclei
38
Epicardium__adult__healthy__D5__cells
31
Epicardium__adult__healthy__D5__nuclei
43
Epicardium__adult__healthy__D6__cells
8
Epicardium__adult__healthy__D6__nuclei
244
Epicardium__adult__healthy__D7__cells
7
Epicardium__adult__healthy__D7__nuclei
86
Epicardium__fetal__healthy__09W0D__cells
177
Epicardium__fetal__healthy__09W6D__cells
163
Epicardium__fetal__healthy__10X127__cells
44
Epicardium__fetal__healthy__10X142__cells
127
Epicardium__fetal__healthy__10X151__cells
66
Epicardium__fetal__healthy__10X158__cells
271
Epicardium__fetal__healthy__10X166__cells
48
Epicardium__fetal__healthy__10X246__cells
9
Epicardium__fetal__healthy__10X253__cells
36
Epicardium__fetal__healthy__10X256__cells
8
Epicardium__fetal__healthy__10X263__cells
17
Epicardium__fetal__healthy__10X274__cells
6
Epicardium__fetal__healthy__10X285__cells
16
Epicardium__fetal__healthy__10X289__cells
122
Epicardium__fetal__healthy__10X299__cells
52
Epicardium__fetal__healthy__10X300__cells
58
Epicardium__fetal__healthy__10X303__cells
34
Epicardium__fetal__healthy__11W1D__cells
181
Epicardium__fetal__healthy__11W4D__cells
53
Epicardium__fetal__healthy__13W3D__cells
176
Epicardium__fetal__healthy__15W0D__cells
69
Epicardium__fetal__healthy__15W2D__cells
60
Epicardium__fetal__healthy__alexsc__cells
27
Epicardium__fetal__healthy__BRC2251__cells
212
Epicardium__fetal__healthy__BRC2252__cells
55
Epicardium__fetal__healthy__BRC2256__cells
142
Epicardium__fetal__healthy__BRC2260__cells
483
Epicardium__fetal__healthy__BRC2262__cells
28
Epicardium__fetal__healthy__BRC2263__cells
19
Epicardium__fetal__healthy__F8W4D__nuclei
413
Epicardium__fetal__healthy__M10W5D__nuclei
548
Epicardium__fetal__healthy__M9W0D__nuclei
113
Fibroblasts__adult__cHF__C1__cells
56
Fibroblasts__adult__cHF__C2__cells
17
Fibroblasts__adult__dHF__D1__cells
73
Fibroblasts__adult__dHF__D2__cells
112
Fibroblasts__adult__dHF__D4__cells
54
Fibroblasts__adult__dHF__D5__cells
3
Fibroblasts__adult__healthy__D2__nuclei

772
Fibroblasts__adult__healthy__D3__cells
77
Fibroblasts__adult__healthy__D3__nuclei
526
Fibroblasts__adult__healthy__D4__cells
27
Fibroblasts__adult__healthy__D4__nuclei
660
Fibroblasts__adult__healthy__D5__cells
355
Fibroblasts__adult__healthy__D5__nuclei
380
Fibroblasts__adult__healthy__D6__cells
342
Fibroblasts__adult__healthy__D6__nuclei
643
Fibroblasts__adult__healthy__D7__cells
180
Fibroblasts__adult__healthy__D7__nuclei
499
Fibroblasts__adult__healthy__N1__cells
26
Fibroblasts__adult__healthy__N10__cells
41
Fibroblasts__adult__healthy__N11__cells
43
Fibroblasts__adult__healthy__N12__cells
23
Fibroblasts__adult__healthy__N13__cells
148
Fibroblasts__adult__healthy__N14__cells
15
Fibroblasts__adult__healthy__N2__cells
26
Fibroblasts__adult__healthy__N3__cells
35
Fibroblasts__adult__healthy__N4__cells
7
Fibroblasts__adult__healthy__N5__cells
55
Fibroblasts__adult__healthy__N6__cells
76
Fibroblasts__adult__healthy__N7__cells
16
Fibroblasts__adult__healthy__N8__cells
19
Fibroblasts__adult__healthy__N9__cells
63
Fibroblasts__adult__healthy__P1__nuclei
3017
Fibroblasts__adult__healthy__P17__nuclei
1954
Fibroblasts__adult__healthy__P7__nuclei
1370
Fibroblasts__adult__healthy__P8__nuclei
3103
Fibroblasts__adult__myocardial infarction_BZ__P12__nuclei
1608
Fibroblasts__adult__myocardial infarction_BZ__P2__nuclei
2132
Fibroblasts__adult__myocardial infarction_BZ__P3__nuclei
2665
Fibroblasts__adult__myocardial infarction_FZ__P14__nuclei
1554
Fibroblasts__adult__myocardial infarction_FZ__P18__nuclei
2121
Fibroblasts__adult__myocardial infarction_FZ__P19__nuclei
1394
Fibroblasts__adult__myocardial infarction_FZ__P20__nuclei
1155
Fibroblasts__adult__myocardial infarction_FZ__P4__nuclei
658
Fibroblasts__adult__myocardial infarction_FZ__P5__nuclei
2941
Fibroblasts__adult__myocardial infarction_IZ__P10__nuclei
1771
Fibroblasts__adult__myocardial infarction_IZ__P13__nuclei
1108
Fibroblasts__adult__myocardial infarction_IZ__P15__nuclei
3481
Fibroblasts__adult__myocardial infarction_IZ__P16__nuclei
665
Fibroblasts__adult__myocardial infarction_IZ__P2__nuclei
188
Fibroblasts__adult__myocardial infarction_IZ__P3__nuclei
23
Fibroblasts__adult__myocardial infarction_IZ__P9__nuclei
2781
Fibroblasts__adult__myocardial infarction_RZ__P11__nuclei
1147
Fibroblasts__adult__myocardial infarction_RZ__P2__nuclei
1277
Fibroblasts__adult__myocardial infarction_RZ__P3__nuclei
4305
Fibroblasts__adult__myocardial infarction_RZ__P6__nuclei
2528
Fibroblasts__adult__myocardial infarction_RZ__P9__nuclei
1955
Fibroblasts__fetal__healthy__09W0D__cells

5981
Fibroblasts__fetal__healthy__09W6D__cells
4924
Fibroblasts__fetal__healthy__10X127__cells
1322
Fibroblasts__fetal__healthy__10X142__cells
798
Fibroblasts__fetal__healthy__10X151__cells
1126
Fibroblasts__fetal__healthy__10X158__cells
4909
Fibroblasts__fetal__healthy__10X166__cells
1030
Fibroblasts__fetal__healthy__10X246__cells
1087
Fibroblasts__fetal__healthy__10X253__cells
1274
Fibroblasts__fetal__healthy__10X256__cells
608
Fibroblasts__fetal__healthy__10X263__cells
1438
Fibroblasts__fetal__healthy__10X274__cells
589
Fibroblasts__fetal__healthy__10X285__cells
685
Fibroblasts__fetal__healthy__10X289__cells
1600
Fibroblasts__fetal__healthy__10X299__cells
586
Fibroblasts__fetal__healthy__10X300__cells
498
Fibroblasts__fetal__healthy__10X303__cells
1386
Fibroblasts__fetal__healthy__11W1D__cells
4592
Fibroblasts__fetal__healthy__11W4D__cells
11506
Fibroblasts__fetal__healthy__13W3D__cells
15973
Fibroblasts__fetal__healthy__15W0D__cells
7515
Fibroblasts__fetal__healthy__15W2D__cells
5332
Fibroblasts__fetal__healthy__alexsc__cells
371
Fibroblasts__fetal__healthy__BRC2251__cells
746
Fibroblasts__fetal__healthy__BRC2252__cells
885
Fibroblasts__fetal__healthy__BRC2256__cells
467
Fibroblasts__fetal__healthy__BRC2260__cells
409
Fibroblasts__fetal__healthy__BRC2262__cells
521
Fibroblasts__fetal__healthy__BRC2263__cells
524
Fibroblasts__fetal__healthy__F8W4D__nuclei
3368
Fibroblasts__fetal__healthy__M10W5D__nuclei
2228
Fibroblasts__fetal__healthy__M9W0D__nuclei
959
Leukocytes__adult__cHF__C1__cells
31
Leukocytes__adult__cHF__C2__cells
15
Leukocytes__adult__dHF__D1__cells
11
Leukocytes__adult__dHF__D2__cells
86
Leukocytes__adult__dHF__D4__cells
11
Leukocytes__adult__dHF__D5__cells
1
Leukocytes__adult__healthy__D2__nuclei
1145
Leukocytes__adult__healthy__D3__cells
483
Leukocytes__adult__healthy__D3__nuclei
712
Leukocytes__adult__healthy__D4__cells
96
Leukocytes__adult__healthy__D4__nuclei
676
Leukocytes__adult__healthy__D5__cells
309
Leukocytes__adult__healthy__D5__nuclei
501
Leukocytes__adult__healthy__D6__cells
448
Leukocytes__adult__healthy__D6__nuclei
650
Leukocytes__adult__healthy__D7__cells
397
Leukocytes__adult__healthy__D7__nuclei
590
Leukocytes__adult__healthy__N1__cells
25
Leukocytes__adult__healthy__N10__cells

12
Leukocytes__adult__healthy__N11__cells
11
Leukocytes__adult__healthy__N12__cells
6
Leukocytes__adult__healthy__N13__cells
8
Leukocytes__adult__healthy__N14__cells
4
Leukocytes__adult__healthy__N2__cells
14
Leukocytes__adult__healthy__N3__cells
2
Leukocytes__adult__healthy__N4__cells
3
Leukocytes__adult__healthy__N5__cells
8
Leukocytes__adult__healthy__N6__cells
38
Leukocytes__adult__healthy__N7__cells
3
Leukocytes__adult__healthy__N8__cells
13
Leukocytes__adult__healthy__N9__cells
14
Leukocytes__adult__healthy__P1__nuclei
1509
Leukocytes__adult__healthy__P17__nuclei
747
Leukocytes__adult__healthy__P7__nuclei
696
Leukocytes__adult__healthy__P8__nuclei
875
Leukocytes__adult__myocardial infarction_BZ__P12__nuclei
1225
Leukocytes__adult__myocardial infarction_BZ__P2__nuclei
748
Leukocytes__adult__myocardial infarction_BZ__P3__nuclei
1480
Leukocytes__adult__myocardial infarction_FZ__P14__nuclei
816
Leukocytes__adult__myocardial infarction_FZ__P18__nuclei
930
Leukocytes__adult__myocardial infarction_FZ__P19__nuclei
3317
Leukocytes__adult__myocardial infarction_FZ__P20__nuclei
1104
Leukocytes__adult__myocardial infarction_FZ__P4__nuclei
112
Leukocytes__adult__myocardial infarction_FZ__P5__nuclei
587
Leukocytes__adult__myocardial infarction_IZ__P10__nuclei
1573
Leukocytes__adult__myocardial infarction_IZ__P13__nuclei
1454
Leukocytes__adult__myocardial infarction_IZ__P15__nuclei
2530
Leukocytes__adult__myocardial infarction_IZ__P16__nuclei
556
Leukocytes__adult__myocardial infarction_IZ__P2__nuclei
119
Leukocytes__adult__myocardial infarction_IZ__P3__nuclei
80
Leukocytes__adult__myocardial infarction_IZ__P9__nuclei
1035
Leukocytes__adult__myocardial infarction_RZ__P11__nuclei
594
Leukocytes__adult__myocardial infarction_RZ__P2__nuclei
937
Leukocytes__adult__myocardial infarction_RZ__P3__nuclei
1465
Leukocytes__adult__myocardial infarction_RZ__P6__nuclei
837
Leukocytes__adult__myocardial infarction_RZ__P9__nuclei
892
Leukocytes__fetal__healthy__09W0D__cells
548
Leukocytes__fetal__healthy__09W6D__cells
459
Leukocytes__fetal__healthy__10X127__cells
132
Leukocytes__fetal__healthy__10X142__cells
187
Leukocytes__fetal__healthy__10X151__cells
66
Leukocytes__fetal__healthy__10X158__cells
681
Leukocytes__fetal__healthy__10X166__cells
96
Leukocytes__fetal__healthy__10X246__cells
251
Leukocytes__fetal__healthy__10X253__cells
103
Leukocytes__fetal__healthy__10X256__cells
65
Leukocytes__fetal__healthy__10X263__cells
111
Leukocytes__fetal__healthy__10X274__cells
49
Leukocytes__fetal__healthy__10X285__cells

92
Leukocytes__fetal__healthy__10X289__cells
81
Leukocytes__fetal__healthy__10X299__cells
60
Leukocytes__fetal__healthy__10X300__cells
63
Leukocytes__fetal__healthy__10X303__cells
75
Leukocytes__fetal__healthy__11W1D__cells
619
Leukocytes__fetal__healthy__11W4D__cells
907
Leukocytes__fetal__healthy__13W3D__cells
1211
Leukocytes__fetal__healthy__15W0D__cells
470
Leukocytes__fetal__healthy__15W2D__cells
581
Leukocytes__fetal__healthy__alexsc__cells
57
Leukocytes__fetal__healthy__BRC2251__cells
470
Leukocytes__fetal__healthy__BRC2252__cells
384
Leukocytes__fetal__healthy__BRC2256__cells
709
Leukocytes__fetal__healthy__BRC2260__cells
1708
Leukocytes__fetal__healthy__BRC2262__cells
463
Leukocytes__fetal__healthy__BRC2263__cells
498
Leukocytes__fetal__healthy__F8W4D__nuclei
256
Leukocytes__fetal__healthy__M10W5D__nuclei
280
Leukocytes__fetal__healthy__M9W0D__nuclei
109
Neurons_Mixed__adult__healthy__D2__nuclei
32
Neurons_Mixed__adult__healthy__D3__nuclei
13
Neurons_Mixed__adult__healthy__D4__nuclei
16
Neurons_Mixed__adult__healthy__D5__cells
2
Neurons_Mixed__adult__healthy__D5__nuclei
13
Neurons_Mixed__adult__healthy__D6__nuclei
9
Neurons_Mixed__adult__healthy__D7__nuclei
27
Neurons_Mixed__adult__healthy__P1__nuclei
60
Neurons_Mixed__adult__healthy__P17__nuclei
52
Neurons_Mixed__adult__healthy__P7__nuclei
97
Neurons_Mixed__adult__healthy__P8__nuclei
119
Neurons_Mixed__adult__myocardial infarction_BZ__P12__nuclei
79
Neurons_Mixed__adult__myocardial infarction_BZ__P2__nuclei
89
Neurons_Mixed__adult__myocardial infarction_BZ__P3__nuclei
82
Neurons_Mixed__adult__myocardial infarction_FZ__P14__nuclei
72
Neurons_Mixed__adult__myocardial infarction_FZ__P18__nuclei
13
Neurons_Mixed__adult__myocardial infarction_FZ__P19__nuclei
21
Neurons_Mixed__adult__myocardial infarction_FZ__P20__nuclei
40
Neurons_Mixed__adult__myocardial infarction_FZ__P4__nuclei
3
Neurons_Mixed__adult__myocardial infarction_FZ__P5__nuclei
93
Neurons_Mixed__adult__myocardial infarction_IZ__P10__nuclei
46
Neurons_Mixed__adult__myocardial infarction_IZ__P13__nuclei
6
Neurons_Mixed__adult__myocardial infarction_IZ__P15__nuclei
21
Neurons_Mixed__adult__myocardial infarction_IZ__P16__nuclei
43
Neurons_Mixed__adult__myocardial infarction_IZ__P2__nuclei
6
Neurons_Mixed__adult__myocardial infarction_IZ__P3__nuclei
5
Neurons_Mixed__adult__myocardial infarction_IZ__P9__nuclei
38
Neurons_Mixed__adult__myocardial infarction_RZ__P11__nuclei
66
Neurons_Mixed__adult__myocardial infarction_RZ__P2__nuclei
28
Neurons_Mixed__adult__myocardial infarction_RZ__P3__nuclei
96
Neurons_Mixed__adult__myocardial infarction_RZ__P6__nuclei

60
Neurons_Mixed__adult__myocardial infarction_RZ__P9__nuclei
289
Neurons_Mixed__fetal__healthy__09W0D__cells
335
Neurons_Mixed__fetal__healthy__09W6D__cells
127
Neurons_Mixed__fetal__healthy__10X127__cells
52
Neurons_Mixed__fetal__healthy__10X142__cells
41
Neurons_Mixed__fetal__healthy__10X151__cells
23
Neurons_Mixed__fetal__healthy__10X158__cells
78
Neurons_Mixed__fetal__healthy__10X166__cells
3
Neurons_Mixed__fetal__healthy__10X246__cells
12
Neurons_Mixed__fetal__healthy__10X253__cells
11
Neurons_Mixed__fetal__healthy__10X256__cells
3
Neurons_Mixed__fetal__healthy__10X263__cells
6
Neurons_Mixed__fetal__healthy__10X274__cells
7
Neurons_Mixed__fetal__healthy__10X285__cells
17
Neurons_Mixed__fetal__healthy__10X289__cells
25
Neurons_Mixed__fetal__healthy__10X299__cells
29
Neurons_Mixed__fetal__healthy__10X300__cells
9
Neurons_Mixed__fetal__healthy__10X303__cells
18
Neurons_Mixed__fetal__healthy__11W1D__cells
91
Neurons_Mixed__fetal__healthy__11W4D__cells
21
Neurons_Mixed__fetal__healthy__13W3D__cells
43
Neurons_Mixed__fetal__healthy__15W0D__cells
20
Neurons_Mixed__fetal__healthy__15W2D__cells
32
Neurons_Mixed__fetal__healthy__BRC2251__cells
832
Neurons_Mixed__fetal__healthy__BRC2252__cells
30
Neurons_Mixed__fetal__healthy__BRC2256__cells
145
Neurons_Mixed__fetal__healthy__BRC2260__cells
57
Neurons_Mixed__fetal__healthy__BRC2262__cells
44
Neurons_Mixed__fetal__healthy__BRC2263__cells
113
Neurons_SchwannGlialCells__adult__healthy__D2__nuclei
98
Neurons_SchwannGlialCells__adult__healthy__D3__cells
4
Neurons_SchwannGlialCells__adult__healthy__D3__nuclei
75
Neurons_SchwannGlialCells__adult__healthy__D4__cells
11
Neurons_SchwannGlialCells__adult__healthy__D4__nuclei
88
Neurons_SchwannGlialCells__adult__healthy__D5__cells
44
Neurons_SchwannGlialCells__adult__healthy__D5__nuclei
59
Neurons_SchwannGlialCells__adult__healthy__D6__cells
75
Neurons_SchwannGlialCells__adult__healthy__D6__nuclei
61
Neurons_SchwannGlialCells__adult__healthy__D7__cells
16
Neurons_SchwannGlialCells__adult__healthy__D7__nuclei
56
Neurons_SchwannGlialCells__adult__healthy__P1__nuclei
80
Neurons_SchwannGlialCells__adult__healthy__P17__nuclei
51
Neurons_SchwannGlialCells__adult__healthy__P7__nuclei
28
Neurons_SchwannGlialCells__adult__healthy__P8__nuclei
87
Neurons_SchwannGlialCells__adult__myocardial infarction_BZ__P12__nuclei
22
Neurons_SchwannGlialCells__adult__myocardial infarction_BZ__P2__nuclei
51
Neurons_SchwannGlialCells__adult__myocardial infarction_BZ__P3__nuclei
35
Neurons_SchwannGlialCells__adult__myocardial infarction_FZ__P14__nuclei
12
Neurons_SchwannGlialCells__adult__myocardial infarction_FZ__P18__nuclei
73
Neurons_SchwannGlialCells__adult__myocardial infarction_FZ__P19__nuclei

13
Neurons_SchwannGlialCells__adult__myocardial infarction_FZ__P20__nuclei
20
Neurons_SchwannGlialCells__adult__myocardial infarction_FZ__P4__nuclei
31
Neurons_SchwannGlialCells__adult__myocardial infarction_FZ__P5__nuclei
11
Neurons_SchwannGlialCells__adult__myocardial infarction_IZ__P10__nuclei
1
Neurons_SchwannGlialCells__adult__myocardial infarction_IZ__P16__nuclei
1
Neurons_SchwannGlialCells__adult__myocardial infarction_IZ__P9__nuclei
3
Neurons_SchwannGlialCells__adult__myocardial infarction_RZ__P11__nuclei
20
Neurons_SchwannGlialCells__adult__myocardial infarction_RZ__P2__nuclei
78
Neurons_SchwannGlialCells__adult__myocardial infarction_RZ__P3__nuclei
51
Neurons_SchwannGlialCells__adult__myocardial infarction_RZ__P6__nuclei
201
Neurons_SchwannGlialCells__adult__myocardial infarction_RZ__P9__nuclei
10
Neurons_SchwannGlialCells__fetal__healthy__09W0D__cells
220
Neurons_SchwannGlialCells__fetal__healthy__09W6D__cells
150
Neurons_SchwannGlialCells__fetal__healthy__10X127__cells
40
Neurons_SchwannGlialCells__fetal__healthy__10X142__cells
35
Neurons_SchwannGlialCells__fetal__healthy__10X151__cells
19
Neurons_SchwannGlialCells__fetal__healthy__10X158__cells
42
Neurons_SchwannGlialCells__fetal__healthy__10X166__cells
18
Neurons_SchwannGlialCells__fetal__healthy__10X246__cells
81
Neurons_SchwannGlialCells__fetal__healthy__10X253__cells
125
Neurons_SchwannGlialCells__fetal__healthy__10X256__cells
54
Neurons_SchwannGlialCells__fetal__healthy__10X263__cells
92
Neurons_SchwannGlialCells__fetal__healthy__10X274__cells
56
Neurons_SchwannGlialCells__fetal__healthy__10X285__cells
30
Neurons_SchwannGlialCells__fetal__healthy__10X289__cells
41
Neurons_SchwannGlialCells__fetal__healthy__10X299__cells
8
Neurons_SchwannGlialCells__fetal__healthy__10X300__cells
16
Neurons_SchwannGlialCells__fetal__healthy__10X303__cells
3
Neurons_SchwannGlialCells__fetal__healthy__11W1D__cells
162
Neurons_SchwannGlialCells__fetal__healthy__11W4D__cells
144
Neurons_SchwannGlialCells__fetal__healthy__13W3D__cells
505
Neurons_SchwannGlialCells__fetal__healthy__15W0D__cells
297
Neurons_SchwannGlialCells__fetal__healthy__15W2D__cells
243
Neurons_SchwannGlialCells__fetal__healthy__BRC2251__cells
234
Neurons_SchwannGlialCells__fetal__healthy__BRC2252__cells
27
Neurons_SchwannGlialCells__fetal__healthy__BRC2256__cells
165
Neurons_SchwannGlialCells__fetal__healthy__BRC2260__cells
124
Neurons_SchwannGlialCells__fetal__healthy__BRC2262__cells
27
Neurons_SchwannGlialCells__fetal__healthy__BRC2263__cells
106
Neurons_SchwannGlialCells__fetal__healthy__F8W4D__nuclei
182
Neurons_SchwannGlialCells__fetal__healthy__M10W5D__nuclei
319
Neurons_SchwannGlialCells__fetal__healthy__M9W0D__nuclei
37
Pericytes__adult__healthy__D2__nuclei
349
Pericytes__adult__healthy__D3__cells
126
Pericytes__adult__healthy__D3__nuclei
250
Pericytes__adult__healthy__D4__cells
115
Pericytes__adult__healthy__D4__nuclei
267
Pericytes__adult__healthy__D5__cells
162
Pericytes__adult__healthy__D5__nuclei
193
Pericytes__adult__healthy__D6__cells

158
Pericytes__adult__healthy__D6__nuclei
188
Pericytes__adult__healthy__D7__cells
161
Pericytes__adult__healthy__D7__nuclei
215
Pericytes__adult__healthy__P1__nuclei
697
Pericytes__adult__healthy__P17__nuclei
599
Pericytes__adult__healthy__P7__nuclei
729
Pericytes__adult__healthy__P8__nuclei
635
Pericytes__adult__myocardial infarction_BZ__P12__nuclei
457
Pericytes__adult__myocardial infarction_BZ__P2__nuclei
832
Pericytes__adult__myocardial infarction_BZ__P3__nuclei
358
Pericytes__adult__myocardial infarction_FZ__P14__nuclei
560
Pericytes__adult__myocardial infarction_FZ__P18__nuclei
172
Pericytes__adult__myocardial infarction_FZ__P19__nuclei
185
Pericytes__adult__myocardial infarction_FZ__P20__nuclei
86
Pericytes__adult__myocardial infarction_FZ__P4__nuclei
109
Pericytes__adult__myocardial infarction_FZ__P5__nuclei
847
Pericytes__adult__myocardial infarction_IZ__P10__nuclei
317
Pericytes__adult__myocardial infarction_IZ__P13__nuclei
235
Pericytes__adult__myocardial infarction_IZ__P15__nuclei
244
Pericytes__adult__myocardial infarction_IZ__P16__nuclei
105
Pericytes__adult__myocardial infarction_IZ__P2__nuclei
75
Pericytes__adult__myocardial infarction_IZ__P3__nuclei
6
Pericytes__adult__myocardial infarction_IZ__P9__nuclei
624
Pericytes__adult__myocardial infarction_RZ__P11__nuclei
459
Pericytes__adult__myocardial infarction_RZ__P2__nuclei
965
Pericytes__adult__myocardial infarction_RZ__P3__nuclei
324
Pericytes__adult__myocardial infarction_RZ__P6__nuclei
1133
Pericytes__adult__myocardial infarction_RZ__P9__nuclei
838
Pericytes__fetal__healthy__10X127__cells
192
Pericytes__fetal__healthy__10X142__cells
95
Pericytes__fetal__healthy__10X151__cells
199
Pericytes__fetal__healthy__10X158__cells
1222
Pericytes__fetal__healthy__10X166__cells
235
Pericytes__fetal__healthy__10X246__cells
396
Pericytes__fetal__healthy__10X253__cells
686
Pericytes__fetal__healthy__10X256__cells
385
Pericytes__fetal__healthy__10X263__cells
418
Pericytes__fetal__healthy__10X274__cells
149
Pericytes__fetal__healthy__10X285__cells
122
Pericytes__fetal__healthy__10X289__cells
121
Pericytes__fetal__healthy__10X299__cells
80
Pericytes__fetal__healthy__10X300__cells
55
Pericytes__fetal__healthy__10X303__cells
99
Pericytes__fetal__healthy__alexsc__cells
138
Pericytes__fetal__healthy__BRC2251__cells
223
Pericytes__fetal__healthy__BRC2252__cells
222
Pericytes__fetal__healthy__BRC2256__cells
123
Pericytes__fetal__healthy__BRC2260__cells
185
Pericytes__fetal__healthy__BRC2262__cells
100
Pericytes__fetal__healthy__BRC2263__cells

76
Pericytes__fetal__healthy__F8W4D__nuclei
321
Pericytes__fetal__healthy__M10W5D__nuclei
168
Pericytes__fetal__healthy__M9W0D__nuclei
68
PlateletsErythrocytes__fetal__healthy__09W0D__cells
2
PlateletsErythrocytes__fetal__healthy__09W6D__cells
5
PlateletsErythrocytes__fetal__healthy__10X127__cells
22
PlateletsErythrocytes__fetal__healthy__10X142__cells
18
PlateletsErythrocytes__fetal__healthy__10X151__cells
11
PlateletsErythrocytes__fetal__healthy__10X158__cells
82
PlateletsErythrocytes__fetal__healthy__10X166__cells
60
PlateletsErythrocytes__fetal__healthy__10X246__cells
25
PlateletsErythrocytes__fetal__healthy__10X253__cells
44
PlateletsErythrocytes__fetal__healthy__10X256__cells
14
PlateletsErythrocytes__fetal__healthy__10X263__cells
38
PlateletsErythrocytes__fetal__healthy__10X274__cells
6
PlateletsErythrocytes__fetal__healthy__10X285__cells
15
PlateletsErythrocytes__fetal__healthy__10X289__cells
38
PlateletsErythrocytes__fetal__healthy__10X299__cells
95
PlateletsErythrocytes__fetal__healthy__10X300__cells
117
PlateletsErythrocytes__fetal__healthy__10X303__cells
9
PlateletsErythrocytes__fetal__healthy__11W4D__cells
2
PlateletsErythrocytes__fetal__healthy__13W3D__cells
4
PlateletsErythrocytes__fetal__healthy__15W0D__cells
2
PlateletsErythrocytes__fetal__healthy__15W2D__cells
2
SMC__adult__cHF__C1__cells
100
SMC__adult__cHF__C2__cells
40
SMC__adult__dHF__D1__cells
101
SMC__adult__dHF__D2__cells
132
SMC__adult__dHF__D4__cells
38
SMC__adult__dHF__D5__cells
10
SMC__adult__healthy__D2__nuclei
120
SMC__adult__healthy__D3__cells
186
SMC__adult__healthy__D3__nuclei
53
SMC__adult__healthy__D4__cells
84
SMC__adult__healthy__D4__nuclei
146
SMC__adult__healthy__D5__cells
192
SMC__adult__healthy__D5__nuclei
45
SMC__adult__healthy__D6__cells
185
SMC__adult__healthy__D6__nuclei
52
SMC__adult__healthy__D7__cells
145
SMC__adult__healthy__D7__nuclei
62
SMC__adult__healthy__N1__cells
56
SMC__adult__healthy__N10__cells
79
SMC__adult__healthy__N11__cells
43
SMC__adult__healthy__N12__cells
5
SMC__adult__healthy__N13__cells
4
SMC__adult__healthy__N14__cells
4
SMC__adult__healthy__N2__cells
46
SMC__adult__healthy__N3__cells
53
SMC__adult__healthy__N4__cells

20
SMC__adult__healthy__N5__cells
86
SMC__adult__healthy__N6__cells
140
SMC__adult__healthy__N7__cells
8
SMC__adult__healthy__N8__cells
29
SMC__adult__healthy__N9__cells
75
SMC__adult__healthy__P1__nuclei
121
SMC__adult__healthy__P17__nuclei
82
SMC__adult__healthy__P7__nuclei
86
SMC__adult__healthy__P8__nuclei
122
SMC__adult__myocardial infarction_BZ__P12__nuclei
79
SMC__adult__myocardial infarction_BZ__P2__nuclei
156
SMC__adult__myocardial infarction_BZ__P3__nuclei
59
SMC__adult__myocardial infarction_FZ__P14__nuclei
228
SMC__adult__myocardial infarction_FZ__P18__nuclei
262
SMC__adult__myocardial infarction_FZ__P19__nuclei
88
SMC__adult__myocardial infarction_FZ__P20__nuclei
97
SMC__adult__myocardial infarction_FZ__P4__nuclei
96
SMC__adult__myocardial infarction_FZ__P5__nuclei
173
SMC__adult__myocardial infarction_IZ__P10__nuclei
48
SMC__adult__myocardial infarction_IZ__P13__nuclei
130
SMC__adult__myocardial infarction_IZ__P15__nuclei
87
SMC__adult__myocardial infarction_IZ__P16__nuclei
30
SMC__adult__myocardial infarction_IZ__P2__nuclei
43
SMC__adult__myocardial infarction_IZ__P3__nuclei
8
SMC__adult__myocardial infarction_IZ__P9__nuclei
164
SMC__adult__myocardial infarction_RZ__P11__nuclei
92
SMC__adult__myocardial infarction_RZ__P2__nuclei
162
SMC__adult__myocardial infarction_RZ__P3__nuclei
96
SMC__adult__myocardial infarction_RZ__P6__nuclei
96
SMC__adult__myocardial infarction_RZ__P9__nuclei
115
SMC__fetal__healthy__09W0D__cells
898
SMC__fetal__healthy__09W6D__cells
876
SMC__fetal__healthy__10X127__cells
465
SMC__fetal__healthy__10X142__cells
73
SMC__fetal__healthy__10X151__cells
197
SMC__fetal__healthy__10X158__cells
1517
SMC__fetal__healthy__10X166__cells
310
SMC__fetal__healthy__10X246__cells
216
SMC__fetal__healthy__10X253__cells
253
SMC__fetal__healthy__10X256__cells
128
SMC__fetal__healthy__10X263__cells
161
SMC__fetal__healthy__10X274__cells
62
SMC__fetal__healthy__10X285__cells
128
SMC__fetal__healthy__10X289__cells
822
SMC__fetal__healthy__10X299__cells
350
SMC__fetal__healthy__10X300__cells
230
SMC__fetal__healthy__10X303__cells
196
SMC__fetal__healthy__11W1D__cells
1872
SMC__fetal__healthy__11W4D__cells
2765
SMC__fetal__healthy__13W3D__cells

2413
SMC__fetal__healthy__15W0D__cells
1143
SMC__fetal__healthy__15W2D__cells
919
SMC__fetal__healthy__alexsc__cells
39
SMC__fetal__healthy__BRC2251__cells
734
SMC__fetal__healthy__BRC2252__cells
186
SMC__fetal__healthy__BRC2256__cells
659
SMC__fetal__healthy__BRC2260__cells
555
SMC__fetal__healthy__BRC2262__cells
203
SMC__fetal__healthy__BRC2263__cells
489
SMC__fetal__healthy__F8W4D__nuclei
10
SMC__fetal__healthy__M10W5D__nuclei
838
SMC__fetal__healthy__M9W0D__nuclei
102
unknown__adult__cHF__C1__cells
6
unknown__adult__cHF__C2__cells
3
unknown__adult__dHF__D1__cells
1
unknown__adult__dHF__D2__cells
3
unknown__adult__dHF__D5__cells
6
unknown__adult__healthy__D2__nuclei
85
unknown__adult__healthy__D3__cells
13
unknown__adult__healthy__D3__nuclei
78
unknown__adult__healthy__D4__cells
8
unknown__adult__healthy__D4__nuclei
45
unknown__adult__healthy__D5__cells
6
unknown__adult__healthy__D5__nuclei
10
unknown__adult__healthy__D6__cells
4
unknown__adult__healthy__D6__nuclei
16
unknown__adult__healthy__D7__cells
15
unknown__adult__healthy__D7__nuclei
25
unknown__adult__healthy__N1__cells
1
unknown__adult__healthy__N10__cells
1
unknown__adult__healthy__N11__cells
14
unknown__adult__healthy__N13__cells
5
unknown__adult__healthy__N14__cells
5
unknown__adult__healthy__N2__cells
1
unknown__adult__healthy__N3__cells
62
unknown__adult__healthy__N5__cells
2
unknown__adult__healthy__N6__cells
9
unknown__adult__healthy__N7__cells
3
unknown__adult__healthy__N8__cells
1
unknown__adult__healthy__N9__cells
10
unknown__adult__healthy__P1__nuclei
92
unknown__adult__healthy__P17__nuclei
67
unknown__adult__healthy__P7__nuclei
37
unknown__adult__healthy__P8__nuclei
28
unknown__adult__myocardial infarction_BZ__P12__nuclei
13
unknown__adult__myocardial infarction_BZ__P2__nuclei
31
unknown__adult__myocardial infarction_BZ__P3__nuclei
47
unknown__adult__myocardial infarction_FZ__P14__nuclei
21
unknown__adult__myocardial infarction_FZ__P18__nuclei
26
unknown__adult__myocardial infarction_FZ__P19__nuclei

23
unknown__adult__myocardial infarction_FZ__P20__nuclei
19
unknown__adult__myocardial infarction_FZ__P4__nuclei
24
unknown__adult__myocardial infarction_FZ__P5__nuclei
17
unknown__adult__myocardial infarction_IZ__P10__nuclei
533
unknown__adult__myocardial infarction_IZ__P13__nuclei
124
unknown__adult__myocardial infarction_IZ__P15__nuclei
998
unknown__adult__myocardial infarction_IZ__P16__nuclei
265
unknown__adult__myocardial infarction_IZ__P2__nuclei
27
unknown__adult__myocardial infarction_IZ__P3__nuclei
18
unknown__adult__myocardial infarction_IZ__P9__nuclei
340
unknown__adult__myocardial infarction_RZ__P11__nuclei
13
unknown__adult__myocardial infarction_RZ__P2__nuclei
30
unknown__adult__myocardial infarction_RZ__P3__nuclei
17
unknown__adult__myocardial infarction_RZ__P6__nuclei
99
unknown__adult__myocardial infarction_RZ__P9__nuclei
14
unknown__fetal__healthy__09W0D__cells
2
unknown__fetal__healthy__09W6D__cells
2
unknown__fetal__healthy__10X127__cells
45
unknown__fetal__healthy__10X142__cells
40
unknown__fetal__healthy__10X151__cells
83
unknown__fetal__healthy__10X158__cells
384
unknown__fetal__healthy__10X166__cells
105
unknown__fetal__healthy__10X246__cells
39
unknown__fetal__healthy__10X253__cells
181
unknown__fetal__healthy__10X256__cells
72
unknown__fetal__healthy__10X263__cells
45
unknown__fetal__healthy__10X274__cells
17
unknown__fetal__healthy__10X285__cells
15
unknown__fetal__healthy__10X289__cells
225
unknown__fetal__healthy__10X299__cells
142
unknown__fetal__healthy__10X300__cells
177
unknown__fetal__healthy__10X303__cells
62
unknown__fetal__healthy__11W1D__cells
9
unknown__fetal__healthy__11W4D__cells
2
unknown__fetal__healthy__13W3D__cells
9
unknown__fetal__healthy__15W0D__cells
4
unknown__fetal__healthy__15W2D__cells
6
unknown__fetal__healthy__alexsc__cells
152
unknown__fetal__healthy__BRC2251__cells
752
unknown__fetal__healthy__BRC2252__cells
909
unknown__fetal__healthy__BRC2256__cells
336
unknown__fetal__healthy__BRC2260__cells
1375
unknown__fetal__healthy__BRC2262__cells
426
unknown__fetal__healthy__BRC2263__cells
548
vCM__adult__cHF__C1__cells
416
vCM__adult__cHF__C2__cells
22
vCM__adult__dHF__D1__cells
46
vCM__adult__dHF__D2__cells
11
vCM__adult__dHF__D4__cells
75
vCM__adult__dHF__D5__cells

139
vCM__adult__healthy__D2__nuclei
653
vCM__adult__healthy__D3__cells
1
vCM__adult__healthy__D3__nuclei
455
vCM__adult__healthy__D4__cells
10
vCM__adult__healthy__D4__nuclei
503
vCM__adult__healthy__D5__cells
4
vCM__adult__healthy__D5__nuclei
520
vCM__adult__healthy__D6__cells
3
vCM__adult__healthy__D6__nuclei
451
vCM__adult__healthy__D7__cells
22
vCM__adult__healthy__D7__nuclei
540
vCM__adult__healthy__N1__cells
267
vCM__adult__healthy__N10__cells
19
vCM__adult__healthy__N11__cells
11
vCM__adult__healthy__N12__cells
1
vCM__adult__healthy__N13__cells
72
vCM__adult__healthy__N14__cells
25
vCM__adult__healthy__N2__cells
290
vCM__adult__healthy__N3__cells
413
vCM__adult__healthy__N4__cells
78
vCM__adult__healthy__N5__cells
347
vCM__adult__healthy__N6__cells
12
vCM__adult__healthy__N7__cells
12
vCM__adult__healthy__N8__cells
12
vCM__adult__healthy__N9__cells
14
vCM__adult__healthy__P1__nuclei
3856
vCM__adult__healthy__P17__nuclei
3982
vCM__adult__healthy__P7__nuclei
6475
vCM__adult__healthy__P8__nuclei
3929
vCM__adult__myocardial infarction_BZ__P12__nuclei
1788
vCM__adult__myocardial infarction_BZ__P2__nuclei
4921
vCM__adult__myocardial infarction_BZ__P3__nuclei
3784
vCM__adult__myocardial infarction_FZ__P14__nuclei
3775
vCM__adult__myocardial infarction_FZ__P18__nuclei
307
vCM__adult__myocardial infarction_FZ__P19__nuclei
400
vCM__adult__myocardial infarction_FZ__P20__nuclei
350
vCM__adult__myocardial infarction_FZ__P4__nuclei
46
vCM__adult__myocardial infarction_FZ__P5__nuclei
2559
vCM__adult__myocardial infarction_IZ__P10__nuclei
996
vCM__adult__myocardial infarction_IZ__P13__nuclei
41
vCM__adult__myocardial infarction_IZ__P15__nuclei
582
vCM__adult__myocardial infarction_IZ__P16__nuclei
40
vCM__adult__myocardial infarction_IZ__P2__nuclei
1139
vCM__adult__myocardial infarction_IZ__P3__nuclei
106
vCM__adult__myocardial infarction_IZ__P9__nuclei
1927
vCM__adult__myocardial infarction_RZ__P11__nuclei
2457
vCM__adult__myocardial infarction_RZ__P2__nuclei
4704
vCM__adult__myocardial infarction_RZ__P3__nuclei
3243
vCM__adult__myocardial infarction_RZ__P6__nuclei
4314
vCM__adult__myocardial infarction_RZ__P9__nuclei

```
8231
vCM__fetal__healthy__09W0D__cells
3651
vCM__fetal__healthy__09W6D__cells
5220
vCM__fetal__healthy__10X127__cells
1385
vCM__fetal__healthy__10X142__cells
353
vCM__fetal__healthy__10X151__cells
1067
vCM__fetal__healthy__10X158__cells
4237
vCM__fetal__healthy__10X166__cells
1146
vCM__fetal__healthy__10X246__cells
289
vCM__fetal__healthy__10X253__cells
593
vCM__fetal__healthy__10X256__cells
175
vCM__fetal__healthy__10X263__cells
36
vCM__fetal__healthy__10X274__cells
201
vCM__fetal__healthy__10X285__cells
171
vCM__fetal__healthy__10X289__cells
2471
vCM__fetal__healthy__10X299__cells
818
vCM__fetal__healthy__10X300__cells
358
vCM__fetal__healthy__10X303__cells
772
vCM__fetal__healthy__11W1D__cells
4276
vCM__fetal__healthy__11W4D__cells
1014
vCM__fetal__healthy__13W3D__cells
8851
vCM__fetal__healthy__15W0D__cells
2449
vCM__fetal__healthy__15W2D__cells
3075
vCM__fetal__healthy__alexsc__cells
443
vCM__fetal__healthy__BRC2251__cells
710
vCM__fetal__healthy__BRC2252__cells
485
vCM__fetal__healthy__BRC2256__cells
405
vCM__fetal__healthy__BRC2260__cells
226
vCM__fetal__healthy__BRC2262__cells
471
vCM__fetal__healthy__BRC2263__cells
235
vCM__fetal__healthy__F8W4D__nuclei
5335
vCM__fetal__healthy__M10W5D__nuclei
12296
vCM__fetal__healthy__M9W0D__nuclei
4793
```

In [342…

```
num_iterations = 20
```

In [24]:

```
# # create i pseudobulk samples for testing
# bulk_samples = list()
# for(i in 1:num_iterations){
#     bulk_samples[[i]] = list()
#     for(j in 1:length(levels(bulk_class))){
#         bulk = levels(bulk_class)[j]
#         # pull out all of condition
#         all_bulk = rownames(metadata)[bulk_class == bulk]
#         # set random seed for our sample i
#         set.seed(i)
#         # generate sample
#         bulk_samples[[i]][[j]] = sample(all_bulk, min(100, length(all_bulk)))
#         }
#     }
```

In [333…

```
# saveRDS(bulk_samples,"bulk_samples_20.rds")
bulk_samples = readRDS("bulk_samples_20.rds")
```

In [26]:

```
# # sum the scRNA-seq libraries in each case to create a pseudobulk size-factor vector
# bulk_size.factors = mclapply(
#     1:num_iterations,
#     function(i){
#             sapply(
#                 1:length(bulk_samples[[i]]),
#                 function(j){
#                     sum(metadata[bulk_samples[[i]][[j]],"total_counts"])


#                     }
#                 )
```

```
#         },mc.cores=4
#     )
```

In [334…

```
# saveRDS(bulk_size.factors,"bulk_size.factors_20.rds")
bulk_size.factors = readRDS("bulk_size.factors_20.rds")
```

In [335…

```
# load global dispersions, pre-calculated using all genes for all sample iterations
global_dispersions = readRDS("global_dispersions.rds")
```

In [29]:

```
# quick sanity check on sampling - columnn 1, pseudobulk 1, is aCM condition
metadata[bulk_samples[[1]][[1]],]
sum(metadata[bulk_samples[[1]][[1]],"total_counts"])
sum(metadata[bulk_samples[[1]][[1]],"total_counts"]) == bulk_size.factors[[1]][[1]]
```

	stage	disease_condition	cell_or_nuclei	n_genes_by_counts	log1p_n_genes_by_counts	total_counts	log1p_total_counts	pct_counts_in_top_50_
	<fct>	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	
SC_96366_22_5	adult	disease	cells	2817	7.943783	22043	10.000795	50.
SC_96366_67_34	adult	disease	cells	2486	7.818832	12340	9.420682	46.
SC_96366_33_0	adult	disease	cells	1783	7.486613	11649	9.363062	54.
SC_96366_57_35	adult	disease	cells	1043	6.950815	4065	8.310415	54.
SC_96359_24_34	adult	disease	cells	3867	8.260493	30422	10.322954	44.
SC_96279_43_46	adult	disease	cells	2197	7.695303	10398	9.249465	47.
SC_96359_27_2	adult	disease	cells	1439	7.272398	3957	8.283494	37.
SC_96366_25_5	adult	disease	cells	1231	7.116394	5413	8.596744	49.
SC_96279_2_24	adult	disease	cells	3010	8.010028	28749	10.266393	56.
SC_96366_71_7	adult	disease	cells	1847	7.521859	8813	9.084097	48.
SC_96366_57_6	adult	disease	cells	1603	7.380256	9114	9.117677	50.
SC_96279_41_13	adult	disease	cells	2114	7.656810	9114	9.117677	52.
SC_96366_30_26	adult	disease	cells	1336	7.198184	6062	8.709960	47.
SC_96366_57_70	adult	disease	cells	1328	7.192182	8605	9.060215	48.
SC_96359_18_3	adult	disease	cells	3485	8.156510	28683	10.264094	42.
SC_96366_64_32	adult	disease	cells	2605	7.865572	17326	9.760021	44.
SC_96366_54_19	adult	disease	cells	2463	7.809541	14952	9.612667	52.
SC_96366_23_28	adult	disease	cells	1550	7.346655	8674	9.068201	54.
SC_96359_58_7	adult	disease	cells	4757	8.467583	55636	10.926603	40.
SC_96366_22_6	adult	disease	cells	3802	8.243546	32707	10.395375	44.
SC_96366_64_8	adult	disease	cells	2081	7.641084	9125	9.118883	43.
SC_96366_65_33	adult	disease	cells	3026	8.015327	18438	9.822224	46.
SC_96359_22_11	adult	disease	cells	3988	8.291296	45099	10.716638	40.
SC_96366_23_1	adult	disease	cells	3338	8.113427	22450	10.019091	42.
SC_96366_64_33	adult	disease	cells	2826	7.946971	18331	9.816403	44.
SC_96366_55_33	adult	disease	cells	1931	7.566311	11368	9.338646	46.
SC_96359_20_4	adult	disease	cells	3220	8.077447	19176	9.861467	39.
SC_96366_32_5	adult	disease	cells	1586	7.369601	6195	8.731659	46.
SC_96366_56_25	adult	disease	cells	2731	7.912789	28292	10.250370	56.
SC_96279_2_0	adult	disease	cells	2044	7.623153	18528	9.827092	46.
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	
SC_96366_69_4	adult	disease	cells	3083	8.033983	24369	10.101109	52.
SC_96366_68_34	adult	disease	cells	1508	7.319202	7061	8.862484	47.
SC_96366_21_2	adult	disease	cells	2528	7.835579	13591	9.517237	46.
SC_96359_23_53	adult	disease	cells	4141	8.328934	43788	10.687138	45.
SC_96366_19_44	adult	disease	cells	1853	7.525101	11677	9.365462	52.
SC_96279_39_11	adult	disease	cells	1966	7.584265	9361	9.144414	55.
SC_96366_24_5	adult	disease	cells	2058	7.629976	15182	9.627932	51.
SC_96366_67_3	adult	disease	cells	2488	7.819636	13389	9.502263	45.
SC_96366_57_33	adult	disease	cells	2249	7.718685	10962	9.302281	47.
SC_96366_23_17	adult	disease	cells	1842	7.519150	7151	8.875147	49.
SC_96366_22_11	adult	disease	cells	1851	7.524021	11499	9.350102	49.
SC_96366_61_71	adult	disease	cells	2109	7.654443	17368	9.762443	48.
SC_96366_33_6	adult	disease	cells	1243	7.126087	5625	8.635154	53.
SC_96366_19_20	adult	disease	cells	1731	7.457032	7565	8.931419	46.
SC_96366_56_10	adult	disease	cells	2543	7.841493	15318	9.636849	45.
SC_96366_28_4	adult	disease	cells	1684	7.429521	7251	8.889032	50.
SC_96366_54_1	adult	disease	cells	1615	7.387709	7741	8.954415	56.
SC_96366_26_10	adult	disease	cells	1252	7.133296	6046	8.707317	60.
SC_96366_63_33	adult	disease	cells	3420	8.137688	22994	10.043032	49.

	stage	disease_condition	cell_or_nuclei	n_genes_by_counts	log1p_n_genes_by_counts	total_counts	log1p_total_counts	pct_counts_in_top_50_
	<fct>	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	
SC_96279_7_29	adult	disease	cells	2741	7.916443	14803	9.602653	46.
SC_96279_36_11	adult	disease	cells	2883	7.966933	13471	9.508368	41.
SC_96366_35_25	adult	disease	cells	3253	8.087640	27373	10.217349	46.
SC_96366_32_0	adult	disease	cells	3764	8.233503	36166	10.495902	46.
SC_96359_23_9	adult	disease	cells	1687	7.431300	8245	9.017484	35.
SC_96279_6_30	adult	disease	cells	1965	7.583756	7923	8.977652	45.
SC_96366_27_8	adult	disease	cells	2692	7.898411	14619	9.590146	44.
SC_96366_55_9	adult	disease	cells	2883	7.966933	14741	9.598455	44.
SC_96366_65_32	adult	disease	cells	1806	7.499423	9249	9.132379	51.
SC_96366_22_3	adult	disease	cells	3426	8.139441	25238	10.136146	47.
SC_96366_24_30	adult	disease	cells	2442	7.800982	13463	9.507774	46.

1824939
TRUE

In [30]:
```
# bulk_data = mclapply(
#     1:num_iterations,
#     function(i){
#         bulk_matrix = sapply(
#         1:length(levels(bulk_class)),
#                 function(j){
#                     return(colSums(x[bulk_samples[[i]][[j]],,drop=F]))
#                 }
#             )
#         },mc.cores = 4
#     )
# for(i in 1:length(bulk_data)){colnames(bulk_data[[i]]) = levels(bulk_class)}
```

In [332…
```
# saveRDS(bulk_data,"bulk_data_20.rds")
 bulk_data = readRDS("bulk_data_20.rds") # This dataset is the non-full dataset
#bulk_data = readRDS("bulk_data_20FULL.rds") # This dataset is a "full" dataset,  - all genes (~15k genes) shared between all datase
```

In [32]:
```
bulk_data[[1]]
```

	aCM__adult__cHF__C1__cells	aCM__adult__dHF__D1__cells	aCM__adult__dHF__D2__cells	aCM__adult__dHF__D4__cells	aCM__adult__healthy__D2__nuclei
TNNT2	17959	42446	23411	18534	1227
MYH7	4684	8064	9938	19359	26
MYH6	29025	51956	22041	21731	1287
NR2F2	98	267	104	61	24
SHOX2	0	0	7	0	0
VSNL1	21	104	274	51	9
HCN4	7	30	20	23	8
MYH11	23	45	253	72	14
ACTA2	109	31	60	18	89
RGS5	19	34	246	9	4
CYGB	0	5	12	2	3
PECAM1	22	11	48	7	1
NPR3	13	67	47	15	12
ADGRG6	0	0	4	0	0
ACKR1	1	1	3	0	0
HEY1	0	13	5	4	0
FBLN5	4	5	5	14	2
RGCC	17	32	86	1	2
TFF3	0	25	0	27	0
NFATC1	42	36	28	15	3
ITLN1	13	1	35	194	2
UPK3B	1	0	0	0	0
LUM	7	6	64	8	0
COL1A1	2	3	14	4	2
NRXN1	12	16	33	7	18
ENO2	8	49	36	12	5
CHGA	1	0	0	4	0
CHGB	114	223	536	784	1
MPZ	0	0	3	0	0
S100B	0	0	0	0	1
PTPRC	1	1	1	0	1
ADIPOQ	0	0	0	0	1
HBB	5	8	5	0	4
ITGB3	0	0	0	0	0
ITGA2B	2	0	6	20	7
EDNRA	130	142	125	92	90
EDNRB	5	8	74	1	5
EDN1	0	0	2	0	0
EDN2	0	0	0	0	1
EDN3	0	0	0	0	0
ECE1	30	34	18	51	22
ECE2	4	11	10	0	0

In [346…
```
meta_condition = paste(metadata$new_clusters, metadata$stage, metadata$disease_subcategory, metadata$donor,  metadata$cell_or_nuclei
```

In [347…
```
bulk_dataset = sapply(
    1:ncol(bulk_data[[1]]),
    function(i){
    unique(metadata[meta_condition == colnames(bulk_data[[1]])[i],"dataset"]     )
        }
    )
```

In [348…
```
bulk_cell_or_nuclei = sapply(
    1:ncol(bulk_data[[1]]),
    function(i){
    unique(metadata[meta_condition == colnames(bulk_data[[1]])[i],"cell_or_nuclei"]     )
        }
```

```
    )
```

In [349…

```
bulk_donor = sapply(
    1:ncol(bulk_data[[1]]),
    function(i){
    unique(metadata[meta_condition == colnames(bulk_data[[1]])[i],"donor"]    )
        }
    )
```

In [350…

```
bulk_disease_subcategory = sapply(
    1:ncol(bulk_data[[1]]),
    function(i){
    unique(metadata[meta_condition == colnames(bulk_data[[1]])[i],"disease_subcategory"]    )
        }
    )
```

In [351…

```
bulk_celltype = sapply(
    1:ncol(bulk_data[[1]]),
    function(i){
    unique(metadata[meta_condition == colnames(bulk_data[[1]])[i],"new_clusters"]    )
        }
    )
```

In [352…

```
bulk_stage = sapply(
    1:ncol(bulk_data[[1]]),
    function(i){
    unique(metadata[meta_condition == colnames(bulk_data[[1]])[i],"stage"]    )
        }
    )
```

In [353…

```
# Sample Metadata
bulk_metadata = list()
for(i in 1:num_iterations){
    bulk_metadata[[i]] = data.frame(
        t(bulk_data[[i]]),
        "disease_subcategory" = gsub("myocardial infarction_", "", bulk_disease_subcategory),
        "celltype" = factor(bulk_celltype,levels(metadata$new_clusters)),
        "cell_number_log" = log(sapply(1:length(bulk_samples[[i]]), function(j){length(bulk_samples[[i]][[j]])})),
        "cell_number" = sapply(1:length(bulk_samples[[i]]), function(j){length(bulk_samples[[i]][[j]])}),
        "donor" = bulk_donor,
        "dataset" = bulk_dataset,
        "size.factors" = bulk_size.factors[[i]],
        "stage" = bulk_stage,
        "cell_or_nuclei" = bulk_cell_or_nuclei


    )
   rownames(bulk_metadata[[i]]) = colnames(bulk_data[[i]])
}
```

In [354…

```
bulk_metadata[[1]]
```

A data.frame: 1382 × 51

	TNNT2	MYH7	MYH6	NR2F2	SHOX2	VSNL1	HCN4	MYH11	ACTA2	RGS5	⋯	ECE2	disease_subcategory	celltype
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	⋯	<dbl>	<chr>	<fct>
aCM__adult__cHF__C1__cells	17959	4684	29025	98	0	21	7	23	109	19	⋯	4	cHF	aCM
aCM__adult__dHF__D1__cells	42446	8064	51956	267	0	104	30	45	31	34	⋯	11	dHF	aCM
aCM__adult__dHF__D2__cells	23411	9938	22041	104	7	274	20	253	60	246	⋯	10	dHF	aCM
aCM__adult__dHF__D4__cells	18534	19359	21731	61	0	51	23	72	18	9	⋯	0	dHF	aCM
aCM__adult__healthy__D2__nuclei	1227	26	1287	24	0	9	8	14	89	4	⋯	0	healthy	aCM
aCM__adult__healthy__D3__cells	66	2	1	0	0	2	0	0	4	0	⋯	0	healthy	aCM
aCM__adult__healthy__D3__nuclei	611	15	577	19	0	8	2	2	20	6	⋯	0	healthy	aCM
aCM__adult__healthy__D4__cells	84	16	11	0	0	0	0	0	5	0	⋯	0	healthy	aCM
aCM__adult__healthy__D4__nuclei	1019	134	637	7	0	22	0	32	129	8	⋯	0	healthy	aCM
aCM__adult__healthy__D5__cells	260	2	17	0	0	0	0	1	28	0	⋯	0	healthy	aCM
aCM__adult__healthy__D5__nuclei	512	20	416	15	0	13	2	5	115	6	⋯	0	healthy	aCM
aCM__adult__healthy__D6__cells	164	2	53	0	0	0	0	0	5	0	⋯	0	healthy	aCM
aCM__adult__healthy__D6__nuclei	607	8	510	16	0	8	1	6	295	7	⋯	0	healthy	aCM
aCM__adult__healthy__D7__cells	316	2	53	0	0	3	0	0	2	0	⋯	0	healthy	aCM
aCM__adult__healthy__D7__nuclei	754	27	564	19	0	15	5	26	317	5	⋯	0	healthy	aCM
aCM__adult__healthy__N10__cells	23886	11323	19378	108	81	56	5	19	109	8	⋯	0	healthy	aCM
aCM__adult__healthy__N11__cells	22531	6932	29677	183	3	44	13	14	53	79	⋯	6	healthy	aCM
aCM__adult__healthy__N12__cells	34588	9781	60226	281	2	105	38	35	159	11	⋯	1	healthy	aCM
aCM__adult__healthy__N13__cells	14462	3950	26520	122	0	187	39	34	161	8	⋯	3	healthy	aCM
aCM__adult__healthy__N14__cells	61379	16024	70188	130	97	84	0	18	467	9	⋯	0	healthy	aCM
aCM__adult__healthy__N6__cells	15967	2737	27779	44	0	38	3	23	80	43	⋯	1	healthy	aCM
aCM__adult__healthy__N7__cells	12169	4991	24524	92	1	45	19	12	46	34	⋯	0	healthy	aCM
aCM__adult__healthy__N8__cells	8681	2235	15262	50	0	69	17	9	33	15	⋯	6	healthy	aCM
aCM__adult__healthy__N9__cells	24308	6524	48168	255	4	96	19	39	74	24	⋯	2	healthy	aCM
aCM__adult__healthy__P1__nuclei	239	167	33	2	0	5	0	1	1	5	⋯	0	healthy	aCM
aCM__adult__healthy__P17__nuclei	206	241	190	0	0	3	2	0	1	4	⋯	0	healthy	aCM
aCM__adult__healthy__P7__nuclei	20	13	3	1	0	0	0	0	0	0	⋯	0	healthy	aCM
aCM__adult__healthy__P8__nuclei	301	348	255	1	0	5	3	4	0	3	⋯	0	healthy	aCM
aCM__adult__myocardial infarction_BZ__P12__nuclei	56	51	3	0	0	2	0	0	0	0	⋯	0	BZ	aCM
aCM__adult__myocardial infarction_BZ__P2__nuclei	1134	305	977	1	0	12	0	11	6	7	⋯	0	BZ	aCM
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱	⋮	⋮	⋮
vCM__fetal__healthy__10X127__cells	2494	2224	114	6	0	4	11	0	65	23	⋯	0	healthy	vCM
vCM__fetal__healthy__10X142__cells	3636	1796	679	36	0	11	6	3	111	42	⋯	0	healthy	vCM
vCM__fetal__healthy__10X151__cells	1729	300	70	7	0	3	1	0	157	29	⋯	0	healthy	vCM
vCM__fetal__healthy__10X158__cells	857	751	141	9	1	2	5	2	60	19	⋯	0	healthy	vCM
vCM__fetal__healthy__10X166__cells	4444	3284	226	4	0	6	10	6	186	29	⋯	0	healthy	vCM
vCM__fetal__healthy__10X246__cells	7557	8149	481	32	0	22	11	0	122	66	⋯	0	healthy	vCM
vCM__fetal__healthy__10X253__cells	4934	4651	201	14	0	12	12	8	107	40	⋯	0	healthy	vCM
vCM__fetal__healthy__10X256__cells	9601	7962	735	53	19	33	29	4	280	100	⋯	0	healthy	vCM
vCM__fetal__healthy__10X263__cells	4272	4256	1034	69	0	29	10	2	370	78	⋯	0	healthy	vCM
vCM__fetal__healthy__10X274__cells	3209	2104	99	7	0	10	12	1	138	53	⋯	0	healthy	vCM
vCM__fetal__healthy__10X285__cells	3342	1315	21	3	0	3	1	0	91	38	⋯	0	healthy	vCM
vCM__fetal__healthy__10X289__cells	1819	1186	77	0	0	6	8	1	11	17	⋯	0	healthy	vCM
vCM__fetal__healthy__10X299__cells	3713	2670	301	17	0	8	14	0	65	44	⋯	0	healthy	vCM
vCM__fetal__healthy__10X300__cells	1352	655	151	13	2	2	2	0	42	21	⋯	0	healthy	vCM
vCM__fetal__healthy__10X303__cells	3074	1563	587	7	4	3	20	9	204	46	⋯	0	healthy	vCM
vCM__fetal__healthy__11W1D__cells	7236	6132	355	19	0	17	16	3	657	118	⋯	0	healthy	vCM
vCM__fetal__healthy__11W4D__cells	6541	6103	284	5	0	8	17	0	164	81	⋯	0	healthy	vCM
vCM__fetal__healthy__13W3D__cells	7490	6678	260	9	0	10	23	1	142	78	⋯	0	healthy	vCM
vCM__fetal__healthy__15W0D__cells	8361	6434	91	8	0	21	16	3	232	50	⋯	0	healthy	vCM

	TNNT2	MYH7	MYH6	NR2F2	SHOX2	VSNL1	HCN4	MYH11	ACTA2	RGS5	⋯	ECE2	disease_subcategory	celltype
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	⋯	<dbl>	<chr>	<fct>
vCM__fetal__healthy__15W2D__cells	7927	5282	96	4	0	17	10	2	191	64	⋯	0	healthy	vCM
vCM__fetal__healthy__alexsc__cells	3835	3948	54	10	0	2	16	0	43	54	⋯	0	healthy	vCM
vCM__fetal__healthy__BRC2251__cells	4897	3168	154	29	7	7	13	2	90	53	⋯	0	healthy	vCM
vCM__fetal__healthy__BRC2252__cells	4456	2730	525	102	35	21	11	5	132	36	⋯	0	healthy	vCM
vCM__fetal__healthy__BRC2256__cells	6650	2951	175	18	2	8	11	3	385	50	⋯	0	healthy	vCM
vCM__fetal__healthy__BRC2260__cells	5179	2639	45	6	0	8	13	5	180	51	⋯	0	healthy	vCM
vCM__fetal__healthy__BRC2262__cells	4359	3327	26	4	0	2	15	3	38	24	⋯	0	healthy	vCM
vCM__fetal__healthy__BRC2263__cells	4992	2800	484	24	10	5	23	1	270	80	⋯	0	healthy	vCM
vCM__fetal__healthy__F8W4D__nuclei	1076	1267	531	31	4	9	29	11	40	31	⋯	0	healthy	vCM
vCM__fetal__healthy__M10W5D__nuclei	1018	2209	309	9	1	9	8	11	41	37	⋯	0	healthy	vCM
vCM__fetal__healthy__M9W0D__nuclei	730	1342	257	7	0	9	9	3	10	14	⋯	0	healthy	vCM

In [355…

```
# saveRDS(bulk_metadata,"bulk_metadata_20.rds")
bulk_metadata = readRDS("bulk_metadata_20.rds")
```

In [356…

```
# Check covariates across design, removing celltypes where no suitable combinations exist
# subset to adult only:
adult_mask = bulk_metadata[[1]]$stage == "adult"

# remove pseudobulk samples with fewer than 10 cells
cell_number_mask = bulk_metadata[[1]]$cell_number >= 10

DESeq_mask = adult_mask & cell_number_mask

conditioncheckTable = table(
    bulk_metadata[[1]]$celltype[DESeq_mask],
    paste(bulk_metadata[[1]]$dataset, bulk_metadata[[1]]$cell_or_nuclei,
          bulk_metadata[[1]]$disease_subcategory,sep="")[DESeq_mask]
)
```

In [44]:

```
conditioncheckTable
```

	KnightSchrijver2022cellshealthy
vCM	2
aCM	1
CCS	0
SMC	5
Pericytes	5
EC_Valves_Pericytes	5
Endocardium	4
EC_Venous	5
EC_Arterial	5
EC_Capillary	5
EC_Lymphatic	2
EC_Mixed	1
Epicardium	2
Fibroblasts	5
Neurons_Mixed	0
Neurons_SchwannGlialCells	4
Leukocytes	5
Adipocytes	0
PlateletsErythrocytes	0
unknown	2

	KnightSchrijver2022nucleihealthy	Kuppe2022nucleiBZ
vCM	6	3
aCM	6	2
CCS	0	0
SMC	6	3
Pericytes	6	3
EC_Valves_Pericytes	6	2
Endocardium	6	0
EC_Venous	6	3
EC_Arterial	6	3
EC_Capillary	6	3
EC_Lymphatic	6	1
EC_Mixed	6	3
Epicardium	6	0
Fibroblasts	6	3
Neurons_Mixed	5	3
Neurons_SchwannGlialCells	6	3
Leukocytes	6	3
Adipocytes	6	1
PlateletsErythrocytes	0	0
unknown	6	3

	Kuppe2022nucleiFZ	Kuppe2022nucleihealthy
vCM	6	4
aCM	3	3
CCS	0	0
SMC	6	4
Pericytes	6	4
EC_Valves_Pericytes	5	4
Endocardium	0	0
EC_Venous	6	4
EC_Arterial	6	4
EC_Capillary	6	4
EC_Lymphatic	6	1
EC_Mixed	6	4
Epicardium	0	0
Fibroblasts	6	4
Neurons_Mixed	5	4
Neurons_SchwannGlialCells	6	4
Leukocytes	6	4
Adipocytes	3	0
PlateletsErythrocytes	0	0
unknown	6	4

	Kuppe2022nucleiIZ	Kuppe2022nucleiRZ
vCM	7	5
aCM	0	4
CCS	0	0
SMC	6	5
Pericytes	6	5
EC_Valves_Pericytes	7	4
Endocardium	0	0
EC_Venous	5	5
EC_Arterial	4	5
EC_Capillary	4	5
EC_Lymphatic	5	3
EC_Mixed	6	5
Epicardium	0	0
Fibroblasts	7	5
Neurons_Mixed	4	5
Neurons_SchwannGlialCells	0	5
Leukocytes	7	5
Adipocytes	3	2
PlateletsErythrocytes	0	0
unknown	7	5

	Wang2020cellscHF	Wang2020cellsdHF
vCM	2	4
aCM	1	3
CCS	2	2
SMC	2	4
Pericytes	0	0
EC_Valves_Pericytes	2	3
Endocardium	0	0
EC_Venous	2	3
EC_Arterial	2	2
EC_Capillary	2	2

```
EC_Lymphatic                             0                0
EC_Mixed                                 2                3
Epicardium                               0                0
Fibroblasts                              2                3
Neurons_Mixed                            0                0
Neurons_SchwannGlialCells                0                0
Leukocytes                               2                3
Adipocytes                               0                0
PlateletsErythrocytes                    0                0
unknown                                  0                0

                          Wang2020cellshealthy
vCM                                         13
aCM                                          9
CCS                                          9
SMC                                         10
Pericytes                                    0
EC_Valves_Pericytes                          6
Endocardium                                  0
EC_Venous                                    8
EC_Arterial                                  6
EC_Capillary                                 8
EC_Lymphatic                                 1
EC_Mixed                                    11
Epicardium                                   0
Fibroblasts                                 13
Neurons_Mixed                                0
Neurons_SchwannGlialCells                    0
Leukocytes                                   7
Adipocytes                                   0
PlateletsErythrocytes                        0
unknown                                      3
```

In [21]:

```
# from above, certain cell types are not present in all conditions so we will remove from analysis..
# unfortunately Endocardium and Epicardium is not really there in the disease dataset(s)

celltypes2ignore = c(
        "CCS",
        "Neurons_Mixed",
        "Endocardium",
        "Epicardium",
        "PlateletsErythrocytes",
        "Adipocytes",
        "unknown"
        )
```

In [22]:

```
celltypes2test = levels(bulk_metadata[[1]]$celltype)
```

In [23]:

```
celltypes2test  = celltypes2test[!celltypes2test %in% celltypes2ignore]
```

In [24]:

```
celltypes2test
```

'vCM' · 'aCM' · 'SMC' · 'Pericytes' · 'EC_Valves_Pericytes' · 'EC_Venous' · 'EC_Arterial' · 'EC_Capillary' · 'EC_Lymphatic' · 'EC_Mixed' · 'Fibroblasts' · 'Neurons_SchwannGlialCells' · 'Leukocytes'

Run DESeq2

In [25]:

```
# load DESeq2 library
library(DESeq2)
```

```
Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: ‘generics’


The following objects are masked from ‘package:base’:

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union



Attaching package: ‘BiocGenerics’


The following object is masked from ‘package:limma’:

    plotMA


The following objects are masked from ‘package:stats’:

    IQR, mad, sd, var, xtabs


The following objects are masked from ‘package:base’:

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min



Attaching package: ‘S4Vectors’


The following objects are masked from ‘package:Matrix’:

    expand, unname


The following object is masked from ‘package:utils’:

    findMatches


The following objects are masked from ‘package:base’:

    expand.grid, I, unname


Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: GenomeInfoDb

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics


Attaching package: ‘MatrixGenerics’


The following objects are masked from ‘package:matrixStats’:

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars


Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
```

```
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: ‘Biobase’


The following object is masked from ‘package:MatrixGenerics’:

    rowMedians


The following objects are masked from ‘package:matrixStats’:

    anyMissing, rowMedians
```

In [50]:
```
# We have run DESeq2 in another script here, containing the full results object.
```

Global model and comparisons

In [51]:
```
# Essentially, to calculate dispersions we really needed the entire dataset
# we have some objects that we can load in to continue the analysis.

# dispersions, calculated on the entire set of genes (bulk_dataFUll)
global_dispersions = readRDS("global_dispersions.rds")

# results, calculated using nbinomlrt() from DESeq2. This object was then subset to only genes_EDN genes.
global_results = readRDS("global_results.rds")
```

In [52]:
```
# We COULD run DESeq2 again, using the pre-calculated dispersions.
# However, the p values appeared to be different.
# We will stick with the subset entire results object for robustness.

# The script below injects the pre-calculated dispersions and re-runs DESeq2
# I will do this to calculate the results object, where we can extract the normalised counts.
# One may examine the results to see that the p values differ.
```

In [53]:
```
# results list object
DESeq_results_global = list()
```

In [75]:
```
# DESeq run for global test in all celltypes
for(i in 1:num_iterations){
    iteration_results = mclapply(
                1:length(celltypes2test),
                function(j){
                    celltype=celltypes2test[j]

                    # First create the DESeq object, adult and celltype only:

                        # subset to adult only:
                        adult_mask = bulk_metadata[[i]]$stage == "adult"

                        # subset to celltype only:
                        celltype_mask = bulk_metadata[[i]]$celltype == celltype

                        # remove pseudobulk samples with fewer than 10 cells
                        cell_number_mask = bulk_metadata[[i]]$cell_number >= 10

                        # total mask
                        DESeq_mask = adult_mask & celltype_mask & cell_number_mask

                        DESeq_metadata = bulk_metadata[[i]][DESeq_mask,]
                        DESeq_data = bulk_data[[i]][,DESeq_mask]
                        DESeq_size.factors = bulk_size.factors[[i]][DESeq_mask]

                        # size-factors
                        # We need to centre the size-factors in each bulk sample around 1.
                            DESeq_size.factors = DESeq_size.factors / exp(mean(log(DESeq_size.factors)))


                    # Set up the DESeq2 object with new group as default design
                        dds <- DESeqDataSetFromMatrix(countData = DESeq_data,
                                                  colData = DESeq_metadata,
                                                  design = ~ cell_number_log + cell_or_nuclei + dataset + disease_subcategory

                        # add in the pre-calculated size.factors from the bulk samples
                        sizeFactors(dds) <- DESeq_size.factors

                        # Bypass the 'estimateDispersions' function, adding in pre-calculated global dispersions data
                        mcols(dds) <- cbind(mcols(dds),global_dispersions[[i]][[j]][genes_EDN,])

                        # re level all to healthy
                        dds$disease_subcategory <- relevel(dds$disease_subcategory, "healthy")

                        dds_lrt <- nbinomLRT(dds,
                                     full = ~ cell_number_log + cell_or_nuclei + dataset + disease_subcategory,
                                     reduced = ~ cell_number_log + cell_or_nuclei + dataset)

                    # results from the LRT
                        res <- results(dds_lrt)
                        LRT_pvalue = res$padj

                        # run DESeq with wald test for between disease cat comparisons
                            #dds_wald <- DESeq(dds)

                            # Get the unique conditions
```

```
                                conditions <- as.character(unique(dds_lrt$disease_subcategory))

                            # Generate all unique combinations (pairs)
                            # This creates a matrix where each column is a pair
                                condition_pairs <- combn(conditions, 2)

                            # Create a list to store the results
                                all_results <- list()

                                # Comparisons Loop through the pairs and run 'results()'
                                    for(k in 1:ncol(condition_pairs)) {
                                        target <- condition_pairs[1, k]
                                        ref <- condition_pairs[2, k]

                                        res_name <- paste0(target, "_vs_", ref)

                                        # Extract results for this specific contrast
                                        all_results[[res_name]] <- results(
                                            dds_lrt,
                                            contrast = c("disease_subcategory", target, ref),
                                            test="Wald"
                                        )
                                    }

                    # add LRT to results
                        for(k in 1:length(all_results)){
                            all_results[[k]][,"LRT_padj"] = LRT_pvalue
                            all_results[[k]] = all_results[[k]][genes_EDN,]
                        }
                    return(
                        list(dds, dds_lrt, all_results)
                    )
                },mc.cores=7
            )
        DESeq_results_global[[i]] = iteration_results
}
```

In [74]:
```
# we can add the global results from the full results object
for(i in 1:length(DESeq_results_global)){
    for(j in 1:length(DESeq_results_global[[i]])){
        DESeq_results_global[[i]][[j]][[4]] = global_results[[i]][[j]]
        }
    }
```

In [26]:
```
#saveRDS(DESeq_results_global, "DESeq_results_global_20.rds")
DESeq_results_global = readRDS("DESeq_results_global_20.rds")
```

In [55]:
```
#########################################
####### Results for global model ########
#########################################
```

In [56]:
```
i = 1
j = 1 # (vCM)
# [[iteration]][[celltype]]  [[  1: DDS,  2: DDS_LRT, 3: ResultsComparisons(subsetGenes) 4: ResultsComparisons(FULL) ]][[comparison i
DESeq_results_global[[1]][[j]][[4]][[1]]["EDNRA",]
```

```
log2 fold change (MLE): disease_subcategory cHF vs dHF
Wald test p-value: disease_subcategory cHF vs dHF
DataFrame with 1 row and 7 columns
       baseMean log2FoldChange     lfcSE      stat    pvalue      padj
      <numeric>      <numeric> <numeric> <numeric> <numeric> <numeric>
EDNRA   24.7344      -0.148133   1.01967 -0.145276  0.884493  0.983206
       LRT_padj
      <numeric>
EDNRA  0.690989
```

In [58]:
```
# Get all unique conditions to generate results matrices
conditions <- gsub("^.*_", "", as.character(unique(metadata$disease_subcategory)))

# Generate all unique combinations (pairs)
# This creates a matrix where each column is a pair
condition_pairs <- combn(conditions, 2)
```

In [59]:
```
condition_pairs
```

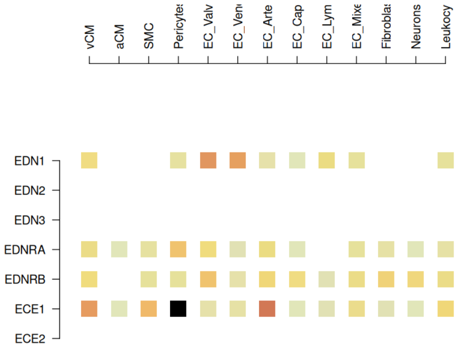
A matrix: 2 × 21 of type chr

healthy	healthy	healthy	healthy	healthy	healthy	cHF	cHF	cHF	cHF	⋯	dHF	dHF	dHF	dHF	FZ	FZ	FZ	BZ	BZ	IZ
cHF	dHF	FZ	BZ	IZ	RZ	dHF	FZ	BZ	IZ	⋯	FZ	BZ	IZ	RZ	BZ	IZ	RZ	IZ	RZ	RZ

In [60]:
```
# Loop through all DESeq2 LRT results and identify consistently differential genes
pdf("figures/panels/results/DESeq2/LRT_Iterations_global.pdf", width=20,height=13.2)
par(mfrow=c(4,5), mar=c(5,5,10,5), oma=c(0,0,2,0))
for(i in 1:num_iterations){
    LRT_celltype_matrix = matrix(NA, length(genes_EDN), length(celltypes2test))
    rownames(LRT_celltype_matrix) = genes_EDN
    colnames(LRT_celltype_matrix) = celltypes2test
        for(j in 1:length(celltypes2test)){
                LRT_celltype_matrix[,j] = round(DESeq_results_global[[i]][[j]][[4]][[1]][,"LRT_padj"],6)
        }
    LRT_celltype_matrix[LRT_celltype_matrix < 0.001] = 0.001
    #LRT_celltype_matrix[LRT_celltype_matrix > 0.1] = 1

    col2plot = matrix(plotcols(c(0.001, 1,-log10(LRT_celltype_matrix)), col=c(Heat, "black"))[-(1:2)],nrow(LRT_celltype_matrix),ncol
    matrix.plot(-log10(LRT_celltype_matrix), round.n=3, ylab="", xlab="", col=col2plot)

    if(i == 1){LRT_celltype_matrix_all = LRT_celltype_matrix}
    if(i >= 2){LRT_celltype_matrix_all = LRT_celltype_matrix_all+LRT_celltype_matrix}
```

```
    }
dev.off()
# check if model is accounting for technical noises:
options(repr.plot.width = 7)
options(repr.plot.height = 7)
matrix.plot(-log10(LRT_celltype_matrix), round.n=3, ylab="", xlab="", col=col2plot)
```

agg_record_1571020054: 2



In [66]:
```
# example of the pvalues on an iteration of the test / samples
LRT_celltype_matrix
```

A matrix: 7 × 13 of type dbl

	vCM	aCM	SMC	Pericytes	EC_Valves_Pericytes	EC_Venous	EC_Arterial	EC_Capillary	EC_Lymphatic	EC_Mixed	Fibroblasts	Neurons_Sc
EDN1	0.392213	NA	NA	0.664435	0.036307	0.043758	0.747758	0.947246	0.408858	0.587081	NA	
EDN2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
EDN3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
EDNRA	0.423385	0.988796	0.669741	0.122858	0.356034	0.890724	0.393346	0.968364	NA	0.614578	0.703510	
EDNRB	0.344932	NA	0.617060	0.573885	0.121163	0.731057	0.283221	0.384740	0.824310	0.429723	0.253210	
ECE1	0.039482	0.999830	0.092071	0.001185	0.764315	0.698335	0.017792	0.955667	0.905637	0.477963	0.928838	
ECE2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	

In [62]:
```
# Run through results, computing the fraction of significant results in each comparison:
significant_fraction_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
rownames(significant_fraction_matrix) = genes_EDN
colnames(significant_fraction_matrix) = celltypes2test
for(i in 1:num_iterations){
    significant_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
    LRT_celltype_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
    rownames(LRT_celltype_matrix) = genes_EDN
    colnames(LRT_celltype_matrix) = celltypes2test
        for(j in 1:length(celltypes2test)){
                LRT_celltype_matrix[,j] = round(DESeq_results_global[[i]][[j]][[4]][[1]][,"LRT_padj"],6)
        }
    significant_matrix[LRT_celltype_matrix < 0.05] = 1
    significant_matrix[LRT_celltype_matrix >= 0.05] = 0
    significant_fraction_matrix = significant_fraction_matrix + significant_matrix
    }
```

In [63]:
```
significant_fraction_matrix = significant_fraction_matrix / num_iterations * 100
significant_fraction_matrix
```

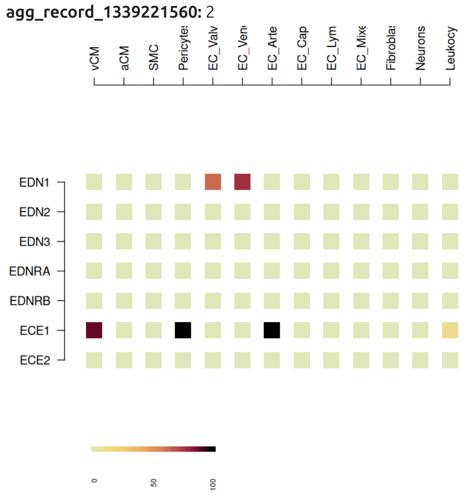
A matrix: 7 × 13 of type dbl

	vCM	aCM	SMC	Pericytes	EC_Valves_Pericytes	EC_Venous	EC_Arterial	EC_Capillary	EC_Lymphatic	EC_Mixed	Fibroblasts	Neurons_SchwannGlialCe
EDN1	0	0	0	0	65	80	0	0	0	0	0	
EDN2	0	0	0	0	0	0	0	0	0	0	0	
EDN3	0	0	0	0	0	0	0	0	0	0	0	
EDNRA	0	0	0	0	0	0	0	0	0	0	0	
EDNRB	0	0	0	0	0	0	0	0	0	0	0	
ECE1	90	0	0	100	0	0	100	0	0	0	0	
ECE2	0	0	0	0	0	0	0	0	0	0	0	

In [64]:
```
# plot % of below alpha global model iterations
pdf("figures/panels/results/DESeq2/LRTmatrix_global.pdf", width=5.5, height=6.3)
par(mar=c(1,5,20,5))
matrix.plot(significant_fraction_matrix, round.n=1, ylab="", xlab="", col=c(Heat, "black"))
dev.off()
matrix.plot(significant_fraction_matrix, round.n=1, col=c(Heat, "black"))
    points(seq(1,5, length=50), rep(-2, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:50, c(Heat, "black")))
    text(
      seq(1,5, length=3),
      rep(-3, 3),
```

```
    seq(signif(min(significant_fraction_matrix, na.rm=T),3), signif(max(significant_fraction_matrix,na.rm=T),3), length=3),
    xpd=NA,
    srt=90,
    adj=1,
    cex=0.6
  )
```

agg_record_1339221560: 2



In [ ]:

In [69]:
```
# Essentially, to calculate dispersions we really needed the entire dataset
# we have some objects that we can load in to continue the analysis.

# dispersions, calculated on the entire set of genes (bulk_dataFUll)
Kuppe2022_dispersions = readRDS("Kuppe2022_dispersions.rds")

# results, calculated using nbinomlrt() from DESeq2. This object was then subset to only genes_EDN genes.
Kuppe2022_results = readRDS("Kuppe2022_results.rds")
```

In [70]:
```
# results list object
DESeq_results_Kuppe2022 = list()
```

In [73]:
```
# DESeq run for Kuppe2022 test in all celltypes
for(i in 1:num_iterations){
    iteration_results = mclapply(
                1:length(celltypes2test),
                function(j){
                    celltype=celltypes2test[j]

                    # First create the DESeq object, adult and celltype only:

                        # subset to adult only:
                        adult_mask = bulk_metadata[[i]]$stage == "adult"

                        # subset to celltype only:
                        celltype_mask = bulk_metadata[[i]]$celltype == celltype

                        # remove pseudobulk samples with fewer than 10 cells
                        cell_number_mask = bulk_metadata[[i]]$cell_number >= 10

                        # subset to dataset only:
                        dataset_mask = bulk_metadata[[i]]$dataset == "Kuppe2022"

                        # remove pseudobulk samples with fewer than 10 cells
                        cell_number_mask = bulk_metadata[[i]]$cell_number >= 10

                        # total mask
                        DESeq_mask = adult_mask & celltype_mask & cell_number_mask & dataset_mask

                        DESeq_metadata = bulk_metadata[[i]][DESeq_mask,]
                        DESeq_data = bulk_data[[i]][,DESeq_mask]
                        DESeq_size.factors = bulk_size.factors[[i]][DESeq_mask]

                        # size-factors
                        # We need to centre the size-factors in each bulk sample around 1.
                            DESeq_size.factors = DESeq_size.factors / exp(mean(log(DESeq_size.factors)))


                    # Set up the DESeq2 object with new group as default design
                        dds <- DESeqDataSetFromMatrix(countData = DESeq_data,
                                                      colData = DESeq_metadata,
                                                      design = ~ cell_number_log + disease_subcategory)

                        # add in the pre-calculated size.factors from the bulk samples
                        sizeFactors(dds) <- DESeq_size.factors

                        # Bypass the 'estimateDispersions' function, adding in pre-calculated global dispersions data
                        mcols(dds) <- cbind(mcols(dds),Kuppe2022_dispersions[[i]][[j]][genes_EDN,])

                        # re level all to healthy
                        dds$disease_subcategory <- relevel(dds$disease_subcategory, "healthy")

                        dds_lrt <- nbinomLRT(dds,
```

```
                                              full = ~ cell_number_log + disease_subcategory,
                                              reduced = ~ cell_number_log)

                    # results from the LRT
                        res <- results(dds_lrt)
                        LRT_pvalue = res$padj

                        # run DESeq with wald test for between disease cat comparisons
                            #dds_wald <- DESeq(dds)

                            # Get the unique conditions
                                conditions <- as.character(unique(dds_lrt$disease_subcategory))

                            # Generate all unique combinations (pairs)
                            # This creates a matrix where each column is a pair
                                condition_pairs <- combn(conditions, 2)

                            # Create a list to store the results
                                all_results <- list()

                                # Comparisons Loop through the pairs and run 'results()'
                                    for(k in 1:ncol(condition_pairs)) {
                                        target <- condition_pairs[1, k]
                                        ref <- condition_pairs[2, k]

                                        res_name <- paste0(target, "_vs_", ref)

                                        # Extract results for this specific contrast
                                        all_results[[res_name]] <- results(
                                            dds_lrt,
                                            contrast = c("disease_subcategory", target, ref),
                                            test="Wald"
                                        )
                                    }

                    # add LRT to results
                        for(k in 1:length(all_results)){
                            all_results[[k]][,"LRT_padj"] = LRT_pvalue
                            all_results[[k]] = all_results[[k]][genes_EDN,]
                        }
                    return(
                        list(dds, dds_lrt, all_results)
                    )
                },mc.cores=7
            )
        DESeq_results_Kuppe2022[[i]] = iteration_results
}
```

```
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```

In [78]:
```
# we can add the global results from the full results object
for(i in 1:length(DESeq_results_Kuppe2022)){
    for(j in 1:length(DESeq_results_Kuppe2022[[i]])){
        DESeq_results_Kuppe2022[[i]][[j]][[4]] = Kuppe2022_results[[i]][[j]]
        }
    }
```

In [27]:
```
#saveRDS(DESeq_results_Kuppe2022, "DESeq_results_Kuppe2022_20.rds")
DESeq_results_Kuppe2022 = readRDS("DESeq_results_Kuppe2022_20.rds")
```

In [76]:
```
#########################################
####### Results for Kuppe2022 model ########
#########################################
```

In [85]:
```
i = 1
j = 1 # (vCM)
```

```
# [[iteration]][[celltype]]  [[  1: DDS,  2: DDS_LRT, 3: ResultsComparisons(subsetGenes) 4: ResultsComparisons(FULL) ]][[comparison i
DESeq_results_Kuppe2022[[1]][[j]][[4]][[1]]["EDNRA",]
```

```
log2 fold change (MLE): disease_subcategory healthy vs BZ
Wald test p-value: disease_subcategory healthy vs BZ
DataFrame with 1 row and 7 columns
       baseMean log2FoldChange     lfcSE      stat    pvalue      padj
      <numeric>      <numeric> <numeric> <numeric> <numeric> <numeric>
EDNRA   43.8482       0.728272  0.876374  0.831005  0.405971         1
        LRT_padj
       <numeric>
EDNRA  0.585081
```

In [86]:
```
# Get all unique conditions to generate results matrices
conditions <- gsub("^.*_", "", as.character(unique(metadata$disease_subcategory)))

# Generate all unique combinations (pairs)
# This creates a matrix where each column is a pair
condition_pairs <- combn(conditions, 2)
```

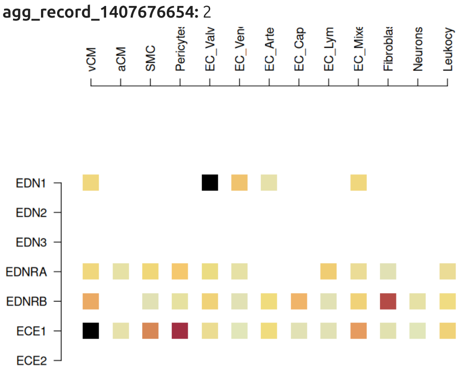
In [87]:
```
condition_pairs
```

A matrix: 2 × 21 of type chr

healthy	healthy	healthy	healthy	healthy	healthy	cHF	cHF	cHF	cHF	⋯	dHF	dHF	dHF	dHF	FZ	FZ	FZ	BZ	BZ	IZ
cHF	dHF	FZ	BZ	IZ	RZ	dHF	FZ	BZ	IZ	⋯	FZ	BZ	IZ	RZ	BZ	IZ	RZ	IZ	RZ	RZ

In [88]:
```
# Loop through all DESeq2 LRT results and identify consistently differential genes
pdf("figures/panels/results/DESeq2/LRT_Iterations_Kuppe2022.pdf", width=20,height=13.2)
par(mfrow=c(4,5), mar=c(5,5,10,5), oma=c(0,0,2,0))
for(i in 1:num_iterations){
    LRT_celltype_matrix = matrix(NA, length(genes_EDN), length(celltypes2test))
    rownames(LRT_celltype_matrix) = genes_EDN
    colnames(LRT_celltype_matrix) = celltypes2test
        for(j in 1:length(celltypes2test)){
            LRT_celltype_matrix[,j] = round(DESeq_results_Kuppe2022[[i]][[j]][[4]][[1]][,"LRT_padj"],6)
        }
    LRT_celltype_matrix[LRT_celltype_matrix < 0.001] = 0.001
    #LRT_celltype_matrix[LRT_celltype_matrix > 0.1] = 1

    col2plot = matrix(plotcols(c(0.001, 1,-log10(LRT_celltype_matrix)), col=c(Heat, "black"))[-(1:2)],nrow(LRT_celltype_matrix),ncol
    matrix.plot(-log10(LRT_celltype_matrix), round.n=3, ylab="", xlab="", col=col2plot)

    if(i == 1){LRT_celltype_matrix_all = LRT_celltype_matrix}
    if(i >= 2){LRT_celltype_matrix_all = LRT_celltype_matrix_all+LRT_celltype_matrix}
    }
dev.off()
# check if model is accounting for technical noises:
options(repr.plot.width = 7)
options(repr.plot.height = 7)
matrix.plot(-log10(LRT_celltype_matrix), round.n=3, ylab="", xlab="", col=col2plot)
```

agg_record_1407676654: 2



In [89]:
```
# example of the pvalues on an iteration of the test / samples
LRT_celltype_matrix
```

A matrix: 7 × 13 of type dbl

	vCM	aCM	SMC	Pericytes	EC_Valves_Pericytes	EC_Venous	EC_Arterial	EC_Capillary	EC_Lymphatic	EC_Mixed	Fibroblasts	Neurons_Sc
EDN1	0.320011	NA	NA	NA	0.001000	0.119384	0.749901	NA	NA	0.328479	NA	
EDN2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
EDN3	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	
EDNRA	0.319525	0.671206	0.267749	0.150034	0.405864	0.707600	NA	NA	0.200175	0.536684	0.924144	
EDNRB	0.055709	NA	0.900529	0.638972	0.240459	0.887282	0.341318	0.070611	0.825665	0.217958	0.006497	
ECE1	0.001000	0.752461	0.020924	0.004280	0.491459	0.993949	0.344823	0.854701	0.920928	0.038312	0.783605	
ECE2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	

In [90]:
```r
# Run through results, computing the fraction of significant results in each comparison:
significant_fraction_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
rownames(significant_fraction_matrix) = genes_EDN
colnames(significant_fraction_matrix) = celltypes2test
for(i in 1:num_iterations){
    significant_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
    LRT_celltype_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
    rownames(LRT_celltype_matrix) = genes_EDN
    colnames(LRT_celltype_matrix) = celltypes2test
        for(j in 1:length(celltypes2test)){
                LRT_celltype_matrix[,j] = round(DESeq_results_Kuppe2022[[i]][[j]][[4]][[1]][,"LRT_padj"],6)
        }
    significant_matrix[LRT_celltype_matrix < 0.05] = 1
    significant_matrix[LRT_celltype_matrix >= 0.05] = 0
    significant_fraction_matrix = significant_fraction_matrix + significant_matrix
    }
```

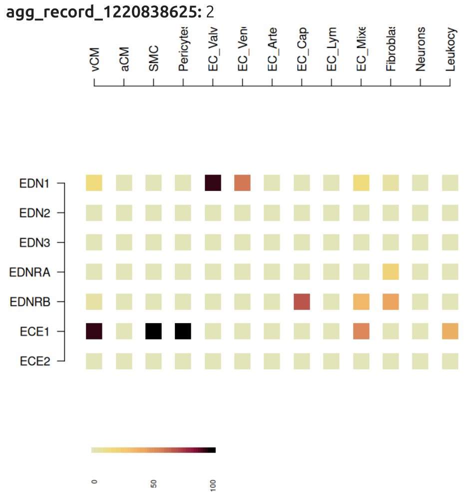
In [91]:
```r
significant_fraction_matrix = significant_fraction_matrix / num_iterations * 100
significant_fraction_matrix
```

A matrix: 7 × 13 of type dbl

	vCM	aCM	SMC	Pericytes	EC_Valves_Pericytes	EC_Venous	EC_Arterial	EC_Capillary	EC_Lymphatic	EC_Mixed	Fibroblasts	Neurons_SchwannGlialCe
EDN1	15	0	0	0	95	60	0	0	0	15	5	
EDN2	0	0	0	0	0	0	0	0	0	0	0	
EDN3	0	0	0	0	0	0	0	0	0	0	0	
EDNRA	0	0	0	0	0	0	0	0	0	0	20	
EDNRB	5	0	0	0	0	0	0	70	0	35	45	
ECE1	95	0	100	100	0	0	0	0	0	55	0	
ECE2	0	0	0	0	0	0	0	0	0	0	0	

In [92]:
```r
# plot % of below alpha global model iterations
pdf("figures/panels/results/DESeq2/LRTmatrix_Kuppe2022.pdf", width=5.5, height=6.3)
par(mar=c(1,5,20,5))
matrix.plot(significant_fraction_matrix, round.n=1, ylab="", xlab="", col=c(Heat, "black"))
dev.off()
matrix.plot(significant_fraction_matrix, round.n=1, col=c(Heat, "black"))
    points(seq(1,5, length=50), rep(-2, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:50, c(Heat, "black")))
    text(
      seq(1,5, length=3),
      rep(-3, 3),
      seq(signif(min(significant_fraction_matrix, na.rm=T),3), signif(max(significant_fraction_matrix,na.rm=T),3), length=3),
      xpd=NA,
      srt=90,
      adj=1,
      cex=0.6
    )
```

agg_record_1220838625: 2



In [ ]:

In [93]:
```r
# Essentially, to calculate dispersions we really needed the entire dataset
# we have some objects that we can load in to continue the analysis.

# dispersions, calculated on the entire set of genes (bulk_dataFUll)
Wang2020_dispersions = readRDS("Wang2020_dispersions.rds")

# results, calculated using nbinomlrt() from DESeq2. This object was then subset to only genes_EDN genes.
Wang2020_results = readRDS("Wang2020_results.rds")
```

In [94]:
```r
# results list object
DESeq_results_Wang2020 = list()
```

In [28]:
```r
wang_ignore_celltypes = c("Pericytes", "EC_Lymphatic", "Neurons_SchwannGlialCells")
```

In [99]:
```r
# DESeq run for Wang2020 test in all celltypes
wang_ignore_celltypes = c("Pericytes", "EC_Lymphatic", "Neurons_SchwannGlialCells")
```

```r
for(i in 1:num_iterations){
    iteration_results = mclapply(
                1:length(celltypes2test),
                function(j){
                    celltype=celltypes2test[j]
                    if(!celltype %in% wang_ignore_celltypes){
                    # First create the DESeq object, adult and celltype only:

                        # subset to adult only:
                        adult_mask = bulk_metadata[[i]]$stage == "adult"

                        # subset to celltype only:
                        celltype_mask = bulk_metadata[[i]]$celltype == celltype

                        # remove pseudobulk samples with fewer than 10 cells
                        cell_number_mask = bulk_metadata[[i]]$cell_number >= 10

                        # subset to dataset only:
                        dataset_mask = bulk_metadata[[i]]$dataset == "Wang2020"

                        # remove pseudobulk samples with fewer than 10 cells
                        cell_number_mask = bulk_metadata[[i]]$cell_number >= 10

                        # total mask
                        DESeq_mask = adult_mask & celltype_mask & cell_number_mask & dataset_mask

                        DESeq_metadata = bulk_metadata[[i]][DESeq_mask,]
                        DESeq_data = bulk_data[[i]][,DESeq_mask]
                        DESeq_size.factors = bulk_size.factors[[i]][DESeq_mask]

                        # size-factors
                        # We need to centre the size-factors in each bulk sample around 1.
                            DESeq_size.factors = DESeq_size.factors / exp(mean(log(DESeq_size.factors)))


                    # Set up the DESeq2 object with new group as default design
                        dds <- DESeqDataSetFromMatrix(countData = DESeq_data,
                                                      colData = DESeq_metadata,
                                                      design = ~ cell_number_log + disease_subcategory)

                        # add in the pre-calculated size.factors from the bulk samples
                        sizeFactors(dds) <- DESeq_size.factors

                        # Bypass the 'estimateDispersions' function, adding in pre-calculated global dispersions data
                        mcols(dds) <- cbind(mcols(dds),Wang2020_dispersions[[i]][[j]][genes_EDN,])

                        # re level all to healthy
                        dds$disease_subcategory <- relevel(dds$disease_subcategory, "healthy")

                        dds_lrt <- nbinomLRT(dds,
                                           full = ~ cell_number_log + disease_subcategory,
                                           reduced = ~ cell_number_log)

                    # results from the LRT
                        res <- results(dds_lrt)
                        LRT_pvalue = res$padj

                        # run DESeq with wald test for between disease cat comparisons
                            #dds_wald <- DESeq(dds)

                            # Get the unique conditions
                                conditions <- as.character(unique(dds_lrt$disease_subcategory))

                            # Generate all unique combinations (pairs)
                            # This creates a matrix where each column is a pair
                                condition_pairs <- combn(conditions, 2)

                            # Create a list to store the results
                                all_results <- list()

                                # Comparisons Loop through the pairs and run 'results()'
                                    for(k in 1:ncol(condition_pairs)) {
                                        target <- condition_pairs[1, k]
                                        ref <- condition_pairs[2, k]

                                        res_name <- paste0(target, "_vs_", ref)

                                        # Extract results for this specific contrast
                                        all_results[[res_name]] <- results(
                                            dds_lrt,
                                            contrast = c("disease_subcategory", target, ref),
                                            test="Wald"
                                        )
                                    }

                    # add LRT to results
                        for(k in 1:length(all_results)){
                            all_results[[k]][,"LRT_padj"] = LRT_pvalue
                            all_results[[k]] = all_results[[k]][genes_EDN,]
                        }
                    return(
                        list(dds, dds_lrt, all_results)
                    )
                    } else {
                        return(
                        list(NULL, NULL, NULL)
                    )
                        }
```

```
            },mc.cores=7
        )
        DESeq_results_Wang2020[[i]] = iteration_results
}
```

```
R_zmq_msg_send errno: 4 strerror: Interrupted system call
R_zmq_msg_send errno: 4 strerror: Interrupted system call
R_zmq_msg_send errno: 4 strerror: Interrupted system call
R_zmq_msg_send errno: 4 strerror: Interrupted system call
```

In [100…

```
# we can add the global results from the full results object
for(i in 1:length(DESeq_results_Wang2020)){
    for(j in 1:length(DESeq_results_Wang2020[[i]])){
        DESeq_results_Wang2020[[i]][[j]][[4]] = Wang2020_results[[i]][[j]]
        }
    }
```

In [29]:

```
#saveRDS(DESeq_results_Wang2020, "DESeq_results_Wang2020_20.rds")
DESeq_results_Wang2020 = readRDS("DESeq_results_Wang2020_20.rds")
```

In [103…

```
#########################################
####### Results for Wang2020 model ########
#########################################
```

In [108…

```
i = 1
j = 1 # (vCM)
# [[iteration]][[celltype]]  [[  1: DDS,  2: DDS_LRT, 3: ResultsComparisons(subsetGenes) 4: ResultsComparisons(FULL) ]][[comparison
DESeq_results_Wang2020[[1]][[j]][[4]][[1]]["EDNRA",]
```

```
log2 fold change (MLE): disease_subcategory cHF vs dHF
Wald test p-value: disease_subcategory cHF vs dHF
DataFrame with 1 row and 7 columns
       baseMean log2FoldChange     lfcSE      stat    pvalue      padj
      <numeric>      <numeric> <numeric> <numeric> <numeric> <numeric>
EDNRA   11.6894       -0.13936   1.11427 -0.125069  0.900469  0.980166
       LRT_padj
      <numeric>
EDNRA  0.952505
```

In [109…

```
# Get all unique conditions to generate results matrices
conditions <- gsub("^.*_", "", as.character(unique(metadata$disease_subcategory)))

# Generate all unique combinations (pairs)
# This creates a matrix where each column is a pair
condition_pairs <- combn(conditions, 2)
```

In [110…

```
condition_pairs
```

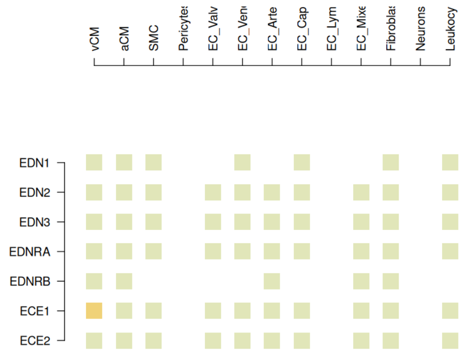
A matrix: 2 × 21 of type chr

healthy	healthy	healthy	healthy	healthy	healthy	cHF	cHF	cHF	cHF	⋯	dHF	dHF	dHF	dHF	FZ	FZ	FZ	BZ	BZ	IZ
cHF	dHF	FZ	BZ	IZ	RZ	dHF	FZ	BZ	IZ	⋯	FZ	BZ	IZ	RZ	BZ	IZ	RZ	IZ	RZ	RZ

In [113…

```
# Loop through all DESeq2 LRT results and identify consistently differential genes
pdf("figures/panels/results/DESeq2/LRT_Iterations_Wang2020.pdf", width=20,height=13.2)
par(mfrow=c(4,5), mar=c(5,5,10,5), oma=c(0,0,2,0))
for(i in 1:num_iterations){
    LRT_celltype_matrix = matrix(NA, length(genes_EDN), length(celltypes2test))
    rownames(LRT_celltype_matrix) = genes_EDN
    colnames(LRT_celltype_matrix) = celltypes2test
        for(j in 1:length(celltypes2test)){
                if(!is.null(DESeq_results_Wang2020[[i]][[j]][[3]][[1]])){
                LRT_celltype_matrix[,j] = round(DESeq_results_Wang2020[[i]][[j]][[3]][[1]][,"LRT_padj"],6)
                    } else {
                LRT_celltype_matrix[,j] = NA
                    }
        }
    LRT_celltype_matrix[LRT_celltype_matrix < 0.001] = 0.001
    #LRT_celltype_matrix[LRT_celltype_matrix > 0.1] = 1

    col2plot = matrix(plotcols(c(0.001, 1,-log10(LRT_celltype_matrix)), col=c(Heat, "black"))[-(1:2)],nrow(LRT_celltype_matrix),ncol
    matrix.plot(-log10(LRT_celltype_matrix), round.n=3, ylab="", xlab="", col=col2plot)

    if(i == 1){LRT_celltype_matrix_all = LRT_celltype_matrix}
    if(i >= 2){LRT_celltype_matrix_all = LRT_celltype_matrix_all+LRT_celltype_matrix}
    }
dev.off()
options(repr.plot.width = 7)
options(repr.plot.height = 7)
matrix.plot(-log10(LRT_celltype_matrix), round.n=3, ylab="", xlab="", col=col2plot)
```

agg_record_1177365135: 2



In [114…

```
# example of the pvalues on an iteration of the test / samples
LRT_celltype_matrix
```

A matrix: 7 × 13 of type dbl

	vCM	aCM	SMC	Pericytes	EC_Valves_Pericytes	EC_Venous	EC_Arterial	EC_Capillary	EC_Lymphatic	EC_Mixed	Fibroblasts	Neurons_Sc
EDN1	0.997398	0.999195	0.998236	NA	NA	0.995704	NA	0.998759	NA	NA	0.998525	
EDN2	0.997398	0.999195	0.998236	NA	1	0.995704	1	0.998759	NA	0.998458	0.998525	
EDN3	0.997398	0.999195	0.998236	NA	1	0.995704	1	0.998759	NA	0.998458	0.998525	
EDNRA	0.997398	0.999195	0.998236	NA	1	0.995704	1	0.998759	NA	0.998458	0.998525	
EDNRB	0.997398	0.999195	NA	NA	NA	NA	1	NA	NA	0.998458	0.998525	
ECE1	0.595068	0.999195	0.998236	NA	1	0.995704	1	0.998759	NA	0.998458	0.998525	
ECE2	0.997398	0.999195	0.998236	NA	1	0.995704	1	0.998759	NA	0.998458	0.998525	

In [116…

```
# Run through results, computing the fraction of significant results in each comparison:
significant_fraction_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
rownames(significant_fraction_matrix) = genes_EDN
colnames(significant_fraction_matrix) = celltypes2test
for(i in 1:num_iterations){
    significant_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
    LRT_celltype_matrix = matrix(0, length(genes_EDN), length(celltypes2test))
    rownames(LRT_celltype_matrix) = genes_EDN
    colnames(LRT_celltype_matrix) = celltypes2test
        for(j in 1:length(celltypes2test)){
            if(!is.null(DESeq_results_Wang2020[[i]][[j]][[3]][[1]])){
                LRT_celltype_matrix[,j] = round(DESeq_results_Wang2020[[i]][[j]][[4]][[1]][,"LRT_padj"],6)
                } else {
                    LRT_celltype_matrix[,j] = NA
                }
        }
    significant_matrix[LRT_celltype_matrix < 0.05] = 1
    significant_matrix[LRT_celltype_matrix >= 0.05] = 0
    significant_fraction_matrix = significant_fraction_matrix + significant_matrix
    }
```

In [117…

```
significant_fraction_matrix = significant_fraction_matrix / num_iterations * 100
significant_fraction_matrix
```

A matrix: 7 × 13 of type dbl

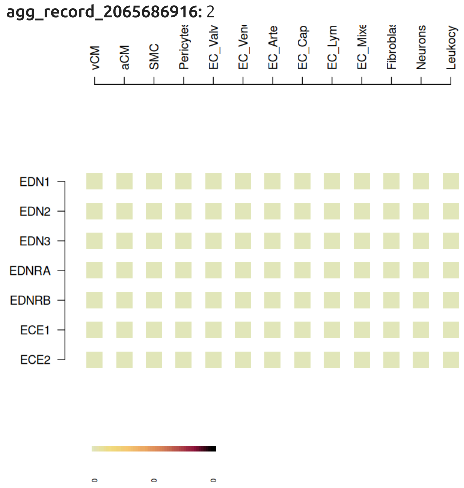
	vCM	aCM	SMC	Pericytes	EC_Valves_Pericytes	EC_Venous	EC_Arterial	EC_Capillary	EC_Lymphatic	EC_Mixed	Fibroblasts	Neurons_SchwannGlialCe
EDN1	0	0	0	0	0	0	0	0	0	0	0	
EDN2	0	0	0	0	0	0	0	0	0	0	0	
EDN3	0	0	0	0	0	0	0	0	0	0	0	
EDNRA	0	0	0	0	0	0	0	0	0	0	0	
EDNRB	0	0	0	0	0	0	0	0	0	0	0	
ECE1	0	0	0	0	0	0	0	0	0	0	0	
ECE2	0	0	0	0	0	0	0	0	0	0	0	

In [118…

```
# plot % of below alpha global model iterations
pdf("figures/panels/results/DESeq2/LRTmatrix_Wang2020.pdf", width=5.5, height=6.3)
par(mar=c(1,5,20,5))
matrix.plot(significant_fraction_matrix, round.n=1, ylab="", xlab="", col=c(Heat, "black"))
dev.off()
matrix.plot(significant_fraction_matrix, round.n=1, col=c(Heat, "black"))
    points(seq(1,5, length=50), rep(-2, 50), pch=15, cex = 1, xpd=NA, col=plotcols(1:50, c(Heat, "black")))
    text(
      seq(1,5, length=3),
      rep(-3, 3),
      seq(signif(min(significant_fraction_matrix, na.rm=T),3), signif(max(significant_fraction_matrix,na.rm=T),3), length=3),
      xpd=NA,
      srt=90,
      adj=1,
```

```
        cex=0.6
    )
```

agg_record_2065686916: 2



In [ ]:

In [ ]:

In [ ]:

In [814…

```
# global results dataframe
all_comparisons_global = data.frame()
for(gene in genes_EDN){
        celltype_comparisons = mclapply(1:length(celltypes2test), function(j) {
        celltype = celltypes2test[j]
        listOfComparisons = names(DESeq_results_global[[i]][[j]][[4]])

           gene_comparison_list = sapply(listOfComparisons, function(comparison) {
                    fraction_significant = sapply(1:20, function(i){
                                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                                }
                                              )
                 if(sum(is.na(fraction_significant)) == 20){
                    p_fraction = 0
                    p_max = 1
                    p_min = 1

                     } else {
                    p_fraction = sum(fraction_significant<0.05, na.rm=T)/20*100
                    p_min = signif(range(fraction_significant, na.rm=T),3)[1]
                    p_max = signif(range(fraction_significant, na.rm=T),3)[2]
                    }

                  fraction_significant_LRT = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"LRT_padj"]
                                }
                              )
                    if(sum(is.na(fraction_significant_LRT)) == 20){
                        lrt_fraction = 0
                        lrt_max = 1
                        lrt_min = 1

                         } else {
                        lrt_fraction = sum(fraction_significant_LRT<0.05, na.rm=T)/20*100
                        lrt_min = signif(range(fraction_significant_LRT, na.rm=T),3)[1]
                        lrt_max = signif(range(fraction_significant_LRT, na.rm=T),3)[2]
                        }


                    l2fc_all = sapply(1:20, function(i){
                                        DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                        }
                                      )
                    l2fc_mean = signif(mean(l2fc_all, na.rm=T),3)

                 return(
                       c(
                            gene,
                            celltype,
                            comparison,
                            lrt_fraction,
                            lrt_min,
                            lrt_max,
                            l2fc_mean,
                            p_fraction,
                            p_min,
                            p_max
                        )
                    )

                    }
                  )
            return(gene_comparison_list)
```

```
        }, mc.cores=7
    )
     gene_comparisons = data.frame()
            for(j in 1:length(celltypes2test)){
                gene_comparisons = rbind(gene_comparisons, t(celltype_comparisons[[j]]))
            }
        all_comparisons_global = rbind(all_comparisons_global, gene_comparisons)
    }

    colnames(all_comparisons_global) = c("gene", "celltype", "comparison", "LRT_percent_signif_iterations", "LRT_padj_min", "LRT_pad
    all_comparisons_global[,4] = as.numeric(all_comparisons_global[,4])
    all_comparisons_global[,5] = as.numeric(all_comparisons_global[,5])
    all_comparisons_global[,6] = as.numeric(all_comparisons_global[,6])
    all_comparisons_global[,7] = as.numeric(all_comparisons_global[,7])
    all_comparisons_global[,8] = as.numeric(all_comparisons_global[,8])
    all_comparisons_global[,9] = as.numeric(all_comparisons_global[,9])
    all_comparisons_global[,10] = as.numeric(all_comparisons_global[,10])
    all_comparisons_global$pairwise_mean_l2fc[is.na(all_comparisons_global$pairwise_mean_l2fc)] = 0
    rownames(all_comparisons_global) = 1:nrow(all_comparisons_global)
```

In [31]:
```
#write.table(all_comparisons_global, file="source_data/all_comparisons_global.csv", sep=",", col.names=T, row.names=T, quote=F)
all_comparisons_global = read.csv("source_data/all_comparisons_global.csv")
```

In [816…
```
# Kuppe2022 results dataframe
all_comparisons_Kuppe2022 = data.frame()
for(gene in genes_EDN){
        celltype_comparisons = mclapply(1:length(celltypes2test), function(j) {
        celltype = celltypes2test[j]
        listOfComparisons = names(DESeq_results_Kuppe2022[[i]][[j]][[4]])

           gene_comparison_list = sapply(listOfComparisons, function(comparison) {
                    fraction_significant = sapply(1:20, function(i){
                                                    DESeq_results_Kuppe2022[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                                    }
                                                  )
                 if(sum(is.na(fraction_significant)) == 20){
                    p_fraction = 0
                    p_max = 1
                    p_min = 1

                     } else {
                    p_fraction = sum(fraction_significant<0.05, na.rm=T)/20*100
                    p_min = signif(range(fraction_significant, na.rm=T),3)[1]
                    p_max = signif(range(fraction_significant, na.rm=T),3)[2]
                    }

                  fraction_significant_LRT = sapply(1:20, function(i){
                                DESeq_results_Kuppe2022[[i]][[j]][[4]][[comparison]][gene,"LRT_padj"]
                                }
                              )
                    if(sum(is.na(fraction_significant_LRT)) == 20){
                        lrt_fraction = 0
                        lrt_max = 1
                        lrt_min = 1

                         } else {
                        lrt_fraction = sum(fraction_significant_LRT<0.05, na.rm=T)/20*100
                        lrt_min = signif(range(fraction_significant_LRT, na.rm=T),3)[1]
                        lrt_max = signif(range(fraction_significant_LRT, na.rm=T),3)[2]
                        }


                    l2fc_all = sapply(1:20, function(i){
                                        DESeq_results_Kuppe2022[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                        }
                                      )
                    l2fc_mean = signif(mean(l2fc_all, na.rm=T),3)

                 return(
                       c(
                            gene,
                            celltype,
                            comparison,
                            lrt_fraction,
                            lrt_min,
                            lrt_max,
                            l2fc_mean,
                            p_fraction,
                            p_min,
                            p_max
                        )
                    )

                    }
                  )
            return(gene_comparison_list)
        }, mc.cores=7
    )
     gene_comparisons = data.frame()
            for(j in 1:length(celltypes2test)){
                gene_comparisons = rbind(gene_comparisons, t(celltype_comparisons[[j]]))
            }
        all_comparisons_Kuppe2022 = rbind(all_comparisons_Kuppe2022, gene_comparisons)
    }

    colnames(all_comparisons_Kuppe2022) = c("gene", "celltype", "comparison", "LRT_percent_signif_iterations", "LRT_padj_min", "LRT_
    all_comparisons_Kuppe2022[,4] = as.numeric(all_comparisons_Kuppe2022[,4])
    all_comparisons_Kuppe2022[,5] = as.numeric(all_comparisons_Kuppe2022[,5])
    all_comparisons_Kuppe2022[,6] = as.numeric(all_comparisons_Kuppe2022[,6])
    all_comparisons_Kuppe2022[,7] = as.numeric(all_comparisons_Kuppe2022[,7])
```

```
    all_comparisons_Kuppe2022[,8] = as.numeric(all_comparisons_Kuppe2022[,8])
    all_comparisons_Kuppe2022[,9] = as.numeric(all_comparisons_Kuppe2022[,9])
    all_comparisons_Kuppe2022[,10] = as.numeric(all_comparisons_Kuppe2022[,10])
    all_comparisons_Kuppe2022$pairwise_mean_l2fc[is.na(all_comparisons_Kuppe2022$pairwise_mean_l2fc)] = 0
    rownames(all_comparisons_Kuppe2022) = 1:nrow(all_comparisons_Kuppe2022)
```

In [32]:
```
#write.table(all_comparisons_Kuppe2022, file="source_data/all_comparisons_Kuppe2022.csv", sep=",", col.names=T, row.names=T, quote=F
all_comparisons_Kuppe2022 = read.csv("source_data/all_comparisons_Kuppe2022.csv")
```

In [67]:
```
# Wang2020 results dataframe
all_comparisons_Wang2020 = data.frame()
for(gene in genes_EDN){
        celltype_comparisons = mclapply(1:length(celltypes2test), function(j) {
        celltype = celltypes2test[j]

        if(!is.null(DESeq_results_Wang2020[[i]][[j]][[1]])){
        listOfComparisons = names(DESeq_results_Wang2020[[i]][[j]][[4]])

           gene_comparison_list = sapply(listOfComparisons, function(comparison) {
                    fraction_significant = sapply(1:20, function(i){
                                                DESeq_results_Wang2020[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                                }
                                              )
                 if(sum(is.na(fraction_significant)) == 20){
                    p_fraction = 0
                    p_max = 1
                    p_min = 1

                     } else {
                    p_fraction = sum(fraction_significant<0.05, na.rm=T)/20*100
                    p_min = signif(range(fraction_significant, na.rm=T),3)[1]
                    p_max = signif(range(fraction_significant, na.rm=T),3)[2]
                    }

                  fraction_significant_LRT = sapply(1:20, function(i){
                                DESeq_results_Wang2020[[i]][[j]][[4]][[comparison]][gene,"LRT_padj"]
                                }
                              )
                    if(sum(is.na(fraction_significant_LRT)) == 20){
                        lrt_fraction = 0
                        lrt_max = 1
                        lrt_min = 1

                         } else {
                        lrt_fraction = sum(fraction_significant_LRT<0.05, na.rm=T)/20*100
                        lrt_min = signif(range(fraction_significant_LRT, na.rm=T),3)[1]
                        lrt_max = signif(range(fraction_significant_LRT, na.rm=T),3)[2]
                        }


                    l2fc_all = sapply(1:20, function(i){
                                        DESeq_results_Wang2020[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                        }
                                      )
                    l2fc_mean = signif(mean(l2fc_all, na.rm=T),3)

                   return(
                       c(
                            gene,
                            celltype,
                            comparison,
                            lrt_fraction,
                            lrt_min,
                            lrt_max,
                            l2fc_mean,
                            p_fraction,
                            p_min,
                            p_max
                        )
                    )
                    }
                  )
            return(gene_comparison_list)
            } else {
                       return(matrix(NA,10,0))
                    }
        }, mc.cores=7
    )
         gene_comparisons = data.frame()
                for(j in 1:length(celltypes2test)){
                        if(!is.null(DESeq_results_Wang2020[[1]][[j]])){
                    gene_comparisons = rbind(gene_comparisons, t(celltype_comparisons[[j]]))
                            } else {
                    gene_comparisons = gene_comparisons
                            }
                }
        all_comparisons_Wang2020 = rbind(all_comparisons_Wang2020, gene_comparisons)
    }

    colnames(all_comparisons_Wang2020) = c("gene", "celltype", "comparison", "LRT_percent_signif_iterations", "LRT_padj_min", "LRT_pa
    all_comparisons_Wang2020[,4] = as.numeric(all_comparisons_Wang2020[,4])
    all_comparisons_Wang2020[,5] = as.numeric(all_comparisons_Wang2020[,5])
    all_comparisons_Wang2020[,6] = as.numeric(all_comparisons_Wang2020[,6])
    all_comparisons_Wang2020[,7] = as.numeric(all_comparisons_Wang2020[,7])
    all_comparisons_Wang2020[,8] = as.numeric(all_comparisons_Wang2020[,8])
    all_comparisons_Wang2020[,9] = as.numeric(all_comparisons_Wang2020[,9])
    all_comparisons_Wang2020[,10] = as.numeric(all_comparisons_Wang2020[,10])
    all_comparisons_Wang2020$pairwise_mean_l2fc[is.na(all_comparisons_Wang2020$pairwise_mean_l2fc)] = 0
    rownames(all_comparisons_Wang2020) = 1:nrow(all_comparisons_Wang2020)
```

In [69]:
```
#write.table(all_comparisons_Wang2020, file="source_data/all_comparisons_Wang2020.csv", sep=",", col.names=T, row.names=T, quote=F)
all_comparisons_Wang2020 = read.csv("source_data/all_comparisons_Wang2020.csv")
```

In [ ]:
```
### Looking at the results of pairwise...
```

In [292…
```
significant_comparisons =
    all_comparisons_global$LRT_percent_signif_iteration >= 70 &
    all_comparisons_global$pairwise_percent_signif_iteration >= 50 &
    abs(all_comparisons_global$pairwise_mean_l2fc) > 1
```

In [293…
```
all_comparisons_global[significant_comparisons,]
```

A data.frame: 6 × 10

	gene	celltype	comparison	LRT_percent_signif_iterations	LRT_padj_min	LRT_padj_max	pairwise_mean_l2fc	pairwise_percent_signif_iterations	pairw
	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<int>	
97	EDN1	EC_Venous	dHF_vs_FZ	80	1.66e-03	0.18600	-5.02	50	
98	EDN1	EC_Venous	dHF_vs_IZ	80	1.66e-03	0.18600	-6.00	100	
1210	ECE1	Pericytes	healthy_vs_IZ	100	9.66e-07	0.00144	-1.54	100	
1213	ECE1	Pericytes	BZ_vs_IZ	100	9.66e-07	0.00144	-1.54	85	
1215	ECE1	Pericytes	FZ_vs_IZ	100	9.66e-07	0.00144	-1.26	95	
1217	ECE1	Pericytes	IZ_vs_RZ	100	9.66e-07	0.00144	1.96	100	

In [294…
```
significant_comparisons =
    all_comparisons_Kuppe2022$LRT_percent_signif_iteration >= 70 &
    all_comparisons_Kuppe2022$pairwise_percent_signif_iteration >= 50 &
    abs(all_comparisons_Kuppe2022$pairwise_mean_l2fc) > 1
```

In [295…
```
all_comparisons_Kuppe2022[significant_comparisons,]
```

A data.frame: 13 × 10

	gene	celltype	comparison	LRT_percent_signif_iterations	LRT_padj_min	LRT_padj_max	pairwise_mean_l2fc	pairwise_percent_signif_iteratio
	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<in
38	EDN1	EC_Valves_Pericytes	healthy_vs_FZ	95	1.83e-05	0.05870	-3.96	7
44	EDN1	EC_Valves_Pericytes	FZ_vs_IZ	95	1.83e-05	0.05870	2.20	5
556	EDNRB	EC_Capillary	healthy_vs_FZ	70	3.21e-04	0.13100	-1.92	6
612	ECE1	vCM	healthy_vs_FZ	95	1.79e-07	0.05380	-1.28	5
613	ECE1	vCM	healthy_vs_IZ	95	1.79e-07	0.05380	-1.68	10
616	ECE1	vCM	BZ_vs_IZ	95	1.79e-07	0.05380	-1.22	6
619	ECE1	vCM	FZ_vs_RZ	95	1.79e-07	0.05380	1.13	5
620	ECE1	vCM	IZ_vs_RZ	95	1.79e-07	0.05380	1.53	10
636	ECE1	SMC	IZ_vs_RZ	100	1.25e-02	0.03470	1.85	10
639	ECE1	Pericytes	healthy_vs_IZ	100	4.49e-06	0.00428	-1.54	9
642	ECE1	Pericytes	BZ_vs_IZ	100	4.49e-06	0.00428	-1.54	6
644	ECE1	Pericytes	FZ_vs_IZ	100	4.49e-06	0.00428	-1.26	8
646	ECE1	Pericytes	IZ_vs_RZ	100	4.49e-06	0.00428	1.96	10

In [296…
```
all_comparisons_Kuppe2022[all_comparisons_Kuppe2022$celltype=="EC_Capillary" & all_comparisons_Kuppe2022$gene=="EDNRB",]
```

A data.frame: 10 × 10

	gene	celltype	comparison	LRT_percent_signif_iterations	LRT_padj_min	LRT_padj_max	pairwise_mean_l2fc	pairwise_percent_signif_iterations	pai
	<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<int>	
555	EDNRB	EC_Capillary	healthy_vs_BZ	70	0.000321	0.131	-0.1800	0	
556	EDNRB	EC_Capillary	healthy_vs_FZ	70	0.000321	0.131	-1.9200	65	
557	EDNRB	EC_Capillary	healthy_vs_IZ	70	0.000321	0.131	-0.0950	0	
558	EDNRB	EC_Capillary	healthy_vs_RZ	70	0.000321	0.131	-0.7820	0	
559	EDNRB	EC_Capillary	BZ_vs_FZ	70	0.000321	0.131	-1.7400	5	
560	EDNRB	EC_Capillary	BZ_vs_IZ	70	0.000321	0.131	0.0851	0	
561	EDNRB	EC_Capillary	BZ_vs_RZ	70	0.000321	0.131	-0.6020	0	
562	EDNRB	EC_Capillary	FZ_vs_IZ	70	0.000321	0.131	1.8200	0	
563	EDNRB	EC_Capillary	FZ_vs_RZ	70	0.000321	0.131	1.1300	0	
564	EDNRB	EC_Capillary	IZ_vs_RZ	70	0.000321	0.131	-0.6870	0	

In [297…
```
significant_comparisons =
    all_comparisons_Wang2020$LRT_percent_signif_iteration >= 70 &
    all_comparisons_Wang2020$pairwise_percent_signif_iteration >= 50 &
    abs(all_comparisons_Wang2020$pairwise_mean_l2fc) > 1
```

In [298…
```
all_comparisons_Wang2020[significant_comparisons,]
```

A data.frame: 0 × 10

gene	celltype	comparison	LRT_percent_signif_iterations	LRT_padj_min	LRT_padj_max	pairwise_mean_l2fc	pairwise_percent_signif_iterations	pairwise_padj_m
<chr>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>	<int>	<db

DESeq2 Global pairwise results

Gobal

1. EDN1 in EC_Venous (80%), 6-fold lower in dHF compared with IZ (100 % of iterations)
2. ECE1 in vCM (85 %) no significance at pairwise levels
3. ECE1 in Pericytes (100 %) Upregulated in IZ compared to healthy, RZ and FZ over 80 % of iterations) with L2FCs of 1.54, 1.96, and 1.26 respectively.
4. ECE1 in Arterial (100 %) no significance at pairwise levels

Kuppe2022 MI only

5. EDN1 in EC_Valves_Pericytes (95 %) Pairwise significance in only 75, and 50 % of iterations, urpegulated in FZ comapred to healthy and IZ (l2fc of 4, and 2 respectively)
6. EDN1 in EC_Venous (60 %) Not significant in pairwise comparisons
7. ECE1 in vCMs (95 %) Upregulated in FZ compared to healthy and RZ (l2fc of 1.28 and 1.13, 55 or 50 iterations only); IZ compared healthy, BZ, and RZ (l2fc of 1.68, 1.22, and 1.52, in 100, 65, and 100 % of iterations)
8. ECE1 in SMCs (100 %) Upregulated in IZ compared with RZ (l2fc, 1.85, 100 % of iterations).
9. ECE1 in Pericytes (100 %) Upregulated in IZ comopared to all. in Kuppe only, IZ vs BZ is significant in 95% of iterations (l2fc 1.54. 95 %)
10. EDNRB up in capillary (70 %) Significant pairwise comparison with 2 fold changes lower in healthy vs FZ in 65 % of iterations.

1. EDN1 in EC_Venous

In [155…

```
# 1. EDN1 in EC_Venous (80 %)
gene = "EDN1"
i = 1 # iteration
j = which(celltypes2test == "EC_Venous") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions

# comparison loop
for(k in 1:length(DESeq_results_global[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

[1] "EC_Venous"

In [156…

```
DESeq_results_global[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_global[[i]][[j]][[4]])[1]
```

0.0299880668760219

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	-0.1240	2.160	-1.98	1.790	-3.27	-0.0518
cHF	0.1240	NA	2.280	-1.86	1.920	-3.15	0.0719
dHF	-2.1600	-2.2800	NA	-4.14	-0.367	-5.43	-2.2100
FZ	1.9800	1.8600	4.140	NA	3.770	-1.29	1.9300
BZ	-1.7900	-1.9200	0.367	-3.77	NA	-5.06	-1.8500
IZ	3.2700	3.1500	5.430	1.29	5.060	NA	3.2200
RZ	0.0518	-0.0719	2.210	-1.93	1.850	-3.22	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	0.960	0.1820	1.000	1.0000	0.1210	1.000
cHF	0.960	NA	0.2910	0.497	0.6580	0.1960	0.991
dHF	0.182	0.291	NA	0.107	0.9610	0.0211	0.500
FZ	1.000	0.497	0.1070	NA	1.0000	NA	1.000
BZ	1.000	0.658	0.9610	1.000	NA	0.0708	1.000
IZ	0.121	0.196	0.0211	NA	0.0708	NA	0.114
RZ	1.000	0.991	0.5000	1.000	1.0000	0.1140	NA

'cHF_vs_dHF'

In [157…
```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [158…
```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

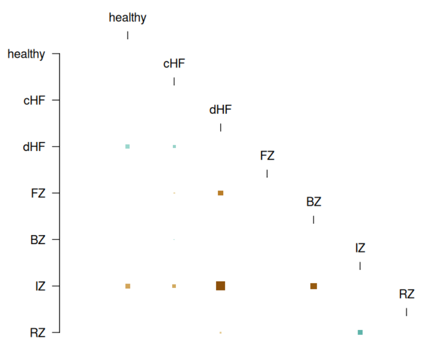
A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	0.12366121	NA	NA	NA	NA	NA	NA
dHF	-2.16102448	-2.28468568	NA	NA	NA	NA	NA
FZ	1.98065730	1.85699609	4.1416818	NA	NA	NA	NA
BZ	-1.79355082	-1.91721203	0.3674737	-3.774208	NA	NA	NA
IZ	3.27056651	3.14690530	5.4315910	1.289909	5.064117	NA	NA
RZ	0.05178708	-0.07187412	2.2128116	-1.928870	1.845338	-3.218779	NA

In [159…
```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [173…
```
pdf(paste("figures/panels/results/DESeq2/Observations_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
    }
dev.off()
options(repr.plot.width = 7)
options(repr.plot.height = 7)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_1811720811: 2



In [174…
```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
```

```
                                                batch = vsd$dataset,
                                                batch2 = vsd$cell_or_nuclei,
                                                design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

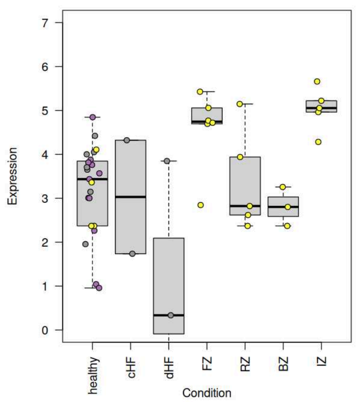
In [175…

```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)

plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df
    )
```

In [176…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/1_EDN1_ECVenous.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
 par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_824012447: 2



In [177…

```
j
celltype
gene
```

6

'EC_Venous'

'EDN1'

2. ECE1 in vCM

In [178…

```
# 2. ECE1 in vCM (85 %)
gene = "ECE1"
i = 1 # iteration
j = which(celltypes2test == "vCM") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions
```

```
# comparison loop
for(k in 1:length(DESeq_results_global[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

```
[1] "vCM"
```

In [179…

```
DESeq_results_global[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_global[[i]][[j]][[4]])[1]
```

0.0320662559485858

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	-2.050	-1.500	-1.080	-0.1310	-1.250	-0.0494
cHF	2.0500	NA	0.551	0.976	1.9200	0.798	2.0000
dHF	1.5000	-0.551	NA	0.425	1.3700	0.247	1.4500
FZ	1.0800	-0.976	-0.425	NA	0.9450	-0.178	1.0300
BZ	0.1310	-1.920	-1.370	-0.945	NA	-1.120	0.0814
IZ	1.2500	-0.798	-0.247	0.178	1.1200	NA	1.2100
RZ	0.0494	-2.000	-1.450	-1.030	-0.0814	-1.210	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	0.101	0.177	0.785	1.000	0.230	1.000
cHF	0.101	NA	0.813	0.666	0.480	0.699	0.397
dHF	0.177	0.813	NA	0.906	0.997	0.919	0.703
FZ	0.785	0.666	0.906	NA	1.000	0.927	0.643
BZ	1.000	0.480	0.997	1.000	NA	0.409	1.000
IZ	0.230	0.699	0.919	0.927	0.409	NA	0.195
RZ	1.000	0.397	0.703	0.643	1.000	0.195	NA

'cHF_vs_dHF'

In [180…

```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [181…

```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	2.0522258	NA	NA	NA	NA	NA	NA
dHF	1.5015565	-0.5506693	NA	NA	NA	NA	NA
FZ	1.0760778	-0.9761480	-0.4254787	NA	NA	NA	NA
BZ	0.1307675	-1.9214583	-1.3707890	-0.9453103	NA	NA	NA
IZ	1.2544640	-0.7977618	-0.2470925	0.1783862	1.12369650	NA	NA
RZ	0.0493911	-2.0028347	-1.4521654	-1.0266867	-0.08137642	-1.205073	NA

In [182…

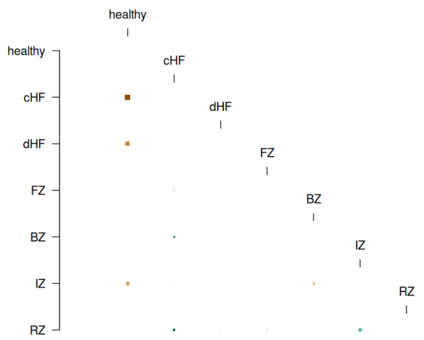
```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [183…

```
pdf(paste("figures/panels/results/DESeq2/Observations_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
    }
dev.off()
```

```
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_2128503839: 2



In [184…

```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
                                          batch2 = vsd$cell_or_nuclei,
                                          design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

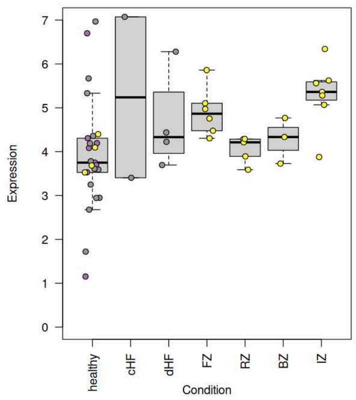
In [185…

```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)

plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df
    )
```

In [186…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/2_ECE1_vCM.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_529770049: 2



In [187…
```
j
celltype
gene
```

1
'vCM'
'ECE1'

In [188…
```
comparison = "cHF_vs_healthy"
fraction_significant = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                }
                              )
signif(range(fraction_significant, na.rm=T),3)
```

0.034 · 0.157

In [189…
```
sum(fraction_significant<0.05)/20*100
```

15

In [190…
```
comparison = "cHF_vs_healthy"
l2fc_all = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                }
                              )
signif(mean(l2fc_all, na.rm=T),3)
```

2.01

In [191…
```
comparison = "healthy_vs_IZ"
fraction_significant = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                }
                              )
signif(range(fraction_significant, na.rm=T),3)
```

0.034 · 0.259

In [194…
```
sum(fraction_significant<0.05)/20*100
```

5

In [195…
```
comparison = "healthy_vs_IZ"
l2fc_all = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                }
                              )
signif(mean(l2fc_all, na.rm=T),3)
```

-1.54

In [196…
```
# notes - vCM, ECE1 appears to be significantly different in LRT (across all groups)
# However, this did not appear to be the result of disease_subcategory in the majority of iterations.

# Example,
# the comparisons between healthy and cHF, or IZ are significant in only 15 %, or 5 % of samples.
# witht he p value ranging between 0.0034 and 0.259.
# the l2fc in both cases is between 2 and 1.5 respectively.
```

3. ECE1 in Pericytes

In [197…
```
# 3. ECE1 in Pericytes (100 %)
gene = "ECE1"
i = 1 # iteration
j = which(celltypes2test == "Pericytes") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
```

```
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions

# comparison loop
for(k in 1:length(DESeq_results_global[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

[1] "Pericytes"

In [525…

```
DESeq_results_global[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_global[[i]][[j]][[4]])[1]
```

0.00143747627011936

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	-0.148	0.0812	-1.37	0.458
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.1480	NA	NA	NA	0.2290	-1.23	0.606
BZ	-0.0812	NA	NA	-0.229	NA	-1.46	0.377
IZ	1.3700	NA	NA	1.230	1.4600	NA	1.830
RZ	-0.4580	NA	NA	-0.606	-0.3770	-1.83	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	0.9090	1.0000	0.0257	1.000
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.9090	NA	NA	NA	1.0000	0.0376	0.629
BZ	1.0000	NA	NA	1.0000	NA	0.0453	1.000
IZ	0.0257	NA	NA	0.0376	0.0453	NA	0.001
RZ	1.0000	NA	NA	0.6290	1.0000	0.0010	NA

'healthy_vs_BZ'

In [198…

```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [199…

```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.14798541	NA	NA	NA	NA	NA	NA
BZ	-0.08120007	NA	NA	-0.2291855	NA	NA	NA
IZ	1.37465750	NA	NA	1.2266721	1.4558576	NA	NA
RZ	-0.45804342	NA	NA	-0.6060288	-0.3768433	-1.832701	NA

In [200…

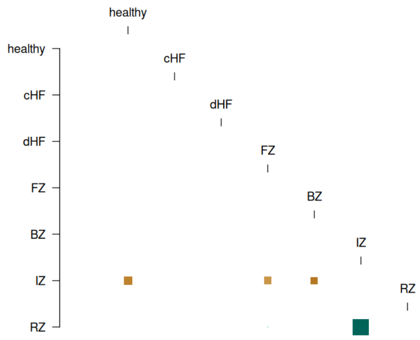
```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [201…

```
pdf(paste("figures/panels/results/DESeq2/Observations_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
```

```
    }
dev.off()

matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_1524842253: 2



In [202…

```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
                                          batch2 = vsd$cell_or_nuclei,
                                          design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

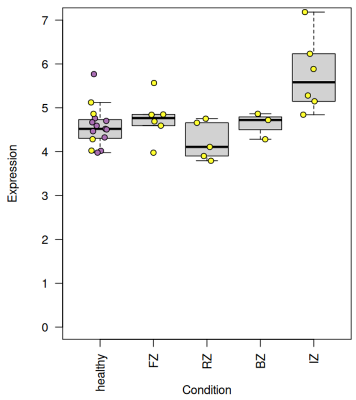
In [203…

```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)

plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df
    )
```

In [204…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/3_ECE1_Pericytes.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_2076957298: 2



In [205…
```
j
celltype
gene
```
4

'Pericytes'

'ECE1'

In [206…
```
comparison = "healthy_vs_IZ"
fraction_significant = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                }
                              )
signif(range(fraction_significant, na.rm=T),3)
```
0.000578 · 0.0318

In [207…
```
sum(fraction_significant<0.05)/20*100
```
100

In [208…
```
l2fc_all = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                }
                              )
signif(mean(l2fc_all, na.rm=T),3)
```
-1.54

In [209…
```
comparison = "FZ_vs_IZ"
fraction_significant = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                }
                              )
signif(range(fraction_significant, na.rm=T),3)
```
0.00401 · 0.0594

In [210…
```
sum(fraction_significant<0.05)/20*100
```
95

In [211…
```
l2fc_all = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                }
                              )
signif(mean(l2fc_all, na.rm=T),3)
```
-1.26

In [212…
```
comparison = "IZ_vs_RZ"
fraction_significant = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                }
                              )
signif(range(fraction_significant, na.rm=T),3)
```
9.21e-06 · 0.00269

In [213…
```
sum(fraction_significant<0.05)/20*100
```
100

In [214…
```
l2fc_all = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                }
                              )
signif(mean(l2fc_all, na.rm=T),3)
```
1.96

In [215…
```
comparison = "BZ_vs_IZ"
fraction_significant = sapply(1:20, function(i){
```

```
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                }
                              )
signif(range(fraction_significant, na.rm=T),3)
```

0.00591 · 0.0662

In [216…
```
sum(fraction_significant<0.05)/20*100
```

85

In [217…
```
l2fc_all = sapply(1:20, function(i){
                                DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"log2FoldChange"]
                                }
                              )
signif(mean(l2fc_all, na.rm=T),3)
```

-1.54

In [218…
```
# notes - ECE1 appears to be significantly different between disease conditiosn in Pericytes (LRT < 0.05).

# In pairwise comparisons, ECE1 is significantly higher by (between 1.26 to 1.96 log 2 fold-change) in IZ compared to BZ, RZ, FZ, an
# 85, 100, 95, nand 100 of psuedobulk sample iterations.
```

4. ECE1 in Arterial

In [231…
```
# 4. ECE1 in EC_Arterial (100 %)
gene = "ECE1"
i = 1 # iteration
j = which(celltypes2test == "EC_Arterial") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions

# comparison loop
for(k in 1:length(DESeq_results_global[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_global[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_global[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_global[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

```
[1] "EC_Arterial"
```

In [232…
```
DESeq_results_global[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_global[[i]][[j]][[4]])[1]
```

0.008818968538619

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	-0.526	-1.630	-0.344	0.0517	-1.280	0.231
cHF	0.5260	NA	-1.100	0.182	0.5780	-0.755	0.758
dHF	1.6300	1.100	NA	1.280	1.6800	0.344	1.860
FZ	0.3440	-0.182	-1.280	NA	0.3960	-0.938	0.575
BZ	-0.0517	-0.578	-1.680	-0.396	NA	-1.330	0.179
IZ	1.2800	0.755	-0.344	0.938	1.3300	NA	1.510
RZ	-0.2310	-0.758	-1.860	-0.575	-0.1790	-1.510	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	0.577	0.0546	1.000	1.000	0.328	1.0000
cHF	0.5770	NA	0.2770	0.904	0.707	0.658	0.5450
dHF	0.0546	0.277	NA	0.248	0.169	0.883	0.0957
FZ	1.0000	0.904	0.2480	NA	1.000	0.998	NA
BZ	1.0000	0.707	0.1690	1.000	NA	0.376	1.0000
IZ	0.3280	0.658	0.8830	0.998	0.376	NA	0.2280
RZ	1.0000	0.545	0.0957	NA	1.000	0.228	NA

'cHF_vs_dHF'

In [233…
```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [234…
```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

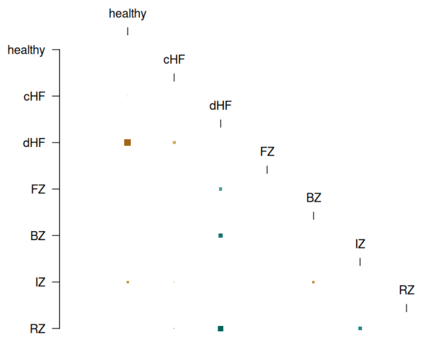
A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	0.5263797	NA	NA	NA	NA	NA	NA
dHF	1.6257128	1.0993330	NA	NA	NA	NA	NA
FZ	0.3442364	-0.1821433	-1.2814763	NA	NA	NA	NA
BZ	-0.0517399	-0.5781196	-1.6774526	-0.3959763	NA	NA	NA
IZ	1.2817474	0.7553677	-0.3439653	0.9375110	1.3334873	NA	NA
RZ	-0.2312390	-0.7576187	-1.8569517	-0.5754754	-0.1794991	-1.512986	NA

In [235…
```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [236…
```
pdf(paste("figures/panels/results/DESeq2/Observations_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
    }
dev.off()

matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_226065466: 2



In [237…
```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
                                          batch2 = vsd$cell_or_nuclei,
                                          design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

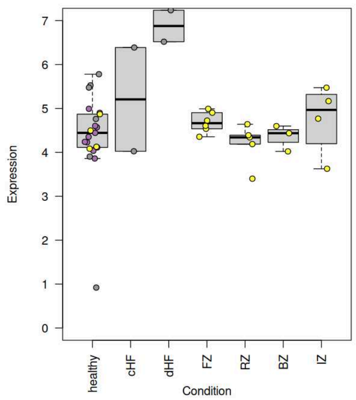
In [238…
```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)
```

```
plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df:
    )
```

In [239…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/4_ECE1_Arterial.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_27777438: 2



In [240…

```
j
celltype
gene
```

7

'EC_Arterial'

'ECE1'

In [241…

```
comparison = "dHF_vs_healthy"
fraction_significant = sapply(1:20, function(i){
                                    DESeq_results_global[[i]][[j]][[4]][[comparison]][gene,"padj"]
                                    }
                                  )
signif(range(fraction_significant, na.rm=T),3)
```

0.0533 · 0.0705

In [242…

```
sum(fraction_significant<0.05)/20*100
```

0

In [245…

```
# make a batch-corrected matrix for this iteration (i)
dds = DESeq_results_global[[i]][[j]][[2]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
                                          batch2 = vsd$cell_or_nuclei,
                                          design = model.matrix(~disease_subcategory, data=colData(vsd)))
```

In [246…

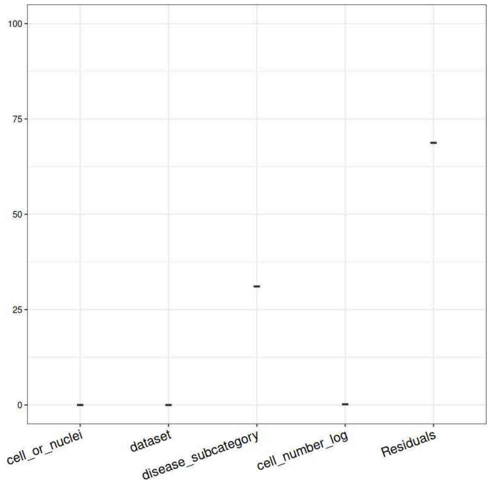
```
# 1. Define your formula
form <- ~ (1|dataset) + (1|cell_or_nuclei) + (1|disease_subcategory) + cell_number_log

# 2. Run the partition (uses your VST counts)
varPart <- fitExtractVarPartModel(
    mat_corrected,
    form,
    colData(dds))
```

```
Warning message in .fitExtractVarPartModel(exprObj, formula, data, REML = REML, :
“Model failed for 2 responses.
  See errors with attr(., 'errors')”
```

In [247…

```
# plot variance results
plotVarPart(varPart["ECE1",,drop=F])
```

```
Warning message:
“Groups with fewer than two datapoints have been dropped.
i Set `drop = FALSE` to consider such groups for position adjustment purposes.”
Warning message:
“Groups with fewer than two datapoints have been dropped.
i Set `drop = FALSE` to consider such groups for position adjustment purposes.”
Warning message:
“Groups with fewer than two datapoints have been dropped.
i Set `drop = FALSE` to consider such groups for position adjustment purposes.”
Warning message:
“Groups with fewer than two datapoints have been dropped.
i Set `drop = FALSE` to consider such groups for position adjustment purposes.”
Warning message:
“Groups with fewer than two datapoints have been dropped.
i Set `drop = FALSE` to consider such groups for position adjustment purposes.”
Warning message:
“Computation failed in `stat_ydensity()`.
Caused by error in `$<-.data.frame`:
! replacement has 1 row, data has 0”
```



In [ ]:

```
# notes - ECE1 appears to be significantly altered across all conditions in EC_Arterial (LRT < 0.05).
# However, no significant pairwise comparisons were observed.
# Likely due to error and large spread of data and latent factors (residuals contributing most to overall variance).
```

5. EDN1 in EC_Valves_Pericytes (95 %)

In [330…

```
# 5. EDN1 in EC_Valves_Pericytes (100 %)
gene = "EDN1"
i = 1 # iteration
j = which(celltypes2test == "EC_Valves_Pericytes") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions

# comparison loop
for(k in 1:length(DESeq_results_Kuppe2022[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_Kuppe2022[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_Kuppe2022[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_Kuppe2022[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

```
[1] "EC_Valves_Pericytes"
```

In [321…

```
DESeq_results_Kuppe2022[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[1]
```

6.8498673479896e-05

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	-3.59	2.07	-1.33	-0.801
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	3.590	NA	NA	NA	5.66	2.26	2.790
BZ	-2.070	NA	NA	-5.66	NA	-3.40	-2.870
IZ	1.330	NA	NA	-2.26	3.40	NA	0.530
RZ	0.801	NA	NA	-2.79	2.87	-0.53	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	0.0111	1.000	0.3940	1.0000
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.0111	NA	NA	NA	NA	0.0179	0.0517
BZ	1.0000	NA	NA	NA	NA	0.4960	1.0000
IZ	0.3940	NA	NA	0.0179	0.496	NA	0.7430
RZ	1.0000	NA	NA	0.0517	1.000	0.7430	NA

'healthy_vs_BZ'

In [322…

```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [323…

```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	3.5893191	NA	NA	NA	NA	NA	NA
BZ	-2.0717677	NA	NA	-5.661087	NA	NA	NA
IZ	1.3308797	NA	NA	-2.258439	3.402647	NA	NA
RZ	0.8010455	NA	NA	-2.788274	2.872813	-0.5298343	NA

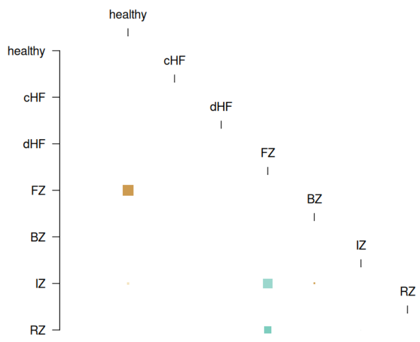
In [324…

```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [325…

```
pdf(paste("figures/panels/results/DESeq2/ObservationsKuppe2022_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
    }
dev.off()

matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_82338927: 2



In [326…

```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
                                          batch2 = vsd$cell_or_nuclei,
                                          design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

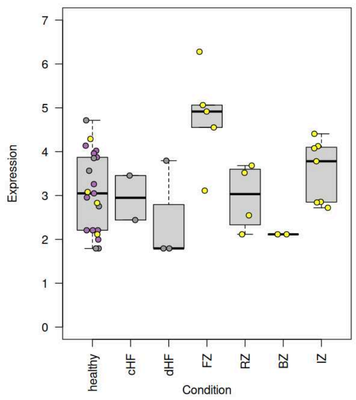
In [327…

```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)

plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df
    )
```

In [328…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/5_EDN1_ValvesPericytes.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_516306940: 2



In [329…

```
j
celltype
gene
```

5

'EC_Valves_Pericytes'

'EDN1'

In [ ]:

```
```

8. ECE1 in SMCs

In [299…

```
# 8. ECE1 in SMCs
gene = "ECE1"
i = 1 # iteration
j = which(celltypes2test == "SMC") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions

# comparison loop
for(k in 1:length(DESeq_results_Kuppe2022[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_Kuppe2022[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_Kuppe2022[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_Kuppe2022[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

```
[1] "SMC"
```

In [300…

```
DESeq_results_Kuppe2022[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[1]
```

0.0199062591799795

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	-0.548	-0.0231	-1.250	0.611
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.5480	NA	NA	NA	0.5250	-0.699	1.160
BZ	0.0231	NA	NA	-0.525	NA	-1.220	0.634
IZ	1.2500	NA	NA	0.699	1.2200	NA	1.860
RZ	-0.6110	NA	NA	-1.160	-0.6340	-1.860	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	0.586	1.000	0.10800	1.00000
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.586	NA	NA	NA	0.742	0.48500	0.14000
BZ	1.000	NA	NA	0.742	NA	0.15300	1.00000
IZ	0.108	NA	NA	0.485	0.153	NA	0.00912
RZ	1.000	NA	NA	0.140	1.000	0.00912	NA

'healthy_vs_BZ'

In [301…

```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [302…

```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.54760169	NA	NA	NA	NA	NA	NA
BZ	0.02307706	NA	NA	-0.5245246	NA	NA	NA
IZ	1.24666153	NA	NA	0.6990598	1.2235845	NA	NA
RZ	-0.61119956	NA	NA	-1.1588012	-0.6342766	-1.857861	NA

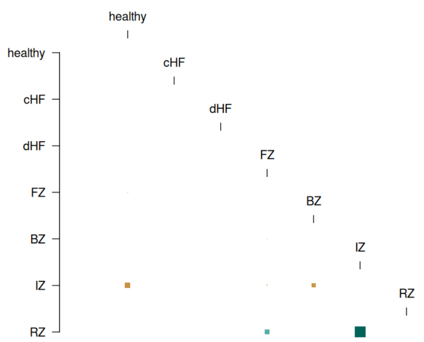
In [303…

```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [304…

```
pdf(paste("figures/panels/results/DESeq2/ObservationsKuppe2022_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
    }
dev.off()

matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_1507926978: 2



In [305…

```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
```

```
                                              batch2 = vsd$cell_or_nuclei,
                                              design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

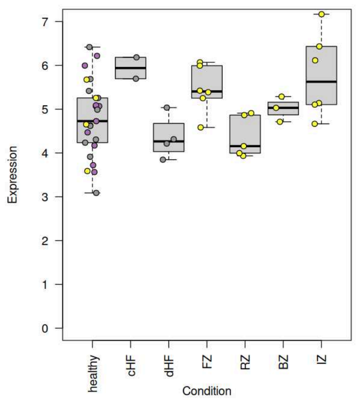
In [306…

```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)

plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df$
    )
```

In [307…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/8_ECE1_SMC.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_2093254060: 2



In [308…

```
j
celltype
gene
```

3

'SMC'

'ECE1'

In [ ]:

11. EDNRB in EC_Capillary

In [309…

```
# 11. EDNRB in EC_Capillary
gene = "EDNRB"
i = 1 # iteration
j = which(celltypes2test == "EC_Capillary") # celltype
celltype=celltypes2test[j]

condition_pair_matrix_pVal = matrix(NA, length(conditions), length(conditions))
rownames(condition_pair_matrix_pVal) = conditions
colnames(condition_pair_matrix_pVal) = conditions

condition_pair_matrix_l2fc = matrix(NA, length(conditions), length(conditions))
```

```
rownames(condition_pair_matrix_l2fc) = conditions
colnames(condition_pair_matrix_l2fc) = conditions

# comparison loop
for(k in 1:length(DESeq_results_Kuppe2022[[i]][[j]][[4]])){
    x_lab = gsub("_vs_.*$", "", names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[k])
    y_lab = gsub("^.*_vs_", "", names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[k])
    condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"padj"]
    condition_pair_matrix_pVal[y_lab,x_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"padj"]

    condition_pair_matrix_l2fc[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]
    condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_Kuppe2022[[i]][[j]][[4]][[k]][gene,"log2FoldChange"]

     #y_lab = gsub("_vs_.*$", "", names(DESeq_results_Kuppe2022[[i]][[j]][[2]])[k])
     #x_lab = gsub("^.*_vs_", "", names(DESeq_results_Kuppe2022[[i]][[j]][[2]])[k])
     #condition_pair_matrix_pVal[x_lab,y_lab] = DESeq_results_Kuppe2022[[i]][[j]][[2]][[k]][gene,"padj"]
     #condition_pair_matrix_l2fc[y_lab,x_lab] = -DESeq_results_Kuppe2022[[i]][[j]][[2]][[k]][gene,"log2FoldChange"]

    }
condition_pair_matrix_pVal[condition_pair_matrix_pVal<0.001] = 0.001
print(celltype)
```

```
[1] "EC_Capillary"
```

In [310…

```
DESeq_results_Kuppe2022[[i]][[j]][[4]][[1]][gene,"LRT_padj"]
signif(condition_pair_matrix_l2fc,3)
signif(condition_pair_matrix_pVal,3)
names(DESeq_results_Kuppe2022[[i]][[j]][[4]])[1]
```

0.00349298122710752

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	-1.79	-0.2740	0.108	-0.2980
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	1.790	NA	NA	NA	1.5200	1.900	1.4900
BZ	0.274	NA	NA	-1.52	NA	0.382	-0.0236
IZ	-0.108	NA	NA	-1.90	-0.3820	NA	-0.4060
RZ	0.298	NA	NA	-1.49	0.0236	0.406	NA

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	0.0141	1.000	0.991	1.0000
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	0.0141	NA	NA	NA	0.167	0.582	0.0558
BZ	1.0000	NA	NA	0.1670	NA	1.000	1.0000
IZ	0.9910	NA	NA	0.5820	1.000	NA	0.9650
RZ	1.0000	NA	NA	0.0558	1.000	0.965	NA

'healthy_vs_BZ'

In [311…

```
for(column  in 1:ncol(condition_pair_matrix_l2fc)){
    condition_pair_matrix_l2fc[column ,column :ncol(condition_pair_matrix_l2fc)] = NA
    }
```

In [312…

```
# cut the matrix into triangles
condition_pair_matrix_l2fc
```

A matrix: 7 × 7 of type dbl

	healthy	cHF	dHF	FZ	BZ	IZ	RZ
healthy	NA	NA	NA	NA	NA	NA	NA
cHF	NA	NA	NA	NA	NA	NA	NA
dHF	NA	NA	NA	NA	NA	NA	NA
FZ	1.7911428	NA	NA	NA	NA	NA	NA
BZ	0.2744968	NA	NA	-1.516646	NA	NA	NA
IZ	-0.1078728	NA	NA	-1.899016	-0.3823696	NA	NA
RZ	0.2981387	NA	NA	-1.493004	0.0236419	0.4060114	NA

In [313…

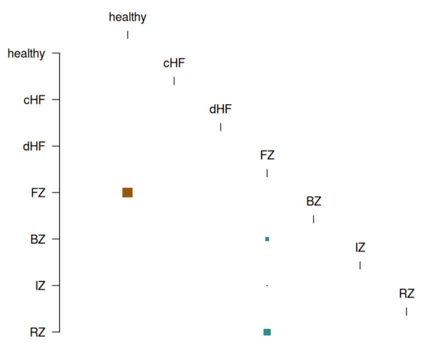
```
bg_matrix = matrix(1, nrow(condition_pair_matrix_l2fc), ncol(condition_pair_matrix_l2fc))
for(column in 1:ncol(bg_matrix)){
    bg_matrix[column ,column :ncol(bg_matrix)] = NA
    }
diag(bg_matrix) = 1
rownames(bg_matrix) = rownames(condition_pair_matrix_l2fc)
```

In [314…

```
pdf(paste("figures/panels/results/DESeq2/ObservationsKuppe2022_", gene, "_", celltype, ".pdf",se=""), width=5, height=5)
par(mar=c(5,5,5,5))
matrix.plot(bg_matrix,round.n=1, cex.multiplier = 3, col="grey", x.axis=F,  pch=0)
par(new=T)
matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=
#matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = 3, col=Earth, x.axis=F, new=T)

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*1.9, colnames(condition_pair_matrix_l2
```

```
    }
dev.off()

matrix.plot(condition_pair_matrix_l2fc,round.n=3, cex.multiplier = -log10(condition_pair_matrix_pVal), col=rev(Earth), x.axis=F, new=

for(column in 1:ncol(condition_pair_matrix_l2fc)){
    axis(3, at = column, labels=(colnames(condition_pair_matrix_l2fc)[column]), line=1-column*3.2, colnames(condition_pair_matrix_l2
    }
```

agg_record_855664340: 2



In [315…

```
corrected_matrix = sapply(1:20, function(i){
dds = DESeq_results_global[[i]][[j]][[1]]

vsd <- varianceStabilizingTransformation(dds, blind=FALSE)

# Remove batch effects for visualization
mat_corrected <- limma::removeBatchEffect(assay(vsd),
                                          batch = vsd$dataset,
                                          batch2 = vsd$cell_or_nuclei,
                                          design = model.matrix(~disease_subcategory, data=colData(vsd)))
    return(mat_corrected[gene, ])
}
       )
```

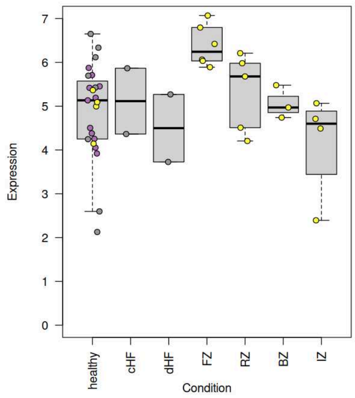
In [316…

```
# Calculate the mean for each sample across the 20 iterations
final_sample_means <- rowMeans(corrected_matrix)

# Create a plotting dataframe
plot_df <- data.frame(
  "Expression" = final_sample_means,
  "Condition" = factor(DESeq_results_global[[i]][[j]][[1]]$disease_subcategory, levels=),
  "Dataset" = DESeq_results_global[[i]][[j]][[1]]$dataset
)

plot_df$Condition = factor(
    as.character(plot_df$Condition),
    levels=c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ")[c("healthy", "cHF", "dHF", "FZ", "RZ", "BZ", "IZ") %in% levels(plot_df
    )
```

In [318…

```
dataset_cols = c(
    "#e41a1cff",
    "#4f8dc0ff",
    "#ac71b5ff",
    "#ffff33ff",
    "#f781bfff",
    "#999999ff"
    )

spread = runif(1000, -0.2, 0.2)
pdf("figures/panels/results/DESeq2/11_EDNRB_Capillary.pdf",width=7, height=7 )
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
dev.off()
    par(mar=c(10,5,2,10))
    boxplot(Expression ~ Condition, plot_df, las=2, outline = F, ylim=c(0,7))
    points(
        as.integer(plot_df$Condition)+sample(spread, nrow(plot_df)),
        plot_df$Expression,
        bg=dataset_cols[c(3,4,6)][plot_df$Dataset],
        pch=21)
```

agg_record_542795345: 2



In [319…
```
j
celltype
gene
```

8
'EC_Capillary'
'EDNRB'

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

sanity checking

In [365…
```
# Just to sanity check the re-clustering. Even if we used the author's original annotation(s)
#sample_check = levels(bulk_class)[grep("^EC_Valves_Pericytes", levels(bulk_class))]
sample_check = colnames(bulk_data[[1]])[which(bulk_metadata[[1]]$dataset == "Kuppe2022" & bulk_metadata[[1]]$celltype == "EC_Valves_I

#bulk_class_condition = sample_check[1]
sample_check_table = sapply(sample_check, function(bulk_class_condition){
        table(
            metadata[bulk_samples[[1]][[which(levels(bulk_class) == bulk_class_condition)]],"cell_type_mid"]
        )
}
)
```

In [369…
```
sample_check_table[rowSums(sample_check_table) > 0,,drop=F]
# all some sort of cardiac endothelial cell label...
# The previous authors did not dive deeped, although the publication does show separation between EC types, including the Endocardiu
```

	EC_Valves_Pericytes__adult__healthy__P1__nuclei	EC_Valves_Pericytes__adult__healthy__P17__nuclei	EC_Valves_Pericytes__adult__healthy__P7__nuclei
cardiac endothelial cell	100	100	10